

Numerical Production Contract for an Adaptive Chiral-Defect CFT

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This document is a preregistered numerical specification for the proposed PRX Quantum manuscript on a postselection-free adaptive class-A fermion circuit. Its stochastic campaigns remain prospective and license no present-tense production claim. Completed deterministic calibration or reconstruction results are labelled explicitly and are not pooled with production data. The contract fixes the physical ensemble, system-size sequences, sample counts, fixed-duration logic, saved data, estimator order, controls, uncertainty analysis, acceptance tests, claim downgrades, and five-figure architecture. Every GPU circuit trajectory must be generated by the canonical `classA_U1FGTN_gpu.run_markov_circuit` entry point on an A100-class 40 GB runtime. The central acceptance standard is deliberately conjunctive: quantitative revalidation of the inherited BPJ bulk, a $c = 1$ and level-one $U(1)$ wall state, physical chiral response, an integer twist spectral flow, a wall-localized tangent sector controlling purification, and universal information recoverable from trajectory-resolved records must all survive thermodynamic, transverse, temporal, and protocol controls.

CONTENTS

		B. Campaign S2: topological-to-trivial sweep	23
I. Scope and evidentiary rules	3	XI. Tangent dynamics and purification	25
II. PRX Quantum thesis and inherited BPJ baseline	3	A. Campaign T1: quenched tangent spectrum and localization	25
III. Canonical protocol and fixed production choices	3	B. Campaign P2: max-mix purification and physical tangent scaling	27
IV. Quantum-simulation implementation and operational observables	4	XII. Exactly three chirality routes	30
V. Reproducibility, sharding, and output schema	5	A. Campaign H1: modular charge drift	30
VI. Master trajectory families and reuse rule	6	B. Campaign H2: physical Born response $\chi(k, \omega)$	32
A. Minimum gate-first execution matrix	7	C. Campaign H3: frozen-record integer twist flow	34
VII. Validation ladder	8	XIII. Trajectory-resolved universality	34
A. Campaign V0: engine and observer validation	8	A. Campaign G5: exact fixed-word many-body singular spectrum	34
B. Campaign V1: reduced serial-order sensitivity study	8	XIV. Mean, postselection, and noise controls	36
VIII. Exact domain-wall calibration	11	A. Campaign M1/L1: exact mean channel and continuous-time generator	36
A. Campaign B0: deterministic analysis-pipeline benchmark	11	B. Campaign M2: postselection and imperfect feedback	38
B. Optional B1: controller-frame surrogate-attraction diagnostic	12	C. Campaign M3: noise and the BPJ bulk transition	39
C. Campaign B2: feedback activity and slow dynamics	14	XV. Figure architecture	40
IX. Topology and wall construction	15	A. Main figures	40
A. Campaign P1: quantitative revalidation of the BPJ bulk baseline	16	B. Supplement and schematic requirements	41
B. Campaign W1: wall realization and transverse-convergence gate	18	XVI. Production order, deterministic CPU work, and A100 resource staging	41
X. Universal prepared-state data	19	A. Colab bundle and Drive-staging contract	41
A. Campaign S1: state CFT and charge sector	19	B. Audited legacy implementation lineage	43
		XVII. Claim-downgrade matrix	46
		XVIII. Deferred work	48

A. Overlap-section winding and the occupied Bloch bundle	48
B. Frozen-record $U(1)$ twist family and wall spectral flow	49
1. Boundary holonomy and twisted controller projectors	49
2. One record, one continuous conditional branch	49
3. Entanglement flow, wall resolution, and crossing count	50
C. Quenched finite-time tangent spectra and localization modes	50
1. Cycle Jacobian, product, and time normalization	50
2. Stabilized QR accumulation and finite-time singular vectors	51
3. Quenched spectrum, mean log strain, and annealed spectrum	51
4. Localization of a nonunique slow band	51
D. Charge full counting statistics and the level-one current algebra	52
E. Common correlation-spectrum origin of entropy and charge fluctuations	52
1. Spectral calculus of the restricted covariance	53
2. CFT coefficient and level-one test	53
3. Cross-boundary coherence and intrinsic mixedness	53
4. Full charge statistics and ensemble decomposition	54
5. Production estimator and acceptance rule	54
F. Circuit depth, Lu bound, wall hybridization, and fixed duration	55
G. Campaign-card checklist	56
References	56

I. SCOPE AND EVIDENTIARY RULES

All stochastic production campaigns below retain status **PLANNED**. The deterministic B0 review calibration and the compact M1/L1 atlas reconstruction are completed and recorded explicitly; the full M1/L1 production matrix remains planned. No saved plot, notebook output, or previous fit is admissible as production evidence. Repository artifacts may be cited only in a field labelled “legacy comparison,” and only when the observable and protocol have been checked at the path level. Such a comparison is never pooled with production data.

The physical object is the Born ensemble of adaptive trajectories. For a record $\xi = (m_1, \dots, m_T)$,

$$\begin{aligned} K_\xi &= K_{m_T|\xi_{<T}} \cdots K_{m_1}, \quad p_\xi = \text{Tr}(K_\xi \rho_0 K_\xi^\dagger), \\ p_\xi &= K_\xi \rho_0 K_\xi^\dagger / p_\xi. \end{aligned} \quad (1)$$

Perfect correction samples every m_t from its conditional Born probability and then applies the prescribed feedback. Forced postselection is a control. The mean channel is a second, inequivalent control. For every nonlinear observable,

$$\mathbb{E}_\xi[O(C_\xi)] \neq O(\mathbb{E}_\xi[C_\xi]) \quad (2)$$

in general. The estimator must therefore be evaluated per trajectory before equal-weight averaging. Since the trajectories are Born sampled, no second probability weight is applied.

The manuscript is ready for a PRX Quantum submission only if the five gates in Table I pass. A campaign may produce an interesting result while failing its predeclared gate; in that case the corresponding weaker wording, not a modified estimator, is used.

II. PRX QUANTUM THESIS AND INHERITED BPJ BASELINE

The numerical program is not designed to re-establish the premises already supplied by the Bhuiyan–Pan–Jian (BPJ) construction [1]. BPJ already provides the mEO/sTM setting, a postselection-free finite-range class-A adaptive instrument, Chern trajectory ensembles, programmable dynamical domain walls, finite-range parallelization, an ancilla–fSWAP implementation route, and quantum-simulation prospects on encoded-qubit or native-fermion hardware. It also reports coherent-noise robustness and distinct trajectory and averaged thresholds. Accordingly, this manuscript shall not advertise postselection freedom, ancilla feedback, Gaussian simulability, finite-range state preparation, or the mere existence of a domain-wall slow mode as its standalone novelty.

The proposed advance is the conjunction of results absent from that baseline. The perfect-correction instrument should program a *universal chiral defect* across typical retained Born trajectories: each trajectory may occupy a

different definite system-charge sector, but the ensemble shares $c_{\text{edge}} = 1$ and level-one $U(1)$ wall data, its sector wandering is localized at the walls, its response and flux flow are oriented, and its slow purification is controlled by a singular conditional tangent sector. The strongest version further recovers defect data from the fixed-word trajectory-operator spectrum and shows precisely what survives or disappears in the unconditional channel. This is the unique intersection to which every headline panel must contribute.

The quantum-simulation proposition is therefore operational rather than computational. It is not a quantum-advantage claim, nor a claim that an adaptive platform can prepare a Chern state for the first time. It is a protocol for realizing and interrogating a programmable topological defect ensemble without rejecting records, with a hierarchy of record-only, fixed-word operator, number-measurement, and covariance-tomography observables. A result that cannot be placed in this hierarchy or tied to one of Gates G1–G5 is supplemental or deferred even if it is numerically interesting.

III. CANONICAL PROTOCOL AND FIXED PRODUCTION CHOICES

Every GPU script or notebook that simulates the circuit shall configure and call `classA_U1FGTN_gpu.run_markov_circuit`. It shall not reproduce the cycle loop or call private local-update helpers. New dynamics functionality belongs in the canonical class. Every output manifest records this entry point explicitly.

The base physical protocol outside the versioned P1 shell sweep is

$$\begin{aligned} n_{\text{shell}} &= 1, \quad \alpha_{\text{top}} = 1, \quad \alpha_{\text{triv}} = 30, \\ \text{perfect_correction} &= \text{True}, \\ \text{postselect} &= \text{False}. \end{aligned} \quad (3)$$

Campaign P1 instead evaluates every declared pure and maximally mixed bulk point at $n_{\text{shell}} = 2, 1, \text{None}$; the first two use the local backend and the untruncated **None** realization uses the dense backend. The production implementation is the ordered serial instrument implemented by the canonical dynamics engine. A fresh random permutation of the active centers is drawn for each trajectory and cycle and stored in metadata; the four controller channels at a center are applied in the fixed declared engine order. Raster and reverse-raster comparisons are not part of the production campaign. Campaign V0 validates exact generation and replay of the declared random-serial schedule; historical raster data remain legacy comparisons only. The candidate quasi-one-dimensional sequence is

$$N_x = 20, \quad N_y \in \{24, 32, 40, 48, 64, 80\}, \quad (4)$$

but $N_x = 20$ is not accepted until Campaign W1 passes. Every stochastic parameter point uses a base batch of $S = 25$ independently initialized trajectories. There is no

TABLE I. Conjunctive manuscript gates. “Pass” means that every campaign-specific finite-size, transverse, temporal, and control test has also passed.

gate	required result	failure consequence
G1	finite-range fresh-ancilla controllers reproduce the BPJ bulk phase, and the explicit-interface and support-terminated constructions converge to the same wall theory	remove the validated-topological-platform claim and/or treat the two wall realizations as distinct programmable defects
G2	state data give $c_{\text{edge}} = 1$, $U(1)$ level $k = 1$, and the expected correlator	say “critical wall,” not “chiral defect CFT”
G3	total Born response is chiral and frozen-record flux has integer flow	do not claim operational chirality or a branchwise integer wall index
G4	a wall-localized tangent gap closes and controls max-mix purification	retain tangent construction only as a finite-time diagnostic
G5	the exact tangent-operator dictionary yields a charge-resolved many-body singular spectrum with $k = 1$ and a stable matched-control Casimir coefficient	report only the finite-word factorized spectrum, without a universal tower claim

separate physical burn-in. Unless a campaign explicitly defines a shorter deterministic validation, each stochastic trajectory runs for a total of $2N_y$ complete cycles from its independent initial state. If a predeclared confidence interval is too broad to decide its gate, launch a second complete $S = 25$ batch for *every* member of the affected comparison family, with a new root seed. Never add samples only to an inconvenient point. The first and second batches remain identified in metadata and are combined only after a heterogeneity check.

The later versioned production decision supersedes the preceding active sampling prescription without altering the historical $S = 25$ evidence quoted elsewhere in this document. Every newly launched ordinary stochastic point now targets

$$S = 10 = 2 \times 5 \quad (5)$$

independent trajectories in two immutable five-trajectory shards. Deterministic and forced-postselection controls remain one-sample cases. Checksum-verified results already completed under the $S = 25$ contract are retained as statistical supersets: all available trajectories are analyzed, no completed trajectory is discarded, and no case with at least ten compatible trajectories is rerun. Compatibility requires identical geometry, physical run parameters, observation contract, sample indices, shard seed, and canonical engine, with only the declared total sample count allowed to differ. New results occupy the separate namespace `classA_final_production_outputs/production_10sample_v2`; the `v1` namespace and the original unversioned output tree are read-only provenance. Compatible reuse searches `v1` first and then the unversioned tree.

Campaign H3 remains a single frozen record from parent shard 0, record index 0, for each of the explicit-interface and matched-trivial protocols. Its closed twist circle now contains 33 points,

$$\phi_j = \frac{2\pi j}{32}, \quad j = 0, \dots, 32, \quad (6)$$

including both endpoints. This is still a representative branch-level spectral-flow diagnostic and not an estimate

of a Born-ensemble frequency. Equations (5) and (6) control every active campaign declaration below; remaining $S = 25$ and 65-point statements are retained only when they describe superseded contracts or historical data.

Dense reduced-Choi covariance tracking is excluded from every production trajectory. All P2 and S2 calls set `track_choi=False`; historical Choi datasets remain valid proxy evidence but do not define a production observable or acceptance gate. All topology, tangent, twist, and fit inputs use `complex128` or `float64`. Figure defaults are 3.375 inches per single-column panel, 300 dpi, CMU Sans Serif, and at least 8 pt text.

IV. QUANTUM-SIMULATION IMPLEMENTATION AND OPERATIONAL OBSERVABLES

The primary model is the fresh-ancilla, perfect-correction idealization of the BPJ adaptive circuit. For a controller mode χ_j with target occupation $s_j \in \{0, 1\}$, measure n_{χ_j} and retain the Born outcome m_j . If $m_j \neq s_j$, prepare a fresh local ancilla mode in occupation s_j , fSWAP it with χ_j , and trace out or reset the outgoing ancilla. Thus the system charge transfer is

$$Y_j = s_j - m_j, \quad (7)$$

and every outcome is a valid retained run. No success flag, trajectory rejection, or reweighting is permitted in the primary data. The older dynamical-bath implementation, where correction succeeds only when a reused ancilla happens to have the needed occupation, is an imperfect-feedback control rather than the definition of perfect correction.

For $n_{\text{shell}} = 1$, an instrument block may be compiled by a number-conserving Gaussian basis rotation on the 3×3 -cell patch that maps χ_j to a readout mode, followed by occupation readout, classical feedforward, target-occupation ancilla preparation, fSWAP, and the required inverse basis rotation. The hardware report shall give the two-mode-gate count and depth of this compilation as d_{inst} , the measurement and feedforward latency, the

number of concurrently live ancillas, reset fidelity, and the colored-sweep bound in Eq. (F6). The same ancilla pool can be recycled between color/channel blocks. Fermionic locality does not automatically imply qubit locality in two dimensions: a qubit implementation must state its Jordan–Wigner, Bravyi–Kitaev, swap-network, or auxiliary-gauge encoding and include that mapping overhead. A native fermionic processor may use the fermionic compilation directly.

Because controllers within each conflict-graph color have disjoint support, their local instruments commute and admit simultaneous execution on a quantum simulator. A numerical evaluation of such a color block may nevertheless apply its instruments sequentially, which samples the identical joint Born law while avoiding a separate correlated multi-mode sampler. This optional finite-color compilation is a quantum-simulation resource advantage, not a requirement of the production numerics. Reordering noncommuting color blocks relative to the production random-serial word would define a different microscopic schedule and is not claimed to leave finite-size trajectories unchanged.

The operational hierarchy is:

1. the measurement/correction record, activity, transfer FCS, and total system-charge history require no state tomography;
2. boundary charge profiles, Eq. (34), and the real-time response require repeated occupation measurements and local source control;
3. the one-body covariance, local marker, and occupation spectrum require a correlation-matrix tomography protocol;
4. Rényi entropies require replicas, randomized measurements, or an equivalent interferometric protocol, while modular-Hamiltonian evolution may remain a trajectory-reconstruction diagnostic rather than a direct hardware primitive.

The simulation package shall attach shot-complexity and calibration estimates to each observable proposed as operational. Since Gaussian circuits are classically simulable, the manuscript does not claim computational quantum advantage. The quantum simulation value is controlled realization of a programmable topological defect and access to typical conditioned states and their associated trajectory records, without conditioning on an exponentially unlikely favorable record.

This implementation inherits rather than supersedes the BPJ hardware proposal. Its concrete new simplification is that a fresh ancilla is prepared in the target occupation s_j for each wrong branch, so the correction is deterministic once the Born outcome is known; no ancillary redistribution layer or finite bath-occupation success probability is part of the default model. Production resource tables must compare this fresh-ancilla cost with

the reused-bath BPJ primitive instead of presenting the whole measurement–fSWAP idea as new.

The ensemble language must also remain precise. Starting from a definite total charge, each conditioned trajectory lies in a definite system-charge sector after every recorded gain/loss event, while the ensemble spans different sectors. The intended result is therefore a Born-sampled ensemble of boundary states with shared infrared data, not one deterministically prepared ket and not an individual charge-indefinite pure state. Campaign S1 must demonstrate that the trajectory-to-trajectory charge wandering is localized at the walls and separately extract the within-trajectory current-algebra coefficient.

V. REPRODUCIBILITY, SHARDING, AND OUTPUT SCHEMA

Each campaign has an immutable JSON manifest containing a schema version, git commit, canonical entry point, environment versions, device, dtype, complete geometry and protocol, observer definitions, root seed, global sample-index allocation, shard RNG seed, and analysis version. The present GPU engine uses one exact generator stream per immutable five-trajectory shard and checkpoints its pre-run and post-run states. It does not claim partition-invariant counter-based randomness: changing the shard partition defines a new versioned ensemble rather than an exact rearrangement of the same 100 random words.

A production directory contains

$$\begin{aligned} &\text{manifest.json, shards/, merged/,} \\ &\text{analysis/, figures/, logs/.} \end{aligned} \quad (8)$$

Sample indices are assigned to shards before execution. The merger rejects duplicate or missing indices and any manifest hash mismatch. Atomic permanent checkpoints are written after each observation block and contain the record cursor, RNG state, accumulated $-\log p$, compact observer accumulators, and, when present, tangent QR log sums and declared low-subspace products. A covariance can exist only in live memory or local scratch during the active stage; it is excluded from the permanent checkpoint schema. Replaying any immutable shard from its saved pre-run RNG state and frozen word must agree within declared tolerances. Arbitrary repartitioning of the current stateful RNG stream is not an exact-reproducibility operation.

Required raw products include per-trajectory scalars, not only ensemble averages; bit-packed replay records for every stochastic family; conditional log probabilities or lossless per-cycle sums; numerical residuals; censor/null masks; and a completion table. Covariances are evaluated on the fly. A parent covariance needed by H1–H3 is a transient shard-local input, not a permanent ensemble product: H1/H2 consume it during the live parent process, while H3 reconstructs it from the saved initial RNG state, ordered word, outcomes, and engine hash. Analysis writes

figure-ready CSV/NPZ files and a fit JSON containing the input data hash, fit window, covariance model, bootstrap distribution, and sensitivity fits. Plotting scripts do not perform unrecorded fits.

Every observable declared anywhere in this contract is retained before ensemble reduction with logical axes (`sample, cycle, ...`) at cycle 0 and after every completed cycle through the declared final cycle. This includes scalar, vector, profile, spectrum, and marker outputs with their natural trailing axes. Means, bootstrap intervals, fits, and plotting-window extracts are derived products and never replace the complete raw array. The merger validates the exact sample IDs and the complete cycle coordinate $0, 1, \dots, T$ for every declared observable, rejecting missing or duplicate entries. This retention rule does not require a dense covariance at every cycle: observers consume the live covariance and save their outputs, while full covariances remain restricted to explicitly declared checkpoints or exact replay. Any intentionally sparse state product must be called *checkpoint-resolved*, with its schedule recorded in the manifest; it must never be described as cycle-resolved.

VI. MASTER TRAJECTORY FAMILIES AND REUSE RULE

A new independent trajectory family is required only when at least one component of the following key changes:

$$\mathfrak{K} = (N_x, N_y, \text{controller construction and parameters,} \\ \text{initial ensemble, serial schedule law,} \\ \text{feedback/noise law, } n_{\text{shell}}).$$

(9)

Changing the observer, fit, source position, analysis window, or deterministic replay does not define a new Born family. All campaigns sharing $\mathfrak{K}$ must consume one common set of independently seeded trajectories rather than launch campaign-specific copies. One trajectory remains the independent sampling unit even when many cycle snapshots, periodic origins, subsystem positions, sources, or twists are derived from it.

For the primary pure-state wall and its matched wall-off control, run one master family per accepted size for exactly $2N_y$ complete cycles from an independently sampled product state. No physical cycles are discarded as burn-in. The same family supplies S1, T1, B2, the late-time states for H1, the unperturbed parents and time origins for H2, and the frozen words for H3 and G5. Response continuations in H2 and twist replays in H3 are derived jobs attached to a parent trajectory ID; they are not new independent samples.

For max-mix initialization, run one separate master family per wall/control geometry and size from $t = 0$ through exactly $2N_y$. It supplies P2 and the max-mix portion of S1. Uniform bulk, secondary-wall, changed- α , changed-noise, imperfect-feedback, postselected, changed- n_{shell} , and changed serial-word controls have different

keys and therefore cannot be manufactured from a baseline trajectory. B0 is deterministic; the required B1 surrogate-attraction test is a separate two-family $S = 25$ calculation executed last. H1 modular evolution and H3 fixed-record flux evolution are deterministic descendants of saved states or records. M1/L1 is instead a sample-free autonomous CPU campaign with its own immutable analytic-channel configuration.

Each pure-state master trajectory saves:

1. its sample seed, initialization hash, controller hash, complete ordered site/channel word, bit-packed outcomes, per-event conditional log probability, X_j , Y_j , and the system charge after every event or a losslessly reconstructible block representation;
2. every declared observable evaluated from the live state and saved separately for every sample at every cycle, including cycle 0; transient parent covariances at the declared diagnostic, tangent-alignment, late-time S1, H2-source, and final checkpoints are consumed shard-locally by H1/H2 and then discarded, while H3 regenerates them recordwise and the permanent archive retains the initial RNG state, ordered record, engine hash, checkpoint cursor, and replay hashes needed to regenerate them;
3. per trajectory, cycle, and Rényi index, the y_0 -averaged full-strip entropy curve, selected y_0 -resolved validation curves, unnormalized wall-window contour sums, subsystem spectra, charge-susceptibility spectra $q_a = \nu_a(1 - \nu_a)$, matched entropy-variance residual curves, interval-charge cumulants, density and charge profiles, structure factors, the full fixed-slice array $C_{G,\xi}(x, r_y)$, its wall/bulk derived curves, topology and purity diagnostics, and the simultaneous $N_{\mathcal{D},\xi}$ needed for sector-resolved fits;
4. tangent QR log increments, null masks, aligned low-subspace projectors and densities, block- R factors at declared checkpoints, and the initial frame seed; and
5. observer residuals, censor masks, completion status, and enough accumulated score data or replay information to reconstruct the H2 conditional and score terms.

The max-mix master obeys the same complete per-sample, per-cycle observable contract and additionally saves its complete start-to-time record and physical tangent QR products on the same realized trajectory. Dense Choi covariance tracking is disabled: no reduced Choi block is propagated, diagonalized, or archived. The purification and tangent products are compared statistically at the trajectory and finite-size levels rather than through an exact Choi Gram identity. Scan-only families may declare a smaller observable set, but every observable they do declare is still saved for every included sample and cycle;

they must not discard either axis or save only ensemble means. Neither family archives a dense covariance history.

Transition rule for active August 2026 bundles. Do not edit an active bundle, its source snapshot, immutable configuration, manifest, or output tree to impose the clarified retention contract. Let `01_bulk_width_gate`, `02_pure_wall_master`, and `06_b1_controller_frame` finish and verify under their launch hashes. The first two retain complete ordered records and replay state, so missing observer products are added afterward by deterministic replay into new versioned output trees, never by overwriting their parent archives. P1 uses the preregistered nested ten-sample subset specified in Campaign P1. The pure-wall master already has $S = 10$, so its completion replays all ten parent trajectories and saves each declared parent-cycle observable at cycles $0, 1, \dots, 2N_y$ without archiving dense covariance histories. Shared checkpoint values must reproduce the parent archives within declared float64 tolerances before the completion products are accepted. The B1 controller-frame bundle already saves each of its declared trajectory scalar histories for all ten samples at every cycle $0, 1, \dots, 2N_y$; its static frame spectrum and residual map are cycle-independent objects and are saved once. B1 therefore needs no retention-completion replay unless its own validator fails.

The first B1 interface shard did fail only after completing its five trajectories and all 80 cycles: an accumulated complex128 charge-trace residual $4.0870418160920963 \times 10^{-10}$ exceeded a mistakenly reused 10^{-10} Ky Fan tolerance. The corrected, versioned B1 runner keeps the Ky Fan tolerance at 10^{-10} , uses a separate absolute charge-integrality tolerance 10^{-8} , and records both tolerances and the realized residual. It also persists the compact ordered record and all per-sample, per-cycle scalar histories before validation, archiving any rejected run under a dedicated rejected-validation tier. Accepted reruns write beneath `06_b1_controller_frame_chargefix_v1`; no physical protocol, seed, estimator, sample count, or cycle count changes.

A. Minimum gate-first execution matrix

The minimum below assumes that B0 accepts $N_x^* = 20$, one active batch of $S = 10$ resolves every gate, the explicit interface is the primary wall, and the support-terminated construction is the secondary wall. It is a count of independently sampled trajectory families, not deterministic exact calculations, observer passes, response continuations, or fixed-record replays. The fixed core is

family block	$S = 10$ families	trajectories
P1 versioned three-shell bulk baseline	48	480
primary pure wall and matched control, six sizes	12	120
primary max-mix wall and matched control, six sizes	12	120
secondary pure wall and matched control, three sizes	6	60
W1 transverse-convergence additions	18	180
core through G1–G4	96	960
G5 deterministic descendants	0	0
core through G1–G5	96	960
explicit pure/max-mix plus support-terminated pure S2 α_{in} scan	96	960
fixed campaign including S2	192	1920
existing versioned R1 record matrix, supplemental	10	100

Here P1 uses pure topological and trivial states at all six square sizes and max-mix states at $L = 20, 28$. Every one of those points is run at $n_{\text{shell}} = 2, 1, \text{None}$, giving $12 \times 3 + 4 \times 3 = 48$ families. The W1 count uses four protocols—two walls and their separate matched controls—at $N_x = 20, 24, 28$ and $N_y = 48, 80$. Of these 24 points, six are already master points, so only 18 are new. The active S2 count is the explicit-interface pure/max-mix pair plus a pure support-terminated mirror, times eight masses and four circumferences, $3 \times 8 \times 4 = 96$ independently sampled families at the active $S = 10$ minimum. The $\alpha_{\text{in}} = 1$ points are retained inside all versioned S2 grids as independently seeded deep-topological scan anchors; they are compared with, but never pooled with, the corresponding S1/P2

master families. G5 launches no Born family: it consumes the five-size explicit-interface and matched-trivial words already saved by the pure-state master and writes independently checksummed deterministic descendants. The ten separately seeded R1 wall/control families already present in the active Bundle 02 configuration are left unchanged for reproducibility, but lie outside the conjunctive gate count and supply only the optional record large-deviation diagnostic.

All other fixed stochastic campaigns reuse those samples. B0 and M1/L1 are deterministic. T1 is an observer of the live pure master record. H1 consumes live stationary states, H2 attaches response continuations to live parent origins, and H3 regenerates a parent from its pre-run RNG state and replays the saved record around the twist

circle. B2 consumes the three common secondary-wall sizes. Raster/reverse-raster production families are omitted. V0 exact checks are small validation jobs. Optional B1 is reported separately from this fixed campaign count and adds exactly two $S = 25$ families, or 50 trajectories, when executed.

The open perturbation grids add

$$N_{100} = 128 + 3n_{\text{imp}} + 17n_{\text{noise}}, \quad (10)$$

where n_{imp} is the number of nonperfect stochastic feedback points in M2 and n_{noise} is the number of nonzero noise values in M3. The deterministic postselected branch adds no stochastic family. Each added M3 noise value contributes five uniform square sizes and two wall geometries at six circumferences. Equation (10) excludes any escalation triggered by a failed gate. A second base batch adds one further $S = 25$ family for every member of the affected comparison block, never only for one selected point.

This minimum is valid only if every master run writes the complete save contract above. In particular the manifest and record must be sufficient to replay the exact ordered instrument; the three stationary live states must be reduced immediately to the declared state, modular, twist, and correlator products; the event stream must retain conditional log probabilities and signed charge transfers for B2, while the ordered word and branch normalization must suffice for the G5 identity-input replay; the tangent QR stream must span the full T1 window; and H2 must retain its declared parent snapshots, score accumulators, and source-time metadata. If one of these products is omitted, reconstructing it later may require a new independent family and invalidates the minimum count.

VII. VALIDATION LADDER

A. Campaign V0: engine and observer validation

Claim and status. Status COMPLETED CPU VALIDATION; GPU NOT RUN. The archived result is CPU_PASS / GPU_NOT_RUN. The CPU gate checks exact local branch normalization, frozen-word replay, conditional probabilities, diagnostic-only observers, covariance physicality, tangent actions, twist closure, checkpoint restart, and serial versus trajectory-sharded execution. This completes the CPU component of V0 only: no CUDA kernel was executed and no CPU-GPU agreement is claimed.

Dependencies and run matrix. None. The completed CPU matrix uses 4×6 and 6×8 , seeds 101, 202, 303, 404, and both three- and ten-cycle records. It compares the production rank-one update with the dense CPU reference under exact replay and exhaustively enumerates the local four-channel branch tree. This is a local normalization test, not an assertion that every exponentially large multi-cycle circuit record was enumerated.

The complete CPU products are under 00_WORKSPACE/CURRENT/validation_campaigns/results/v0_v1_cpu_20260816_183244/v0/.

Initialization, duration, and observation. No stochastic preparation interval is needed. Compare every elementary update and every cycle boundary. The base protocol is $n_{\text{shell}} = 1$, $\alpha_{\text{top}} = 1$, $\alpha_{\text{triv}} = 30$, perfect correction, and no postselection.

Raw saves and estimator. The archive retains branch/event records, schedules, RNG and checkpoint states, covariance diagnostics, compact spectra, direct observer comparisons, and the five-scale centered-difference tangent test. Exact active-site coverage, periodic wrapping, and the fixed channel order are saved with the replay data. Every tolerance is recorded as $\max(10^{-12}, 2048\epsilon_{64}N\sqrt{n_{\text{block}}})$ for the corresponding case.

Controls, statistics, and gate. Schedules and outcomes agree bitwise under replay; the serial/shard covariance residual is zero. Direct observer and probability residuals are zero, the largest dense/rank-one residual is 7.39×10^{-12} , the largest restart residual is 3.52×10^{-15} , and the largest charge-bookkeeping residual is 4.88×10^{-12} , all below their recorded float64-scaled bounds. The five-scale tangent comparison reaches a minimum relative error of 7.37×10^{-11} , and the 2π projector-relabel closure residual is 2.22×10^{-16} . Thus the CPU correctness prerequisite for the reduced V1 study passes.

Downgrade, figure, sharding, and legacy. There is no main-text figure. The GPU inspection is recorded separately as AUDIT_ONLY / NOT_VALIDATED: all deployable copies matched the canonical file and the canonical entry point was present, but PyTorch/CUDA was unavailable, reverse-raster support and exact replay were absent, and projector signatures were not fully recorded. These audit facts must not be converted into a GPU-validation or CPU-GPU agreement claim.

B. Campaign V1: reduced serial-order sensitivity study

Claim and status. The full $S = 100$ production-scale study remains deferred. A reduced CPU study is complete with status STAGED_INCONCLUSIVE and categorical_change=false. It tests whether topology and wall diagnostics on one finite system are artifacts of the serial site order. It is a legacy diagnostic, not production evidence and not a full V1 pass.

Dependencies and run matrix. The completed archive under 00_WORKSPACE/CURRENT/validation_campaigns/results/v1_cpu_N16x16_S10_20260816_225400/ uses $N_x = N_y = 16$, ten independently initialized trajectories for each of random, raster_y, and reverse_raster_y, and 64 cycles per trajectory. All 30 trajectory shards completed. The campaign used 28 one-thread CPU workers and the canonical classA_U1FGTN.run_markov_circuit entry point; it launched no GPU work and disabled automatic size or sample-count escala-

tion.

Initialization, duration, and observation. The initial burn limit is 32 cycles and the retained observation window is 32 cycles. The stationarity time T_ϵ is evaluated per trajectory from the streamed charge, purity, covariance-change, and wall-profile diagnostics. All three schedules satisfy the staged rule $T_{\epsilon,95}^{\text{upper}} \leq 32$: the upper bounds are 12, 22, and 15 cycles for random, raster, and reverse-raster order, respectively, with median $T_\epsilon = 2$ in each family.

Raw saves and estimator. The archive saves per-trajectory events, schedules, checkpoints, cycle observables, three stationary covariances, merged manifests, and fit products. Scalar intervals are 95% percentile intervals from 2,000 resamples of whole trajectories. In Table II, f_{wall} is the wall-localized excess fraction, a_S and a_Q are the log-chord entropy and interval-charge-variance coefficients, and v_L, v_R are conditional paired-response velocities. The latter omit the Born likelihood-score term and therefore are not physical Born-response velocities.

Controls, statistics, and gate. All three bulk Chern-marker intervals are positive and close to one, so changing the serial order does not change the finite-system topological category. The entropy and charge coefficients vary quantitatively with schedule; in particular, the raster and reverse-raster intervals are separated. No schedule establishes the preregistered $f_{\text{wall}} > 1/2$ criterion, and every conditional-velocity interval contains zero. Consequently wall localization and conditional chirality are unresolved at $S = 10$.

Downgrade, figure, sharding, and legacy. The supported statement is that the finite-size topological phase is robust to the three tested serial orders, while nonuniversal amplitudes are schedule sensitive. The study is underpowered for wall localization and chirality. Full Born-response chirality, larger sizes, and an $S = 100$ production comparison remain open; the three schedule families are kept distinct and are never relabelled or pooled.

TABLE II. Completed reduced V1 CPU schedule comparison at 16×16 , with ten independent trajectories per schedule and 64 cycles per trajectory. Entries other than $T_{\epsilon,95}^{\text{upper}}$ are mean [95% CI] from 2,000 whole-trajectory bootstrap resamples. The velocities are conditional diagnostics without the Born score.

schedule	$T_{\epsilon,95}^{\text{upper}}$	bulk Chern marker	f_{wall}	(a_S, a_Q)	(v_L, v_R)
random	12	0.9936	0.443[0.386, 0.509]	0.591[0.366, 0.862]	0.351[−0.075, 0.764]
		[0.9861, 0.9989]		0.188[0.105, 0.278]	0.423[−0.133, 0.941]
raster	22	0.9981	0.454[0.404, 0.4996]	0.311[0.133, 0.471]	0.352[−0.213, 0.856]
		[0.9969, 0.9991]		0.089[0.032, 0.136]	0.135[−0.452, 0.692]
reverse raster	15	0.9994	0.474[0.359, 0.598]	0.792[0.608, 0.966]	0.256[−0.262, 0.766]
		[0.9991, 0.9997]		0.255[0.196, 0.309]	−0.306[−0.875, 0.249]

VIII. EXACT DOMAIN-WALL CALIBRATION

A. Campaign B0: deterministic analysis-pipeline benchmark

Claim and status. Status COMPLETED REVIEW CALIBRATION. The common analysis pipeline recovers the known chiral wall behavior of an exact class-A domain-wall Hamiltonian. This calibrates geometry and fit windows; it is not evidence about the adaptive circuit.

Dependencies and run matrix. Depends on V0. Use `classA_U1FGTN_domain_wall_hamiltonian` and `G_CI_domain_wall`. At $N_y = 48$, scan $N_x = 12, 16, 20, 24, 28, 32$. At the selected width scan the sequence in Eq. (4). Run both the exact coupled spatial-mass interface and `triv_region_local_mode`, a decoupled-exterior hard analogue. Neither is assumed identical to controller `dw_truncation`.

Initialization, duration, and observation. There is no stochastic preparation interval. Enforce exact half filling. Resolve zero-energy crossings with a small boundary twist and continuous overlap tracking; test both twist signs and the zero-twist limit. Never fill states solely by $E < -\epsilon$.

Raw saves and estimator. Save Hamiltonians or their reproducible construction metadata, low spectra and vectors, wall weights, entanglement spectra, the complete fixed- x correlator tensor for every x and $1 \leq r_y \leq \lfloor N_y/2 \rfloor$, modular responses, density responses, and twist-overlap matrices. Recompute the wall, exterior-neighbor, topological-bulk, trivial-bulk, wall-window, and full- x correlation curves and their fit tables at every benchmark size; no archived curve or fitted exponent is imported. Apply without alteration the estimators later used in S1 and the three chirality campaigns.

Controls, statistics, and gate. Extract edge velocity, transverse localization length, finite-width hybridization, $c_{\text{edge}} = 1$ entropy coefficients in the one- and two-wall geometries, an r^{-2} squared correlator, opposite modular/physical velocities, and unit twist flow. Select N_x only when the hybridization mass satisfies

$$m(w) \ll \frac{2\pi v}{N_y}. \quad (11)$$

Failure means the analysis pipeline or geometry must be repaired before stochastic runs.

Downgrade, figure, sharding, and legacy. The calibration becomes Fig. S1, not a primary result. Deterministic sizes may be sharded independently. Path-audited visual references are collected under `exact_DW_benchmark_notes/DW_GS_Benchmark_Figs/`. They are not production inputs.

Current audited campaign. The authoritative version-2 calculation is campaign `00_WORKSPACE/CURRENT/experiment_review/b0_exact_domain_wall/results/20260816_191957/`; all entries in its machine-readable acceptance ledger pass. It preserves but supersedes version-1 campaign `20260816_184251` for interpretation. The first

run used a normalized packet at the center of the retained half-cylinder and fitted an absolute center of mass. That estimator mixed oppositely directed modular components and returned small same-sign slopes. It also selected $N_x = 12$ from the hybridization mass alone, even though the coupled local correlator and corrected modular response remain underconverged at $N_x = 12, 16$. The discrepancy is classified as an estimator mismatch together with a finite-width convergence effect, not as a failure of modular charge spreading.

Version 2 retains the mass gate but requires every single-geometry calibration to pass at both the accepted width and the next larger scanned width. This selects $N_x^* = 20$ because $N_x = 20$ and 24 pass whereas 12 and 16 fail the coupled local checks. For modular flow it evolves the $y = 0$ and $y = N_y/2 - 1$ sources separately at each physical wall and fits

$$D_w(t) = \frac{1}{2} [\bar{y}_{w,0}(t) + \bar{y}_{w,L-1}(t) - (L-1)], \quad (12)$$

$$L = N_y/2,$$

on $0.1 \leq t \leq 2.0$. One- and three-column sources and three alternate time windows give the same sign. Table III records the accepted-width values; modular slopes use modular-time units and are not compared in magnitude with physical velocities.

Transverse-width plateau audit. The version-2 selector above did not preregister a successive-width percentage-change threshold: it required absolute single-geometry gates to pass at $N_x = 20$ and at the next width $N_x = 24$. Thus its recorded choice $N_x^* = 20$ must not be described as the output of a percentage-plateau test. To expose the distinction, Table IV reports the direct fractional change from $N_x = 20$ to 28 at fixed $N_y = 48$, normalized to the $N_x = 20$ value. Every listed continuous observable changes by less than one percent; the largest change is the coupled lower-wall modular slope, 0.684%. The signed twist flows remain exactly $(-1, +1)$ in both constructions.

A post hoc one-percent criterion applied only to the direct $20 \rightarrow 28$ comparison would therefore accept $N_x = 20$. A stricter criterion requiring each adjacent increment to be below one percent, and treating the nonuniversal modular-slope magnitude as a convergence observable, would not: the lower-wall modular magnitude changes by 1.27% and 1.15% from $N_x = 20$ to 24 in the coupled and hard-exterior constructions, respectively. The corresponding $24 \rightarrow 28$ changes are 0.58% and 0.61%, and all listed $28 \rightarrow 32$ changes are below 0.063%. Under that stricter definition the smallest supported width would be $N_x = 24$; under a two-percent adjacent-width threshold, or when modular flow is tested only for its preregistered stable sign, $N_x = 20$ remains supported. Relative changes of the hybridization ratio and contour factor-of-two residual are omitted from the table because both are already near numerical-zero floors; their relative percentages are not stable measures of physical convergence. The production choice must therefore state the percentage threshold and observable set before invoking marginal gains.

TABLE III. Audited version-2 B0 results at $N_x = 20, N_y = 48$. The complete campaign also contains the accepted-width sequence $N_y = 24, 32, 40, 48, 64, 80$, all raw packet fields, and the 33/65/129-point twist controls.

construction	representative values	audited implication
Coupled spatial-mass interface	$c_1 = 1.0001, \beta = 2.07447$; $v_{\text{phys}} = (-0.87203, +0.87203)$; $v_{\text{mod}} = (-2.93045, +4.53563)$	The three-column modular source retains at least 0.3245 charge in the primary fit window, all sensitivity signs agree, and the norm drift is 3.3×10^{-15} . The wall twist flows are $(-1, +1)$.
Hard-exterior analogue	$c_1 = 1.0001, \beta = 2.07437$; $v_{\text{phys}} = (-0.87203, +0.87203)$; $v_{\text{mod}} = (-2.59985, +4.40578)$	The minimum modular retention is 0.3349, all sensitivity signs agree, and the norm drift is 4.0×10^{-15} . The wall twist flows are again $(-1, +1)$.
Legacy-protocol reproduction at 20×40	Aggregate initial charges 8 and 24; $0 \leq t \leq 40$ with $\Delta t = 0.05$	The complete aggregate $N_{xy}(t)$ fields are saved with norm drift 1.4×10^{-14} , while the separate-source data make handedness identifiable.

TABLE IV. Post-campaign transverse-width plateau audit from $N_x = 20$ to 28 at $N_y = 48$. Entries are $100|O(28) - O(20)|/|O(20)|$. “Numerical zero” denotes changes below $10^{-6}\%$. Modular slopes are quoted only as diagnostics in modular-time units.

observable	coupled interface	hard exterior
spectral $ v $	numerical zero	numerical zero
transverse localization length $\xi_\perp$	0.147%	numerical zero
von Neumann coefficient c_1	0.00044%	0.00044%
Rényi coefficient c_2	0.000085%	0.000049%
Rényi coefficient c_3	0.00324%	0.000010%
wall correlator exponent β	0.00910%	numerical zero
physical-response $ v $	numerical zero	numerical zero
modular-slope magnitude, lower wall	0.684%	0.536%
modular-slope magnitude, upper wall	0.0675%	0.0470%

Legacy comparison. **Assessment: promising and already a substantial single-geometry calibration, but not a B0 pass.** A repository-wide audit finds quantitative exact-state entropy and correlator fits for the coupled interface, corresponding fit tables for `triv_region_local_mode`, and raw exact modular-evolution arrays. The strongest path-audited results are summarized in Table V. Here m_{full} is the two-wall full-width von Neumann log-chord slope, m_{wall} is the unnormalized single-wall contour-window slope, and β is defined by $C_G \sim \ell_y^{-\beta}$.

The principal defect is the zero-mode prescription. The archived periodic spectrum has two modes at numerical zero, whereas `G.Cl_domain_wall` occupies only $E < -\text{tol}$. Reconstructing the archived 20×40 , $x = 6, \dots, 14$ geometry with that audited rule selects 799 modes for the coupled interface rather than the required 800. For the hard-exterior Hamiltonian it selects 1239 modes because the decoupled exterior orbitals lie at energy -1 . Thus the excellent legacy slopes do not yet constitute the exact-half-filled, twist-tracked benchmark demanded above. The artifacts also predate the present domain-wall convention: they use walls at $x = 6, 14$, and some notebooks use the historical reversed `alpha.1/alpha.2` naming, whereas the current source uses a wider slab and explicit topological/trivial aliases.

At the time of the legacy audit, no completed artifact was found for the required $N_y = 48$ scan over $N_x = 12, 16, 20, 24, 28, 32$, the accepted-width scan over Eq. (4),

a one-wall entropy geometry, edge velocity, transverse localization length, hybridization mass, physical density response, twist-overlap tracking, or unit twist flow. An exact entropy size-sweep driver exists under `scripts/`, but its declared output is absent and it saves no numerical fit table, so it is not counted as a completed legacy run.

B. Optional B1: controller-frame surrogate-attraction diagnostic

Claim and status. Status PLANNED final-priority appendix diagnostic. This calculation asks one question: whether the adaptive circuit approaches the artificial ground-state manifold of the signed controller-frame operator over the finite interval $0 \leq t \leq 2N_y$. It is not a manuscript gate, an equilibrium model of the circuit, evidence for topology or a wall CFT, or a claim about infinite-time relaxation.

Formal object. For controller targets $s_j \in \{0, 1\}$ and normalized rank-one projectors P_j , define

$$H_{\text{frame}} = \sum_j (1 - 2s_j) P_j. \quad (13)$$

The static target cost over rank- N Slater projectors is

$$\mathcal{F}[C] = \sum_j s_j + \text{Tr}(H_{\text{frame}} C), \quad C^2 = C, \quad \text{Tr} C = N. \quad (14)$$

TABLE V. Legacy B0 evidence. These deterministic results motivate the production fit windows but are not pooled with the preregistered campaign.

artifact and coverage	strongest quantitative indication	implication and limitation
Coupled spatial-mass interface in the exact-DW material under <code>notebooks/</code> and <code>exact_DW_benchmark_notes/</code> , 20×40 , two walls at $x = 6, 14$	$m_{\text{full}} = 0.343924$ ($3m_{\text{full}} = 1.03177$, $R^2 = 0.999080$); $m_{\text{wall}} = 0.168600$ – 0.168642 ($R^2 \simeq 0.999875$); $\beta_{\text{wall}} = 2.024641$ ($R^2 = 0.999835$) on $r_y = 3, \dots, 15$. The saved spectrum shows two low-energy crossing branches, while the marker and modular current maps are wall localized.	Strong evidence that the existing geometry and estimators recover the expected $c = 1$ and r^{-2} structure at one size. It is a two-wall calculation only and has no finite-width or longitudinal-size extrapolation.
Exact fit tables under <code>notebooks/flattened_hamiltonian_analysis/data/</code> , 20×40	For <code>triv_region_local_mode</code> , $m_{\text{full}} = 0.343923$, $m_{\text{wall}} = 0.168600$ – 0.168642 , and $\beta_{\text{wall}} = 2.024671$ with $R^2 = 0.999835$. The coupled and hard-exterior values differ by less than 2×10^{-6} in the entropy slope and 3.1×10^{-5} in β .	Very promising coupled-versus-hard agreement, but only for the same 20×40 geometry. The table preserves fits rather than Hamiltonians, eigenvectors, or complete spectral data.
Current-normalization correlation archive in <code>00_WORKSPACE/COLAB/colab_small_system_testing/analysis_outputs/pure_state_square_correlations_cpu/</code> , 16×30 and 16×40	Complete scalar $C_G(x, r_y)$ curves are saved for every x and $0 \leq r_y \leq N_y/2$. The wall exponents are respectively $\beta = 2.042461$ and 2.040144 , with $R^2 = 0.999742$ and 0.999753 on $r_y = 3, \dots, 10$; the full- x averages give $\beta = 2.11628$ and 2.10899 .	Confirms modest N_y stability using the normalization of Eq. (31) and supplies reusable raw curves and fit rows. It still lacks the B0 N_x scan and the preregistered $r_y = 3, \dots, 15$ model comparison.
Exact modular data under <code>notebooks/modular_charge_spreading/exact_dw_gs/</code> and static-map material under <code>exact_DW_benchmark_notes/DW_GS_Benchmark_Figs/</code> , 20×40	The static J_y map has opposite signs at the two interfaces. Two saved exact evolutions contain the full $N_{xy}(t)$ array for $0 \leq t \leq 40$ with total-charge drift below 4.4×10^{-7} .	Provides raw material for a modular-drift calibration, but no wall velocity was fitted; there is no corresponding physical density-response velocity or orientation-reversal control.

Ky Fan minimization therefore gives

$$C_\star = \sum_{a=1}^N |u_a\rangle\langle u_a|, \quad F_\star = \sum_j s_j + \sum_{a=1}^N \epsilon_a^\dagger(H_{\text{frame}}). \quad (15)$$

This evaluates all constraints on one common covariance. The actual circuit is instead a stochastic ordered sequence of Born measurements and feedback corrections and is not generated by H_{frame} . The object called the flattened Hamiltonian in the legacy notebooks is therefore only a static surrogate; its spectrum is not generally ± 1 , and it is neither the conditioned tangent operator nor the mean channel.

Dependencies and run matrix. Depends only on V0 and executes after every manuscript-relevant campaign. At $N_x = 20$, $N_y = 48$, and $n_{\text{shell}} = 1$, run the explicit topological-trivial interface and its full-support all-trivial matched control. Each arm uses $S = 25$ independent default half-filled initializations in five immutable five-trajectory shards, the random schedule, perfect correction, complex128, exactly $2N_y$ physical cycles, and no burn-in. There is no width, support, parameter, or construction sweep.

Raw saves and estimator. Construct H_{frame} once per arm and save its frame hash, spectrum, nominal half-filled rank N_0 , Fermi degeneracy cluster, selection gap, $F_\star(N_0)$, and half-filled residual map. Dense operators, minimizer projectors, and trajectory covariances are transient and are not archived. With $C_t = (G_t + \mathbf{1})/2$, stream the direct cost $F[C_t]$, the integer total charge, the rank-matched Ky

Fan excess

$$\Delta F(t) = F[C_t] - F_\star(N(t)), \quad (16)$$

and the minimum normalized Frobenius distance to the nominal half-filled minimizing manifold $\mathcal{M}_{N_0}$,

$$D_{\text{frame}}(t) = \left[\frac{1}{d} \min_{P \in \mathcal{M}_{N_0}} \|C_t - P\|_F^2 \right]^{1/2}. \quad (17)$$

The minimization is performed invariantly within a degenerate Fermi subspace. Also save purity defect, successive-cycle covariance change, and the compact replay record; wrong-outcome count is contextual metadata rather than a response variable.

Controls, statistics, and gate. There is no charge-sector frequency, empirical weight, training/test split, shifted profile, or fitted activity predictor. Instantaneous $N(t)$ appears only in the exact rank-matched lower bound in Eq. (16). Report ensemble means, medians, trajectory quantiles, and whole-trajectory bootstrap intervals. Compare cycle zero with $3N_y/2 \leq t \leq 2N_y$ and fit per-trajectory slopes on $N_y \leq t \leq 2N_y$. Call the result a “finite-window attraction signal” only if both ΔF and D_{frame} decrease and have bootstrap-supported negative late slopes; otherwise report “no evidence of attraction over $0 \leq t \leq 2N_y$.” This classification has no downstream consequence.

Execution, figure, and preserved CPU pilot. The self-contained A100 implementation is the final optional bundle `00_WORKSPACE/CURRENT/final_production_ready_`

`figure_scripts/06_b1_controller_frame/`. It produces appendix-only spectrum/residual, convergence, late-distribution, and wall-versus-control figures plus a standalone report. CPU campaign 20260816.225717 used the superseded $N_y = 24, 40, 4N_y$ -cycle, $2N_y$ -burn-in activity-prediction protocol. It was intentionally stopped after eight static cases and eight completed trajectories; its files remain immutable partial provenance and are not pooled with the GPU data.

Current legacy controller-form-factor result. The deterministic reconstruction under `00_WORKSPACE/LEGACY/form_factor_analysis/` supplies a useful support-design check adjacent to B1, but not a surrogate-attraction result. For the uniform $\alpha = 1$ two-band projector and X trial spinor, the finite OW overlap is

$$f_n^{\text{eff}}(\mathbf{k}') = \frac{1}{N} \sum_{\mathbf{k}} \tilde{g}_n(\mathbf{k}' - \mathbf{k}) f(\mathbf{k}) \langle u_-(\mathbf{k}') | u_-(\mathbf{k}) \rangle. \quad (18)$$

α	C	ν_{none}	ν_0	ν_1	ν_2	$\min f_{\text{none}} $	$\min f_0 $	$\min f_1 $	$\min f_2 $
1	+1	-1	-1	-1	-1	0	0	0	0
1.5	+1	-1	-1	-1	-1	0	0	0	0
3	0	0	0	0	0	0.500000	0.652414	0.525837	0.499765
30	0	0	0	0	0	0.694808	0.706710	0.694824	0.694800

where “none” means no real-space truncation. The zero locations, ordered as (none, 0, 1, 2), are

$$\begin{aligned} \alpha = 1 : \quad \frac{k_x^{(0)}}{\pi} &= (0.500000, 0.209346, \\ &\quad 0.492186, 0.492488), \\ \alpha = 1.5 : \quad \frac{k_x^{(0)}}{\pi} &= (0.333333, 0.064606, \\ &\quad 0.346511, 0.345054), \end{aligned} \quad (20)$$

all on $k_y = 0$ within the numerical refinement tolerance. Thus the on-site $n_{\text{shell}} = 0$ mask does not evade the obstruction: its compact spinor has $C_{\psi_0} = 0$, while its overlap phase has turn number $\nu_0 = -1$ in both topological cases. At $\alpha = 3, 30$ the positive global minima show that all four overlaps are gapped, so no local vortex is assigned and the total winding is zero. Appendix A gives the bundle derivation and numerical algorithm.

This result quantitatively validates the practical finite-shell mode as a probe of the target-band obstruction, but it contains no wall trajectory and does not test approach to the signed controller-frame minimizing manifold. It therefore remains an adjacent static result rather than B1 production data.

Available legacy inputs are under `notebooks/flattened_hamiltonian_analysis/` and `00_WORKSPACE/CURRENT/topological_frustration_diagnostics/docs/`; the complementary finite-support audit is under `00_WORKSPACE/LEGACY/form_factor_analysis/`. They guide sizes and estimator debugging

The $n_{\text{shell}} = 1$ three-by-three mask retains 0.992155 of the projected-mode norm, moves the $k_y = 0$ overlap zero only from $k_x/\pi = 0.5$ to 0.492186, and preserves local winding $\nu_n = -1$; $n_{\text{shell}} = 2$ retains 0.999137 with its zero at 0.492488. By contrast, the on-site mask retains 0.620624 and moves the zero to 0.209346. The associated scales are $\xi_{\text{top}} = 0.56419a$, $\xi_{\text{rms}} = 0.67352a$, and $\xi_{\text{exp}} = 1.44270a$. The overlap winding is not the Chern number of the smooth compact-support spinor, whose stable FHS result is $C_{\psi_n} = 0$ for $n_{\text{shell}} \leq 2$; near-singular coarse-grid values at $n_{\text{shell}} = 4, 8$ are not interpreted.

The winding is now evaluated explicitly rather than inferred from the visible zero. The overlap is a section of the dual occupied-band bundle, so its isolated-zero indices obey $\sum_a \nu_a = -C$. A periodic 181×181 zero search, two-dimensional refinement, and 721-point local phase loop give

only. Legacy postselected Frobenius convergence does not validate the Born-sampled perfect-correction circuit.

C. Campaign B2: feedback activity and slow dynamics

Claim and status. Status PLANNED. This campaign tests the dynamical association

$$\begin{aligned} &\text{Born feedback activity} \\ &\longleftrightarrow \end{aligned}$$

wall-localized slow tangent and purification sectors, (21)

rather than treating several wall-centered plots as independent observations. Optional B1 is an optional comparison and is not required for this campaign.

Exact branch relation. Immediately before channel j , the covariance is C_{j-} and the wrong-outcome indicator and signed system transfer are

$$X_j = |m_j - s_j|, \quad Y_j = s_j - m_j. \quad (22)$$

For the perfect-correction primitive,

$$\mathbb{E}[X_j | C_{j-}] = s_j + (1 - 2s_j) \text{Tr}(P_j C_{j-}), \quad |Y_j| = X_j \quad (23)$$

event by event. This identity, evaluated on the actual pre-event covariance, is the primary microscopic statement. No simultaneous-frame cost or static variational bound is assumed for the ordered sweep.

Dependencies and run matrix. Depends on T1 and P2. Reuse the production serial schedule and controller hashes at the common secondary-wall sizes $N_y = 32, 48, 64$, $S = 25$, for both wall constructions and their separate geometry- and update-count-matched wall-off controllers. These are the S1/W1 master families, so B2 launches no independent production trajectories. P1 supplies the uniform bulk reference. Changed n_{shell} , support, or slab width is an escalation only if the primary wall-versus-control association is unresolved; such an escalation receives its own preregistered grid and is not part of the minimum count.

Raw saves and estimator. Save every X_j , Y_j , controller label, position, pre-branch probability, schedule position, and observation-block index. Measure stationary signed and absolute activity profiles, wall/interior rates, inter-event correlations, and the joint fluctuations of activity, total system charge, tangent gap, and residual max-mix entropy. Compare the activity profile directly with the globally normalized low-tangent-subspace density and purification residual using profile overlap, Spearman rank correlation, and independently fitted localization lengths. Condition the ordered within-cycle record on its instantaneous N_j - where useful. The B1 controller-frame diagnostic is not used as a B2 predictor or launch gate. Correlations across different times from one trajectory do not increase the trajectory sample count.

The empirical activity function

$$\theta_R(s, T) = T^{-1} \log \mathbb{E}_\xi [e^{-s \sum_{j \in R, T} X_j}] \quad (24)$$

may be reported with log-sum-exp stabilization and reweighting effective sample size. It is a finite-time annealed record statistic, not the G5 many-body singular spectrum, not a trajectory-resolved CFT partition function, and not a tilted-channel eigenvalue unless the tilted operator is explicitly constructed and its long-time limit is converged.

Controls, statistics, and gate.

Validate Eq. (23) branch by branch and the mean profile against the M1/L1 exact autonomous averaged-channel prediction. Require a stationary interface excess over the matched active-channel control. To support a dynamical association, the same parameter sweep must link increased activity to the closing tangent sector and delayed purification with a preregistered monotone or scaling relation. Wall colocalization alone does not pass, and causation is not inferred from correlation.

Downgrade, figure, sharding, and legacy. If only Eq. (23) and a wall excess pass, report boundary-localized controller work without claiming that activity causes the CFT or tangent closing. The result is supplemental unless it quantitatively explains a Fig. 5 trend. Legacy parameter guidance is available in `00_WORKSPACE/CURRENT/topological_frustration_diagnostics/results/campaigns/20260813\_121340\_Nx20\_perfect\_Ny20\_22\_24/`; its $N_x = 20$, $N_y = 20, 22, 24$ perfect-correction activity and spectral runs use only 25 trajectories, raster order, and short

size-dependent times, so they are not production evidence. The corresponding formal reference is under `00_WORKSPACE/CURRENT/topological_frustration_diagnostics/docs/`. Local click words and regularized-Choi endpoint spectra are not part of B2.

Current legacy result. Assessment: a quantitatively promising precursor, but not a B2 pass. The completed CPU campaign at the path above used the canonical `classA_U1FGTN.run_markov_circuit` entry point for a hard-truncated wall and a uniform reference at $N_x = 20$, $N_y = 20, 22, 24$, with 25 trajectories per geometry, raster ordering, a burn-in of N_y cycles, and a total duration of $2N_y$ cycles. The saved records verify $|Y_j| = X_j$ in every run. Across the three sizes, the normalized all-channel wrong-outcome rate is 0.01691–0.01710 for the wall geometry and 0.00353–0.00382 for the uniform reference. Within the wall geometry the interface rate is 0.04698–0.04721, compared with 0.00355–0.00374 in the interior. The mean final Lyapunov gap is simultaneously much smaller for the wall, 0.1014–0.1125, than for the uniform reference, 0.8936–0.8964, and the associated wall slow mode has 0.9922–0.9958 of its weight in the interface region. Table VI records what this calculation establishes and what remains outside its scope.

These results show boundary-localized controller work and a boundary-localized slow one-body sector in the same finite-size range. They do not yet establish the B2 association in Eq. (21): activity and spectra were collected in separate trajectory suites, the uniform reference is not a geometry- and update-count-matched active controller, and no purification residual or joint-fluctuation observable was saved. Thus their coexistence is suggestive rather than a measured trajectory-level or scaling relation.

IX. TOPOLOGY AND WALL CONSTRUCTION

Logic of the first gate. Campaigns P1 and W1 jointly constitute Gate G1. They answer two logically separate questions that must not be inferred from one another. First, P1 asks whether the finite-range controllers actually retain the intended bulk topology: the practical $n_{\text{shell}} = 1, 2$ instruments are compared with the untruncated $n_{\text{shell}} = \text{None}$ reference on uniform periodic topological and trivial systems. A chiral-looking wall observable alone cannot establish this bulk premise. Second, after that premise is verified, W1 asks whether two microscopically different ways of programming the wall converge to the same long-distance defect theory.

Two programmable wall constructions. For the *explicit-interface* construction, `dw.truncation=False`: the physical lattice remains periodic, controller orbitals centered on both sides retain their finite-range tails across the interface, and a wall is created by changing the controller from the topological parameter $\alpha_{\text{top}} = 1$ to the trivial parameter $\alpha_{\text{triv}} = 30$. The interface is therefore the dynamical analogue of a coupled topological-trivial material

TABLE VI. Completed legacy work relevant to B2. Rates are normalized per active channel. The wall and uniform entries are finite-size parameter pilots and are not pooled with the preregistered production campaign.

artifact and coverage	strongest quantitative result	B2 implication and limitation
Perfect-correction activity records, $N_x = 20$, $N_y = 20, 22, 24$, $S = 25$	The branch identity $ Y_j = X_j$ passes in all six geometry-size runs. Wall and uniform all-channel rates are respectively 0.01691–0.01710 and 0.00353–0.00382; the wall interface rates lie in [0.04698, 0.04721], while interior rates lie in [0.00355, 0.00374].	Establishes a stationary wall-localized excess for this hard-truncated raster pilot. The uniform reference does not match the wall geometry or active update count, and the records use neither the production random-serial schedule nor $S = 25$.
Lyapunov tangent records at the same three sizes and sample count	The wall final-gap means are 0.1125, 0.1083, and 0.1014, versus 0.8936, 0.8964, and 0.8960 for the corresponding uniform runs. Wall interface weights are 0.9937, 0.9958, and 0.9922.	Establishes a strongly wall-localized slow one-body sector. It was measured in a separate trajectory suite, so no activity-gap covariance, profile overlap, rank correlation, or common-sweep scaling relation has been tested.
Auxiliary regularized-Choi particle-transfer records	The wall finite-sector gaps are 0.04165, 0.04424, and 0.03773, with interface weights 0.8646, 0.8975, and 0.8587; the uniform endpoints saturate before the final cycle.	Supplies an independent slow-sector diagnostic for method development. Endpoint spectra are explicitly outside B2 and do not substitute for the required max-mix purification residual.
Unrun B2 production and cross-analysis	No accepted $N_y = 32, 48, 64$, $S = 25$ master family or unified activity-tangent-purification record is present.	Still required are both wall constructions, separate geometry- and update-count-matched active controls, the random-serial schedule, purification residuals, and preregistered profile-overlap, Spearman, localization-length, joint-fluctuation, and monotone/scaling tests.

boundary. For the *support-terminated* construction, `dw.truncation=True`: each controller Wannier orbital is masked to the domain-wall sector containing its center and renormalized, while the excluded exterior is prepared according to the active-slab convention. This terminates the one-particle controller frame at the interface. It is therefore analogous to imposing open boundary conditions on the controller/Wannier orbitals, even though the underlying physical lattice is still a torus. On a periodic geometry the support mask necessarily creates a pair of such boundaries.

Meaning of construction independence. The two prescriptions are not identical circuits and need not agree at the lattice scale. If, after the transverse-width and opposite-wall leakage tests, they give compatible bulk markers, wall localization lengths, entanglement coefficients, charge data, correlators, and tangent observables, the result is evidence that the emergent chiral wall theory is independent of whether the defect was produced by a topological mass interface or by an OBC-like termination of controller support. This is the programmable-wall universality claim tested by W1. If they do not converge, the correct conclusion is that the circuit supports two distinct programmable wall realizations, not that one dataset may be silently substituted for the other.

Existing methodology comparison. The executed notebook `notebooks/flattened_hamiltonian_analysis/flattened_hamiltonian_analysis.ipynb` already compares both constructions at $N_x \times N_y = 20 \times 40$ for $n_{\text{shell}} = 0, 1, 2, \text{None}$. For the topological-trivial benchmark at $n_{\text{shell}} = 1$, the untruncated and support-terminated full-width log-chord slopes are 0.333938 and 0.333983, while the corresponding one-wall-window slopes are 0.166756 and 0.166784. At $n_{\text{shell}} = 2$ the full-

width slopes are 0.340314 and 0.340096 and the one-wall-window slopes are 0.168203 and 0.168129. These near-equalities are strong static evidence for a common infrared wall theory once $n_{\text{shell}} \geq 1$. The focused notebook `notebooks/flattened_hamiltonian_analysis/flattened_hamiltonian_alpha2_sweep.ipynb` provides the clean OBC test: it sets $\alpha_1 = \alpha_2$ so that no mass interface exists, retains `dw.truncation=True`, and finds $c_{\text{est}} \simeq 2$ throughout the uniform topological regime before the signal collapses toward zero in the uniform trivial regime. Thus the support boundary alone can expose topological edge modes.

These are flattened-Hamiltonian/controller-frame comparisons, not fresh-ancilla Born production. Moreover, their finite- n_{shell} static Chern estimator is approximately 0.795 for both constructions at $n_{\text{shell}} = 1, 2$, rather than quantized to one; only the full-support support-terminated case reaches one in that diagnostic. The notebooks therefore motivate the two-construction methodology and its OBC interpretation, but they do not replace P1’s dynamical finite- n_{shell} bulk-topology gate or W1’s stochastic construction-to-construction comparison.

A. Campaign P1: quantitative revalidation of the BPJ bulk baseline

Claim and status. Status PLANNED. The fresh-ancilla $n_{\text{shell}} = 2, 1, \text{None}$ sweep quantitatively tests reproduction of the postselection-free Chern-one bulk phase inherited from BPJ. This licenses the platform used by the wall campaigns; it is not the new universal result.

Dependencies and run matrix. Depends on V0. Use uniform periodic $L \times L$ systems with $L =$

12, 16, 20, 24, 28, 32, topological and trivial controller parameters, pure product-state initialization, $n_{\text{shell}} = 2, 1, \text{None}$, and $S = 10$ per point. At $L = 20, 28$ also run max-mix initialization at all three shell choices. This is the versioned 48-family P1 block in the minimum execution matrix. Additional max-mix sizes are launched only if these two sizes or the wall purification campaign expose a bulk finite-size ambiguity.

Initialization, duration, and observation. Pure-state and max-mix runs begin from their independently sampled declared initial states at $t = 0$ and run for $2L$ cycles. There is no discarded physical burn-in. Time-resolved marker and purity diagnostics determine whether the final-cycle estimator is usable. For every included trajectory, every declared P1 observable is evaluated on the initial state and after each completed cycle, so its saved cycle coordinate is exactly $t = 0, 1, \dots, 2L$. A thinned checkpoint schedule is not an admissible substitute.

Raw saves and estimator. Save purity spectra, local Chern markers, real-space Chern estimators, Bott data, density, and convergence diagnostics for every included sample and every cycle, retaining their natural array dimensions and without first averaging trajectories. Save full covariances only at explicitly declared state checkpoints and retain the RNG state, ordered schedule, and outcomes needed for exact intervening-cycle replay. For mixed covariances define $P_\xi = \Theta(C_\xi - \frac{1}{2})$ only if

$$\Delta_{\text{pur},\xi} = \min_j |\nu_{j,\xi} - \frac{1}{2}| \quad (25)$$

exceeds the declared numerical and finite-size threshold. Compute each invariant per trajectory and cycle and only then average its reported scalar/distribution. For a family of S trajectories, the completion validator requires exactly $S(2L + 1)$ rows, or the equivalent dense array, for each scalar observable, with every sample ID and every cycle $0, 1, \dots, 2L$ present once.

Controls, statistics, and gate. Bootstrap trajectories. Compare at least two finite-size correction forms. Require the topological bulk marker to extrapolate to one, the trivial marker to zero, an open purity gap, and Bott index +1 in every accepted large-size trajectory in the initial production batch. Report the actual denominator and an exact binomial interval; the $S = 10$ campaign does not by itself license a claim that the population success probability exceeds 95%. Require agreement across the $n_{\text{shell}} = 2, 1, \text{None}$ sweep within the declared finite-size corrections; any shell-dependent discrepancy must be reported rather than pooled.

Downgrade, figure, sharding, and legacy. If any shell choice approaches a noninteger plateau, report the shell-resolved plateau; do not claim convergence to one or change the primary cutoff after seeing the result. Results map to Fig. 1(c,d). Shard by trajectory. Path-audited visual references are under `00_WORKSPACE/COLAB/colab_lyapunov/figs/`; they are reference only.

Frozen-v1 qualification and temporal completion. The frozen `production_25sample_v1` archive remains the $S =$

25 final-state P1 gate dataset, but its Chern, purity, density, and related state observables were retained only at $\{L/4, L/2, 3L/4, L, 3L/2, 2L\}$. Those arrays are checkpoint-resolved and cannot support a cycle-zero or early-time convergence claim. The archive does retain the full ordered site word, bit-packed outcomes, pre-run RNG state, sample IDs, and canonical engine hash for every cycle and sample. Therefore the missing temporal observables are recovered by deterministic replay rather than by drawing a new Born ensemble.

Create a new immutable `production_25sample_v2.cycle_complete` output tree and preserve `v1` unchanged. For the temporal completion, preregister the nested subset of `v1` global sample IDs $0, \dots, 9$, denoted $S_{\text{time}} = 10$, in every P1 family. Replay those exact trajectories and save every declared observable at cycles $0, 1, \dots, 2L$; do not save dense covariance histories. Validate exact schedule and outcome replay, require the replayed values at the six shared checkpoints to reproduce `v1` within declared float64 tolerances, and keep the $S = 25$ `v1` final-state estimates and gates unchanged. Every temporal figure and caption reports $S_{\text{time}} = 10$; every final-state gate and finite-size figure reports its actual $S = 25$ denominator.

Current legacy result. Assessment: compelling single-width evidence for the intended Chern-one bulk, but not a P1 pass. The strongest path-audited precursor is `00_WORKSPACE/COLAB/colab_lyapunov/gpu_data/lyapunov_spectra/campaigns/N20_Ny30-50_nsh1_a1-1_S25_cyclesNy/`. It used the canonical `classA_U1FGTN_gpu.run_markov_circuit` entry point with a uniform $\alpha_1 = \alpha_2 = 1$ controller, $n_{\text{shell}} = 1$, default product-state initialization, and perfect correction. At $N_x = 20$ it ran 25 trajectories for $N_y = 30, 40, 50$, with depth $C = N_y$. The saved Chern arrays contain a radius-eight tripartition estimator $C_R(t)$ for every trajectory and saved cycle. As summarized in Table VII, its final mean is within 3.2×10^{-4} of one at all three sizes, and even the smallest final trajectory value is 0.992548.

The data therefore show rapid and numerically stable convergence of this uniform topological controller to a Chern-one tripartition estimator over the tested rectangular geometries. They do not establish the full P1 claim. The archive has no uniform trivial controller, no production square sequence $L = 12, 16, 20, 24, 28, 32$, no $n_{\text{shell}} = 2$ comparison, and no max-mix purity-spectrum run. It also contains neither the required local-marker profile nor a Bott-index implementation or output. Consequently no topological-versus-trivial extrapolation, open-purity-gap test, bootstrapped finite-size comparison, or production Bott acceptance test can be performed from this legacy campaign.

TABLE VII. Legacy uniform-topological real-space Chern data relevant to P1. The first reported value is the first saved post-cycle estimator, not a $t = 0$ invariant. Means, sample standard deviations, and ranges are taken directly over the 25 trajectories in each saved array.

N_y	(S, C)	$\langle \mathcal{C}_R(1) \rangle$	$\langle \mathcal{C}_R(C) \rangle \pm \text{SD}$	final trajectory range
30	(25, 30)	0.762876	0.999995 ± 0.000012	[0.999959, 1.000014]
40	(25, 40)	0.761324	0.999687 ± 0.001488	[0.992548, 1.000007]
50	(25, 50)	0.784492	0.999927 ± 0.000299	[0.998506, 1.000000]

B. Campaign W1: wall realization and transverse-convergence gate

Claim and status. Status PLANNED. An explicit topological-trivial controller interface and a support-terminated controller realize the same long-distance wall theory after transverse convergence.

Dependencies and run matrix. Depends on V0, B0, and P1; B1 is a nonblocking appendix diagnostic. At $N_y = 48, 80$, use $N_x = 20, 24, 28$, $S = 10$, and compare: an explicit topological-trivial controller interface with both sides active and no support mask; a support-terminated `dw_truncation` controller with the exterior prepared and excluded according to the active-slab convention; a *separate* geometry- and update-count-matched trivial controller for each construction. The exact coupled spatial-mass Hamiltonian and P1 uniform topological runs are common benchmarks, not extra stochastic W1 protocols. Six of these 24 points coincide with accepted-width master families, leaving 18 new families. If $N_x = 20$ fails, extend the same grid upward in increments of four rather than preregistering unused widths.

Initialization, duration, and observation. Use independent product-state starts and evolve each trajectory for a total of $2N_y$ cycles, with checkpoints every $N_y/4$. No separate physical burn-in or subsequent observation extension is used.

Raw saves and estimator. Save bulk-marker profiles, one- and two-wall entropy curves, charge FCS, correlators, tangent wall weights, and opposite-wall overlap. Compare asymptotic coefficients, not pointwise covariances.

Controls, statistics, and gate. The candidate $N_x = 20$ passes only if all headline coefficients agree with the next two widths within joint 95% intervals at both tested N_y , the fitted transverse localization length is well below wall separation, and opposite-wall low-mode weight is below 5%. The exact benchmark must also satisfy the hybridization inequality in Appendix F. Passing at $N_y = 48$ alone does not license a fixed- N_x thermodynamic claim because the required separation can grow as $\xi_\perp \log N_y$.

Downgrade, figure, sharding, and legacy. If $N_x = 20$ fails, all later campaigns use the smallest passing width. If no fixed width passes, use a joint N_x, N_y scaling. If the explicit-interface and support-terminated walls disagree, present them as distinct programmable walls rather than one universality class. Results map to Fig. 1(b,e) and Fig. S2.

Current legacy result. Assessment: the support-terminated wall and exact transverse benchmark are well supported, and a correct-feedback untruncated-interface purification comparison is now identified, but the full W1 construction-to-construction gate has not been run. The completed B0 exact campaign supplies the common benchmark required by W1. At $N_y = 48$ its coupled spatial-mass and hard-exterior constructions both pass the consecutive-width gate and select $N_x = 20$; the fitted localization lengths are 0.29557 and 0.222754, respectively. The corresponding hybridization ratios at $N_x = 20$ are 1.63×10^{-8} and 1.51×10^{-8} , far below the preregistered 0.1 threshold. This establishes transverse convergence for the exact models, not for either stochastic controller.

The strongest correct-feedback stochastic precursor is the support-terminated $N_x = 20$ campaign in `00_WORKSPACE/COLAB/colab_lyapunov/gpu_data/lyapunov_spectra/campaigns/N20_Ny30-50_nsh1_a1-1_S25_cyclesNy/`. For $N_y = 30, 40, 50$, 25 perfect-correction trajectories give final mean tangent gaps 0.02698, 0.02431, and 0.01968 and final mean tripartition Chern estimates 0.9781, 0.9543, and 0.9535. The minimum-gap vector places 0.9326, 0.9349, and 0.9435 of its norm directly on the two wall columns; including one neighboring column on each side raises these values to 0.9611, 0.9672, and 0.9708. Because hard support truncation forces exactly zero exterior weight, these numbers demonstrate wall localization but do not supply the fitted leakage or localization-length comparison required by W1.

Independent support-terminated entropy runs at $N_x = 20$, $S = 25$, and fixed depth $C = 50$ give full-transverse log-chord slopes 0.35572, 0.35092, and 0.34920 for $N_y = 30, 40, 50$, approaching the two-wall target $1/3$. A larger $N_y = 80$ pilot gives 0.35316, but uses only ten trajectories and depth $C = 40$. Table VIII separates these useful one-construction results from the missing W1 comparison.

The Colab-wide audit identifies an additional correct-feedback precursor in `00_WORKSPACE/COLAB/colab_charge_fluctuations/cpu_data/purification_charge_sharpening_alpha_sweep/`. It keeps `DW=True` and $\alpha_{\text{out}} = 30$ while setting `dw_truncation=False`, so the spatial interface remains and the controller projectors overlap it. This is not a homogeneous `DW=False` control. At $N_x = 16$, $N_y = 16, 24, 32$, and $T = 2N_y$, the completed perfect-correction arm contains ten trajectories at each of nine α_{in} values, for 27 configurations and 270 trajectories. At $N_y = 32$, its final entropy is 0.10488, 0.26286, and

6.06×10^{-5} at $\alpha_{\text{in}} = 1, 1.5, 1.9$, compared with 0.18509, 0.45161, and 0.06416 in the support-terminated arm. The corresponding untruncated intrinsic charge variances are 0.03279, 0.07807, and 4.98×10^{-6} , versus 0.04626, 0.13334, and 0.02241. Both constructions reach the numerical purity floor near $\alpha_{\text{in}} = 2$, but the untruncated interface generally purifies faster below the transition. Because it saves full-layer purification and charge observables rather than Chern, log-chord, tangent, or leakage data, it upgrades the W1 construction inventory without satisfying the W1 equivalence gate.

Consequently W1 remains planned. The newly identified untruncated campaign supplies the second stochastic construction for purification and charge sharpening at smaller sizes, but no archive provides both constructions and their separate geometry- and update-count-matched trivial controls at $N_y = 48, 80$, $N_x = 20, 24, 28$, and $S = 25$. The required independent starts, total $2N_y$ trajectories, windows, joint 95% coefficient comparisons, stochastic localization-length fits, and opposite-wall low-mode weight below 5% therefore remain untested. The legacy record supports the support-terminated wall as a strong production candidate, but does not license a common stochastic universality class or a fixed- N_x W1 pass.

X. UNIVERSAL PREPARED-STATE DATA

A. Campaign S1: state CFT and charge sector

Claim and status. Status PLANNED. Stationary Born trajectories carry wall-localized complex-fermion state data with $c_{\text{edge}} = c_L + c_R = 1$ and a level-one $U(1)$ current algebra. This campaign does not establish handedness; chirality is licensed only after H1–H3.

Dependencies and run matrix. Depends on W1. Use the accepted N_x , Eq. (4), and $S = 25$. The primary wall and its matched control use both an onsite pure product state and max-mix initialization, the latter entering only after purification. At $N_y = 32, 48, 64$ also run the secondary wall construction and its separate matched control from pure product states. These are respectively the 12, 12, and 6 master-family blocks in the minimum matrix.

Retired tripartite-information scope. Tripartite mutual information (TMI), conditional mutual information, and the historical contour-decomposed $I_2 - I_3$ estimator are retired from the production campaign. No new TMI trajectory family, subsystem scan, covariance replay, post-processing job, acceptance gate, or manuscript panel is to be run. Campaign S1 is restricted to the strip entropies, matched charge fluctuations, contours, and boundary–bulk correlators declared below. The archived TMI material summarized later in this section is provenance only and is not an executable campaign requirement.

Initialization, duration, and observation. Pure and max-mix runs begin independently at $t = 0$ and last a total of $2N_y$ cycles. There is no separate physical

burn-in. Save the three preregistered late-time snapshots at $t = N_y, 3N_y/2, 2N_y$ (rounded to whole cycles when necessary). Max-mix snapshots enter this analysis only if the P2 purity and stability diagnostics pass.

Raw saves and estimator. For

$$X_L(\ell) = \log \left[\frac{L}{\pi} \sin \frac{\pi \ell}{L} \right], \quad (26)$$

fit the two-wall full-width interval to

$$\mathbb{E}[S_q(\ell)] = \frac{c_{\text{edge}}}{6} \left(1 + \frac{1}{q} \right) X_L(\ell) + b_q + \dots, \quad (27)$$

$q = 1, 2, 3,$

using the full trajectory-bootstrap covariance across ℓ . Because one realization prepares one conditioned state, define $c_{\xi,q}$ trajectory by trajectory using the same preregistered fit window, design matrix, and linear estimator for every sample. Two co-primary summaries are reported: the coefficient c_q^{Born} fitted from the Born-averaged conditional entropy curve $\mathbb{E}_\xi[S_{\xi,q}(\ell)]$, and the full Born distribution of $c_{\xi,q}$ with its mean and whole-trajectory bootstrap interval. With a shared linear estimator, $c_q^{\text{Born}} = \mathbb{E}_\xi[c_{\xi,q}]$ algebraically, while displaying both the mean curve and the individual distribution separately tests ensemble scaling and trajectory typicality. Resolve the latter by the definite trajectory charge $N_{\mathcal{D},\xi}$ and report sector-conditioned means $c_q(N)$ wherever the sector count is sufficient. A broad $P(N)$ together with mutually compatible $c_q(N)$ is the direct test of a charge-fluctuating ensemble of critical states with shared infrared data. Do not average results from trajectory-dependent fit windows or model selections, and never extract a central charge from the entropy of the averaged covariance. Average multiple stationary snapshots within a trajectory before fitting or bootstrapping so they do not become independent samples.

Legacy-compatible entropy processing and production upgrade. The established repository estimator uses full-transverse strips

$$A(y_0, A_y) = [0, N_x] \times \{y_0, y_0 + 1, \dots, y_0 + A_y - 1\} \quad (28)$$

with periodic y , computes the entropy from the eigenvalues of the restricted one-body correlation matrix, and averages over all strip origins within each trajectory,

$$\bar{S}_{q,\xi}(A_y, t) = \frac{1}{N_y} \sum_{y_0=0}^{N_y-1} S_{q,\xi}[A(y_0, A_y), t]. \quad (29)$$

The legacy full-strip size campaign then averaged Eq. (29) over samples before an ordinary least-squares fit with an intercept against $X_{N_y}(A_y)$ on $A_y = 8, \dots, \lfloor N_y/2 \rfloor$ at a fixed final cycle. Its target von Neumann slope was $m_1 = 1/3$. Earlier wall-window contour scripts similarly summed the unnormalized contour over two-cell windows centered on each wall, averaged those curves over samples, and fitted the two complementary A_y branches separately.

TABLE VIII. Completed and implemented legacy work relevant to W1. Stochastic entries use the canonical perfect-correction dynamics unless explicitly marked otherwise. Exact B0 results are common benchmarks and are not counted as stochastic W1 protocols.

artifact and coverage	strongest quantitative result	W1 implication and limitation
Exact B0 coupled and hard-exterior walls, $N_y = 48$, $N_x = 12\text{--}32$	Both constructions pass at $N_x = 20$ and the next larger width. At $N_x = 20$, $\xi_\perp = 0.29557$ and 0.222754 and the hybridization ratios are 1.63×10^{-8} and 1.51×10^{-8} .	Passes W1's exact-model transverse benchmark and supports $N_x = 20$ as a candidate. It does not establish convergence or equivalence of the stochastic controllers.
Support-terminated tangent and Chern data, $N_x = 20$, $N_y = 30, 40, 50$, $S = 25$	Final mean gaps are 0.02698, 0.02431, and 0.01968; final mean Chern estimates are 0.9781, 0.9543, and 0.9535. Direct wall-column weights are 0.9326, 0.9349, and 0.9435.	Establishes a topological strip with a sharply wall-localized slow sector for one construction. There is no N_x scan, stochastic localization fit, or valid opposite-wall leakage gate.
Support-terminated entropy data, $N_x = 20$, $N_y = 30, 40, 50$, $S = 100$, $C = 50$	Full-transverse log-chord slopes are 0.35572, 0.35092, and 0.34920, versus the two-wall target $1/3$; all three fits have $R^2 > 0.99997$.	Supplies a strong asymptotic-coefficient precursor for one wall realization, but not the independently initialized, fixed- $2N_y$ comparison at $N_y = 48, 80$ across $N_x = 20, 24, 28$.
Support-terminated large-circumference pilot, $N_x = 20$, $N_y = 80$, $S = 10$, $C = 40$	The log-chord slope is 0.35316 with $R^2 = 0.999992$.	Reaches one W1 circumference, but the sample count, depth, observation window, width coverage, and missing comparison arm exclude it from the W1 gate.
Correct-feedback untruncated overlapping interface, $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, $T = 2N_y$	With $\alpha_{\text{out}} = 30$ and nine α_{in} values, all 270 trajectories are complete. At $N_y = 32$, entropy drops from 0.26286 at $\alpha_{\text{in}} = 1.5$ to 6.06×10^{-5} at 1.9 and reaches the numerical floor at 2.	Establishes a genuine overlapping-interface perfect-correction precursor. It has no N_x scan, $N_y = 48, 80$ arm, Chern or entropy coefficient, tangent localization length, or opposite-wall overlap, so it does not pass W1.

Production retains this geometry, origin average, legacy fit window, intercept, and full-width target as the primary compatibility analysis, but preserves the sample axis before fitting. This follows the newer streaming repository path whose saved entropy array has axes (`sample`, `cycle`, `Ay`) after the y_0 average. For each q and trajectory, first average Eq. (29) over the declared stationary snapshots, then perform the common linear fit

$$\bar{S}_{q,\xi}(A_y) = m_{q,\xi} X_{N_y}(A_y) + b_{q,\xi}, \quad c_{\xi,q} = \frac{6m_{q,\xi}}{1 + 1/q}. \quad (30)$$

Thus $c_{\xi,1} = 3m_{1,\xi}$. Save $m_{q,\xi}$, $b_{q,\xi}$, $R_{q,\xi}^2$, residual vectors, and the number of fit points, along with the full per-trajectory entropy curves. The regression standard error across A_y used in legacy tables diagnoses deviation from the fitted shape; it is not the sampling uncertainty of the Born ensemble. All quoted uncertainties on c_q^{Born} , sector-conditioned $c_q(N)$, or distributional statistics come from resampling whole trajectories. Periodic origins and stationary snapshots remain correlated averages internal to one trajectory.

Repeat the fits after moving the lower endpoint from $A_y = 8$ by the B0-declared sensitivity increments, deleting the largest included A_y , and including the preregistered finite-size correction model. Check $S_{q,\xi}(A_y) = S_{q,\xi}(N_y - A_y)$ within tolerance for pure snapshots; fit the two branches separately only as a diagnostic. The primary wall-window result uses per-trajectory unnormalized contour sums before Born averaging, upgrading the older sample-averaged contour procedure without changing its spatial windows. The audited implementation references are under `00_WORKSPACE/COLAB/colab_large_entanglement_scaling_N20/src/` and `00_WORKSPACE/`

`COLAB/colab_charge_fluctuations/src/`; their legacy outputs remain comparison data rather than production inputs.

Production boundary-bulk correlator campaign. All correlator observables and fits are recalculated from the newly generated B0 covariances and S1 trajectory snapshots. Prior repository results motivate the observable and expose likely fit-window sensitivity, but none of their curves, averages, or fitted exponents enter the production estimates. This recalculation requires no additional Born families because it is performed on every covariance snapshot already generated for S1. Use the fixed-slice squared Green function

$$C_{G,\xi}(x, r_y) = \frac{1}{2N_y} \sum_{y,\mu,\nu} |C_{\xi;(x,y,\mu),(x,y+r_y,\nu)}|^2 \quad (31)$$

for every transverse slice x and every independent separation $1 \leq r_y \leq \lfloor N_y/2 \rfloor$. The exact B0 calculation and the stochastic S1 calculation use identical normalization, slice definitions, chord coordinate, candidate models, and fit windows.

For every trajectory retain the full $C_{G,\xi}(x, r_y)$ array and derive, without discarding the sample axis, the two individual wall curves, exterior-neighbor curves, distance-from-wall profiles, declared topological- and trivial-bulk averages, fixed wall-window sums, and the full- x average. Save all per-trajectory curves as well as the Born means, covariance across r_y , fit parameters, residuals, R^2 , point counts, and model-comparison scores. Source-position and stationary-snapshot averages remain correlated averages internal to a trajectory.

B0 fixes the primary distance window before the stochastic data are examined, using $r_y \in [3, 15]$ where the cir-

cumference permits. Fit the wall data to chord-power, exponential, chord-power-plus-floor, and exponential-plus-floor models; apply the same models to matched bulk data rather than assuming exponential decay. Endpoint and window sweeps are sensitivity diagnostics and may not select the reported exponent. Report the Born-averaged fit and the complete trajectory-level exponent distribution, including its dependence on the trajectory charge sector, exactly as for the central-charge analysis. The primary conclusion is spatial: stable algebraic behavior should be wall localized and absent from the bulk. Quote a precision exponent near two only if it is stable under the preregistered window and model checks; otherwise retain the freshly calculated boundary–bulk contrast as qualitative support and let entropy and charge fluctuations carry the universal claim. Bootstrap whole trajectories.

The audited exact references are under `exact_DW_benchmark_notes/`. Earlier stochastic correlator diagnostics are under `00_WORKSPACE/COLAB/colab_charge_fluctuations/analysis_outputs/`. They document what was tried before and preregister failure modes; every plotted curve and quoted number in the new manuscript is regenerated under the production protocol.

Test whether the contour integrated over a declared single-wall window carries half that coefficient. This window sum is a diagnostic decomposition of the full-strip entropy, not the entropy of an independent wall subsystem. Save every trajectory’s subsystem spectra, entropy curves, unnormalized spatial contour $s_{A,\xi}(x,y)$, charge-generating function, wall-projected connected density structure factor, and globally normalized real-space correlator. Form $s_A(x,y) = S^{-1} \sum_{\xi} s_{A,\xi}(x,y)$ without rescaling individual contours, and verify $\sum_{x,y} \overline{s_A} = \mathbb{E}[S_A]$. Average S_q , quantum cumulants, structure factors, and squared correlations only after their per-trajectory evaluation. The interval variance and its small- k structure factor are primary current diagnostics; the real-space squared Green function is an appendix cross-check. Appendix D fixes the FCS conventions.

Separate quantum charge fluctuations from classical record-to-record profile fluctuations. For every interval A save

$$F_{\xi}^q(A) = \langle N_A^2 \rangle_{\xi} - \langle N_A \rangle_{\xi}^2 = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})], \quad (32)$$

$$F^{\text{cl}}(A) = \text{Var}_{\xi}(\langle N_A \rangle_{\xi}), \quad (33)$$

and verify the law of total variance

$$\text{Var}(N_A) = \mathbb{E}_{\xi}[F_{\xi}^q(A)] + F^{\text{cl}}(A). \quad (34)$$

Only the logarithmic coefficient of $\mathbb{E}_{\xi}[F_{\xi}^q(A)]$ is used to infer the $U(1)$ current level. The classical term is not purely sector wandering. Let $\mathcal{D}$ denote the declared dynamical subsystem—the active slab for support termination and the full system for the explicit interface—and let $N_{\mathcal{D}}$ be its charge. For a definite-charge input, each pure conditioned trajectory has a definite $N_{\mathcal{D}}$ and

$$F^{\text{cl}}(A) = \mathbb{E}_{N_{\mathcal{D}}} [\text{Var}_{\xi|N_{\mathcal{D}}}(\langle N_A \rangle_{\xi})] + \text{Var}_{N_{\mathcal{D}}}(\mathbb{E}_{\xi|N_{\mathcal{D}}}[\langle N_A \rangle_{\xi}]). \quad (35)$$

The first term is fixed-sector shape variation and the second is the contribution from sector wandering. Max-mix trajectories enter this decomposition only after both the global purity deficit and the within-trajectory quantum variance of $N_{\mathcal{D}}$ fall below declared tolerances.

To test where the total-sector fluctuations couple spatially, define

$$q_{\xi x} = \sum_{(y,\mu):(x,y,\mu) \in \mathcal{D}} \langle n_{xy\mu} \rangle_{\xi}, \quad N_{\mathcal{D},\xi} = \sum_x q_{\xi x}. \quad (36)$$

Save the complete transverse covariance matrix and its signed additive row sum,

$$\begin{aligned} K_{xx'}^{\text{cl}} &= \text{Cov}_{\xi}(q_{\xi x}, q_{\xi x'}), \\ \kappa_{\text{cl}}(x) &= \sum_{x'} K_{xx'}^{\text{cl}} = \text{Cov}_{\xi}(q_{\xi x}, N_{\mathcal{D},\xi}), \\ \sum_x \kappa_{\text{cl}}(x) &= \text{Var}_{\xi}(N_{\mathcal{D},\xi}). \end{aligned} \quad (37)$$

Also save the nonnegative onsite-heterogeneity profile $v_{\text{cl}}(x) = \sum_{y,\mu} \text{Var}_{\xi}(\langle n_{xy\mu} \rangle_{\xi})$. Report signed κ_{cl} without taking its absolute value, its signed partial sums in fixed wall windows, and its absolute leakage outside those windows. Full-system charge bookkeeping is saved separately. The profile v_{cl} omits inter-site covariances, so it neither sums to $\text{Var}(N_{\mathcal{D}})$ nor defines a variance weight; use it only as a descriptive localization control. Match every system-charge change to the signed gain/loss record. These quantities test whether sector wandering is coupled to the walls; they are not themselves universal CFT coefficients.

Matched entropy–variance spectral test. For every trajectory, stationary snapshot, origin, and interval used in the entropy fit, compute the quantum interval variance from the same restricted covariance and perform the same internal snapshot and origin averages. With

$$\bar{F}_{\xi}^q(A_y) = f_{\xi} X_{N_y}(A_y) + b_{F,\xi} + \dots, \quad (38)$$

define the paired trajectory statistic and its Born mean by

$$\delta_{SF,\xi} = m_{1,\xi} - \frac{\pi^2}{3} f_{\xi}, \quad \delta_{SF}^{\text{Born}} = \mathbb{E}_{\xi}[\delta_{SF,\xi}] = a_S - \frac{\pi^2}{3} a_F. \quad (39)$$

The primary analysis fits the paired residual curve

$$\begin{aligned} R_{SF}(A_y) &= \mathbb{E}_{\xi} \left[\bar{S}_{1,\xi}(A_y) - \frac{\pi^2}{3} \bar{F}_{\xi}^q(A_y) \right] \\ &= \delta_{SF}^{\text{Born}} X_{N_y}(A_y) + b_{SF} + \dots \end{aligned} \quad (40)$$

with the full paired covariance across A_y and a whole-trajectory bootstrap. Report a_S/a_F only as a secondary statistic because a ratio is unstable when its denominator is uncertain. For Rényi index q , also test the free-fermion prediction

$$a_{S_q} - \frac{\pi^2}{6} \left(1 + \frac{1}{q} \right) a_F = 0. \quad (41)$$

Appendix E derives these relations and their limits. This is an observer analysis of the existing S1 master covariances and launches no new trajectory family.

Controls, statistics, and gate. Use a trajectory bootstrap with the full ℓ covariance matrix. Predeclare fit windows from B0 and repeat fits after deleting each endpoint window and the smallest size. Require $c_{\text{edge}} = 1$ for all three Rényi indices, the factor-of-two relation between one and two walls, and level $k = 1$ from both interval variance and the small- k structure factor. Using the same fit windows and finite-size model, require the thermodynamic extrapolation of $\delta_{SF}^{\text{Born}}$ to be compatible with zero at 95% confidence and stable under the declared endpoint and smallest-size deletions. Apply the same paired test to the exact-Hamiltonian benchmark and require no false logarithmic signal in the matched trivial control. The wall squared-Green-function exponent must be compatible with two on the benchmark-selected window and remain stable under the declared sensitivity checks; no matched bulk slice may support a comparably stable algebraic fit. If the exponent remains window sensitive, downgrade it to a boundary-localization diagnostic rather than tuning the window. Require a log-chord model to outperform an area-law constant in the wall geometry and not in the trivial control.

Archived TMI evidence (provenance only; no new computation). A repository-wide audit found a clean deterministic calibration of the contour-decomposed tripartite-information estimator, but no production-compatible stochastic S1 result. In the exact-DW benchmark material under `00_WORKSPACE/LEGACY/exact_DW_benchmark_notes/`, the subsystem is first defined across the full transverse width and only the final entanglement-contour sum is restricted in x . For the exact 20×40 domain-wall ground state, $L_A = L_C = 3$, and fixed $y_0 = 0$, the two three-column wall windows each give a slope $0.166967668 \pm 0.000027005$ in $I_2 - I_3$ versus $\log[1/(1-x)]$, with $R^2 = 0.99999945$. The full-width sum gives $0.334232464 \pm 0.000095377$, with $R^2 = 0.99999829$. These values validate the expected $1/6$ single-wall coefficient, $1/3$ two-wall coefficient, and factor of two within the deterministic analysis pipeline. Equivalently, the direct TMI central-charge estimates are

$$\begin{aligned} c_{\text{TMI}}^{\text{wall}} &= 6m = 1.0018060(1620), \\ c_{\text{TMI}}^{\text{full}} &= 3m = 1.0026974(2861), \end{aligned} \quad (42)$$

making this one of the cleanest archived $c = 1$ calibrations.

The strongest admissible stochastic input is the full-covariance history under `00_WORKSPACE/LEGACY/sample_average_testing/`. It contains $S = 100$ trajectories on 12×31 through $C = 100$ and evaluates the entropy combinations trajectory first, followed by the y_0 and Born averages. The matched wall/full curves and slopes quoted here come from the PDF whose name ends in `_xsum_window_integrated_contour_and_mi_fits_v2.pdf`; this reduction retains the three-column wall sum and the full-width sum together. The similarly named

`_sample_average_y0_average_integrated_contour_and_mi_fits_v2.pdf` output is not the source of this matched pair. At $S = 100$ and $L_A = L_C = 4$, the three-column wall sum has slope 0.2487 ± 0.0014 and the full-width sum has slope 0.5112 ± 0.0014 , both with R^2 rounded to 1.000; the separate $L_A = L_C = 3$ full-width analysis gives 0.5251 ± 0.0016 with $R^2 = 0.9999$. The approximate factor of two is visible, but both coefficients lie roughly 50% above the exact targets. In direct central-charge units these are 1.4922(84), 1.5336(42), and 1.5753(48), respectively. Moreover this run uses $n_{\text{shell}} = \text{None}$, nonperfect stochastic feedback, the legacy `dw_symmetric` schedule, one circumference, and weak source provenance, so it is a reusable estimator pilot rather than evidence for the S1 production model.

The atlas truth notebook now re-exports the exact left-wall, right-wall, and full-width curves together with the historical $S = 100$ matched wall/full curves in a compact, machine-readable cache under `experiment_review/legacy_evidence_figure_atlas/data/`. The paired `result_05_tripartite_information_reduction.npz` and `result_05_tripartite_information_reduction.json` artifacts record the curve rows, fit summaries, protocol metadata, and source fingerprints used by the atlas. This provenance repair makes the surviving curves reusable; it does not upgrade the historical stochastic protocol or its central-charge estimate.

A second, previously omitted Colab archive provides a useful physical-state cross-check. The canonical GPU covariance snapshots and downstream CPU analyses under `00_WORKSPACE/COLAB/colab_small_system_testing/` cover $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, $S = 10$, and $C = 50$ with perfect correction. Fitting the final full-width entropy over $8 \leq A_y \leq N_y/2$ gives

	$n_{\text{shell}} = 1$	$n_{\text{shell}} = 2$
$N_y = 30$	$c_{\text{eff}} = 1.04583(13)$	$1.05513(16)$
$N_y = 40$	$c_{\text{eff}} = 1.03244(22)$	$1.08650(21)$

(43)

with $R^2 \geq 0.99995$; the parentheses here are regression standard errors across the A_y fit points, not trajectory bootstrap errors. On the fixed correlator window $2 \leq r_y \leq N_y/4$, the two-wall-window squared-correlator exponents are 2.154(36), 2.126(33), 2.191(36), and 2.092(34) in the same order, while the matched non-wall slices are much steeper, 4.11–6.37. These numbers independently support a wall-localized algebraic sector and show that it persists at both shell depths. They are not a precision exponent determination: the dedicated window audit under `00_WORKSPACE/COLAB/colab_charge_fluctuations/analysis_outputs/correlator_scaling_diagnostics/` finds order-one drift when the short- and long-distance endpoints are varied. This is a fit-window/finite-size limitation, not a contradiction with the stable selected-window fits.

The same raw cycle-50 covariances also contain a previously unreported charge-sector observable. Reconstructing $C = (G + 1)/2$ and evaluating the full- x interval

quantum variance trajectory by trajectory,

$$F_{A,\xi}^q = \text{Tr}[C_{A,\xi}(\mathbf{1}_A - C_{A,\xi})], \quad (44)$$

gives the two-wall log-chord fit

$$F_A^q = \frac{k_{\text{eff}}}{\pi^2} \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + b. \quad (45)$$

Over $8 \leq A_y \leq N_y/2$, the fitted levels are

	$n_{\text{shell}} = 1$	$n_{\text{shell}} = 2$
$N_y = 30$	1.0409(166)	1.0468(106)
$N_y = 40$	1.0328(346)	1.0853(449)

(46)

where the parentheses are standard errors over ten trajectory-resolved coefficients. The ensemble-mean curves have $R^2 > 0.99996$, and the four values track the matched entropy coefficients above. This upgrades the archive from an entropy-only $c \simeq 1$ precursor to a matched $c \simeq k \simeq 1$ precursor on the same conditioned ensembles. It remains finite-size evidence rather than a thermodynamic current-algebra measurement.

Table IX records these non-executable historical results, the compact atlas re-export, and the source-output mismatch that is otherwise easy to mistake for stronger evidence. The saved $S = 250$ plots in the legacy analysis material under `notebooks/` do not match the current source configuration and are not treated as reproducible results.

Thus the exact estimator and its factor-of-two decomposition are established, one longer stochastic history is preserved at curve level as a wrong-protocol pilot, and a separate canonical perfect-correction archive already supplies matched von Neumann entropy and within-trajectory quantum charge variance. No admissible adaptive dataset combines the production channel word and random-serial schedule, the accepted circumference sweep and sample count, stationary snapshots, $q = 1, 2, 3$ entropies, and the full matched observable bundle. S1 therefore remains planned for its entropy, charge, contour, and correlator products only; the TMI estimator is not to be recomputed.

Downgrade, figure, sharding, and legacy. If only entropy is compatible with one, say “consistent with a $c = 1$ wall” and do not claim a charge-resolved CFT. If c and k are individually stable but Eq. (39) fails, report them separately and do not identify the wall as a minimal charged $U(1)_1$ sector. This campaign supplies all of Fig. 2. Shard by trajectory; source-position averages are internal correlated averages. Path-audited visual references are under `00_WORKSPACE/COLAB/colab_large_entanglement_scaling_N20/docs/figures/`. The contour is a localization and sum-rule check, not a standalone novelty claim; BPJ already used entanglement contours to identify slow domain-wall structure. The archived TMI pilots in Table IX are provenance only and are excluded from the production gate and execution matrix. The newly reduced $N_x = 16$ pure-state covariance archive does contain both full-strip entropy and matched within-trajectory

interval variance, but only at $N_y = 30, 40$ with ten trajectories and without the production schedule, Rényi bundle, one-wall decomposition, or thermodynamic extrapolation. The saved max-mix total variance remains a purification observable rather than the S1 current-algebra variance.

B. Campaign S2: topological-to-trivial sweep

Claim and status. Status PLANNED. The logarithmic wall sector and near-marginal tangent sector occur when the two sides have different bulk topology.

Dependencies and run matrix. Calibrate α_c from the exact uniform Hamiltonian. Fix $\alpha_{\text{out}} = 30$ and take

$$\alpha_{\text{in}} = 1, 1.5, 1.75, 1.875, 2.125, 2.25, 2.5, 3, \quad (47)$$

at $N_y = 24, 32, 48, 64$, the accepted N_x , and the active $S = 10 = 2 \times 5$ minimum. The explicit-interface construction has independently randomized half-filled pure Gaussian and $C(0) = 1/2$ arms. A third, pure-only arm repeats every point with support termination: `DW=True, dw.truncation=True, meas.slabs_only=True`, $n_{\text{shell}} = 1$, $\alpha_1 = \alpha_{\text{in}}$, and $\alpha_2 = 30$. No support-terminated max-mix arm is introduced. The singular critical projector at $\alpha_{\text{in}} = 2$ is bracketed but not used as a wall-CFT data point. The three arms share the mass grid, schedule law, duration, tangent rank, and output conventions, but are distinct Born ensembles and are never pooled trajectory by trajectory.

Initialization, saves, estimator, and controls. Use a total duration of $2N_y$ with no discarded physical burn-in in either arm. The pure arm is the stationary conditioned-spectrum read: initialize every trajectory independently, align its physical tangent frame during cycles $1, \dots, N_y$, and save the late state, entropy, charge, topology, correlator, and tangent products at $N_y, 3N_y/2, 2N_y$. The max-mix arm is the purification-spectrum read: begin every record at $C(0) = 1/2$, save every early cycle through $\min(16, N_y/2)$ and a rounded logarithmic late grid, and retain the same late checkpoints. Both arms save the bulk marker and purity gap, entropy and charge coefficients, correlator, ordered compact record, stabilized tangent QR products, and physical low-subspace localization. Do not fit the bracketed bulk-critical point to a pure wall CFT. Compare deep topological, crossover, and deep trivial regimes separately in each initialization arm.

For the two pure constructions, fit each trajectory at $t = 2N_y$ over $2 \leq A_y \leq N_y/2 - 1$ to

$$S(A_y) = b + m \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right], \quad c_{\text{eff}} = 3m, \quad (48)$$

and to the nested constant area-law model. Report $\Delta \text{AICc} = \text{AICc}_{\text{area}} - \text{AICc}_{\text{log}}$, retain every per-trajectory fit, and bootstrap whole trajectories with 2000 deterministic resamples. Repeat the fit at N_y and $3N_y/2$ as a checkpoint-stability diagnostic. The figure shows representative endpoint curves, $c_{\text{eff}}(\alpha_{\text{in}})$ at all four circumferences for both constructions, and ΔAICc or the bulk marker.

TABLE IX. Archived tripartite-information evidence retained for provenance only. No row in this table defines or motivates a new production run. Slopes refer to the repository’s contour-decomposed $I_2 - I_3$ fit against $\log[1/(1-x)]$; a wall-window sum is a decomposition of a full-width subsystem entropy, not an independent wall-subsystem entropy. The exact and historical $S = 100$ curve rows are re-exported by the atlas truth notebook under `experiment_review/legacy_evidence_figure_atlas/data/`. The excluded source–output mismatch does not enter the scientific summary.

Artifact and protocol	Result	S1 interpretation
Exact DW ground state, 20×40 , $L_A = L_C = 3$; raw left/right/full curves re-exported in the atlas compact cache	Each wall: $c_{\text{TMI}} = 1.0018060(1620)$; full width: $c_{\text{TMI}} = 1.0026974(2861)$, with $R^2 > 0.999998$.	Strong deterministic estimator calibration; passes the one-wall/two-wall coefficient and factor-of-two checks, but contains no Born dynamics.
Canonical pure-state covariance snapshots, $16 \times (30, 40)$, $n_{\text{shell}} = 1, 2$, $S = 10$, $C = 50$	Late-window entropy gives $c_{\text{eff}} = 1.032\text{--}1.086$; matched interval fluctuations give $k_{\text{eff}} = 1.033\text{--}1.085$; fixed-window wall exponents are $2.092\text{--}2.191$, versus nonwall values $4.11\text{--}6.37$.	Strong perfect-correction matched entropy–charge cross-check. Only two sizes and ten trajectories are present, and the correlator exponent is window sensitive.
Full covariance history, 12×31 , $C = 100$, $S = 100$, $n_{\text{shell}} = \text{None}$, nonperfect feedback; matched wall/full <code>xsum.window.integrated</code> reduction re-exported in the atlas compact cache	For $L_A = L_C = 4$, one wall gives $c_{\text{TMI}} = 1.4922(84)$ and full width gives $1.5336(42)$; the $L_A = L_C = 3$ full-width value is $1.5753(48)$.	Best reusable dynamical TMI input, but wrong protocol, one small geometry, and coefficients well above the exact values.
Saved dynamic TMI output under <code>notebooks/</code>	Plot output claims $S = 250$, while the current source/configuration does not reproduce that sample contract.	Excluded as a stale source–output mismatch.

Statistics and acceptance gate. Within each initialization arm, bootstrap whole trajectories to locate jointly the topological-marker change, disappearance of the logarithmic or purification residual, and opening of the tangent gap. The pure-state tangent gap is the primary stationary phase-boundary result. The max-mix physical-tangent and purification products are the primary mechanistic result. Require each arm to resolve a common finite-size crossover interval among its own observables, then test overlap of the two arm-level intervals; do not require identical finite-size pseudocritical points or equality of finite-time gaps. A max-mix late-state CFT interpretation additionally requires its declared purity and stability gates.

For the two-construction entropy claim, additionally require both pure arms to approach $c_{\text{eff}} = 1$ at $\alpha_{\text{in}} = 1$, approach zero at $\alpha_{\text{in}} = 3$, prefer log-chord and constant scaling on the corresponding sides, and locate overlapping finite-size crossover brackets within $1.875 < \alpha_c < 2.125$. Pointwise agreement near the finite-size crossover is not required. Failure is reported as distinct programmable-wall transitions or as inconclusive, never silently promoted to equivalence.

Current legacy result. The exact uniform Bloch Hamiltonian places the positive-mass transition at $\alpha_c = 2$. Existing legacy sweeps consistently resolve a dynamical crossover there, but none measures the complete S2 observable bundle. The closest production-path input is the canonical GPU particle–Choi transfer campaign under `00_WORKSPACE/COLAB/colab_no_feedback_alpha_sweep_transfer/`. At $N_x = 20$, $N_y = 20, 30, 40$, $S = 10$, $C = 2N_y$, $\alpha_{\text{out}} = 30$, and $\alpha_{\text{in}} = 1, 1.5, 1.8, 1.9, 2, 2.1, 2.2, 2.5, 3$, the no-feedback branch is complete at all three sizes. Its mean smallest finite trans-

fer exponent is respectively $0.01215(229)$, $0.00808(182)$, and $0.00249(90)$ at $\alpha_{\text{in}} = 1.9$, but rises to $0.04366(577)$, $0.03440(527)$, and $0.03683(450)$ at 2.1 . At 2.2 the corresponding values are $0.12059(1268)$, $0.09505(841)$, and $0.08394(908)$, with only eight of ten finite samples at $N_y = 40$; by 2.5 no finite transfer mode remains. Thus a finite particle–Choi sector softens with circumference on the topological side and is lost across the crossover. A missing finite mode is a censored transfer observable, not an infinite measured gap.

The perfect-correction branch gives the same qualitative boundary more sharply. At $\alpha_{\text{in}} = 1.5$ its gap decreases from $0.03065(805)$ to $0.02244(316)$ to $0.01847(199)$ across $N_y = 20, 30, 40$, whereas at 1.9 it is already $0.10730(938)$, $0.11146(838)$, and $0.09887(581)$; the last value has eight finite samples. The $N_y = 40$ perfect-correction sweep stops at 1.9 , and at smaller sizes the finite sector is almost entirely absent at and above 2 . These are singular exponents of the particle–Choi transfer construction, not the physical covariance tangent exponents required by S2.

The executed finite-mode analysis in the same Colab folder adds spatial information that was absent from the original scalar summary. At $N_y = 40$ and $\alpha_{\text{in}} = 1$, the slowest finite particle–Choi mode has mean weight 0.45187 within one column of the reduced-slab edges without feedback, versus 0.87933 under perfect correction. Its mean edge distance correspondingly falls from 1.9944 to 0.5745 cells, and its site participation ratio falls from $156.24(728)$ to $56.72(278)$. Approaching the transition on the perfect-correction branch, the edge weight decreases to 0.65428 at $\alpha = 1.5$ and 0.46673 at 1.9 , while the mean edge distance grows to 1.2694 and 1.9432 . There are no parse failures in the $7,151$ -mode analysis. This is a strong localization diagnostic for the conditioned transfer

construction, but the reduced Choi-slab basis and finite-sector censoring prevent its reinterpretation as a physical tangent eigenmode.

A complementary canonical CPU max-mix campaign under `00_WORKSPACE/COLAB/colab_charge_fluctuations/cpu_data/` completes all 63 combinations of $N_x = 16$, $N_y = 16, 24, 32$, 21 masses from 1 to 3 in steps of 0.1, and $S = 10$. At $N_y = 32$ the final slab entropy and total charge variance are 0.1954 and 0.04793 at $\alpha_{\text{in}} = 1$, and 0.1244 and 0.03447 at 1.8, but fall to approximately 2.04×10^{-8} and 7.9×10^{-12} at 2. The saved projector diagnostics flag zero minimum Bloch norm at the critical point itself, so that point is not interpreted as a wall-CFT fit. Moreover these are purification observables from a maximally mixed initial state, not stationary pure-state wall entropy and current coefficients.

Table X includes the exact and CPU particle-Choi cross-checks. The exact flattened-Hamiltonian sweep fixes $\alpha_{\text{in}} = 1$ and varies α_{out} , so it calibrates the hard-wall interpretation rather than executing the S2 scan. The CPU Choi campaign completes 27 runs at $N_x = 16$, $N_y = 16, 24, 32$, and the nine masses above. At $N_y = 32$ its mean finite exponent grows from 0.04936(560) at 1.8 to 0.08948(524) at 1.9; at 2 no sample has a finite mode, while the apparent values near 0.21 at 2.1 and 2.2 each come from only two of ten samples and are therefore censored tails.

Downgrade, figure, sharding, and legacy. Preemptive wall crossover is allowed and must be reported. This is Fig. 2(e) if compact; otherwise Fig. S3. Shard by trajectory. The legacy results in Table X locate a common crossover near $\alpha = 2$, but no saved dataset provides the paired pure/max-mix physical-tangent grids together with the bulk marker and purity gap, stationary wall entropy and charge coefficients, correlator, and purification products at $S = 25$ and $N_y = 24, 32, 48, 64$. S2 therefore remains planned.

XI. TANGENT DYNAMICS AND PURIFICATION

A. Campaign T1: quenched tangent spectrum and localization

Claim and status. Status PLANNED. The differential of the normalized realized covariance map has a wall-localized near-marginal sector whose gap closes with N_y although exact projectors make a conventional square transfer product singular.

Dependencies and run matrix. Depends on W1. Use the accepted N_x , Eq. (4), $S = 25$, wall and matched trivial controls, and enough tangent vectors to retain at least 12 physical modes nearest zero. At $N_y = 32, 48$, double the tangent-frame rank and repeat at least four independently randomized initial tangent frames on the same replayed physical records.

Initialization, duration, and observation. Initialize the physical trajectory and a common physical tangent subspace together at $t = 0$. Evolve for a total of $2N_y$ cycles. Use the first N_y cycles only to align the tangent frame; discard their QR log sums but do not call or treat these physical cycles as burn-in. Without resetting the physical trajectory or aligned frame, accumulate the reported exponents over the final N_y cycles, with cumulative checkpoints at $N_y/4, N_y/2, 3N_y/4, N_y$ within that accumulation window. Fixed-size finite-time drift across these checkpoints is reported as a systematic diagnostic. Production is not automatically extended beyond $2N_y$; failure to stabilize is a failed temporal gate and requires a separately approved amendment.

Raw saves and estimator. Save per-cycle QR log increments, finite/null masks, gauge directions, aligned low vectors, orthogonality residuals, and stable cocycle snapshots. A snapshot consists of the physical covariance at its left endpoint, the ordered realized branch/channel word for the window, the aligned input and output QR frames, block- R factors with their diagonal log scales, and a replay hash. On validation sizes also save the explicit window product and its SVD; at production sizes reconstruct it from the stabilized factors rather than multiplying a long ill-conditioned dense word. Dense Choi frames are outside the production save and runtime contracts. For trajectory s , define the one-leg half-Jacobian growth rates

$$\begin{aligned} \ell_{j,s}(T) &= \frac{1}{T} \sum_{\tau=1}^T \log |(R_{\tau,s})_{jj}|, \\ \Delta_{1\text{leg}}(T) &= \min_{j \in \text{physical}} |\bar{\ell}_j(T)|. \end{aligned} \quad (49)$$

The primary quenched spectrum averages log stretches at continuously tracked Oseledec index before taking the gap. If decay-rate notation is desired, use $\gamma_j = -\ell_j$ and state the sign convention. The full covariance-tangent map is a congruence, so its exponents are pair sums $\ell_a + \ell_b$ (or decay rates $\gamma_a + \gamma_b$), not the one-leg spectrum itself. The mean and median of per-trajectory minimum gaps are reported as sensitivity diagnostics. Exact null and normalization directions are removed before defining the physical reference exponent. Appendix C fixes the cycle-time normalization, the equivalent finite-time log-strain operator, the stabilized reconstruction of singular modes, and the distinction between a quenched spectrum, an averaged log-strain operator, and an annealed second-moment object.

For a near-degenerate low set $\mathcal{L}_s$ of rank r_s , localize its projector rather than a basis-dependent QR vector. The primary density is

$$\rho_s(x, y) = \frac{1}{r_s} \sum_{a \in \mathcal{L}_s, \mu} |v_{s,a,\mu}(x, y)|^2, \quad \sum_{x,y} \rho_s(x, y) = 1. \quad (50)$$

Never normalize the orbital components independently. Resolve both walls by diagonalizing a projected transverse-position operator within $\mathcal{L}_s$ or by reporting left/right-

TABLE X. Legacy evidence relevant to the S2 topological-to-trivial sweep. Parentheses give the standard error in the last quoted digits. Particle–Choi gaps are reported only over finite samples; loss of the finite sector is recorded separately and is not converted into a numerical gap.

Artifact and protocol	Result	S2 interpretation
Exact flattened Hamiltonian, $\alpha_{\text{in}} = 1$, exterior-mass sweep	The physical-wall estimate crosses from $c_{\text{est}} = 2.006$ at $\alpha_{\text{out}} = 1$ to 1.079 at 2 and 1.004 at 3.	Exact calibration of $\alpha_c = 2$ and of the hard-wall limit; the sweep direction differs from S2.
Canonical GPU particle–Choi transfer, $20 \times (20, 30, 40)$, $C = 2N_y$, $S = 10$	No feedback: the gap changes from 0.01215, 0.00808, 0.00249 at $\alpha = 1.9$ to 0.04366, 0.03440, 0.03683 at 2.1. Perfect correction shows a sharper rise by 1.9 and extensive censoring at and above 2.	Strongest direct dynamical transition precursor; canonical source and useful size trend, but a particle–Choi proxy rather than the S2 tangent/entropy/marker bundle.
Finite particle–Choi mode profiles from the same GPU sweep	At $N_y = 40$, $\alpha = 1$, perfect correction raises one-column edge weight from 0.45187 to 0.87933 and reduces mean edge distance from 1.9944 to 0.5745 cells; localization weakens again by $\alpha = 1.9$.	Spatially resolves the feedback-dependent conditioned transfer mode. It remains a reduced-basis proxy and is not a physical covariance-tangent localization result.
Canonical CPU max-mix purification, $16 \times (16, 24, 32)$, 21 masses, $C = 2N_y$, $S = 10$	All 63 runs complete. At $N_y = 32$, entropy/variance change from 0.1244/0.03447 at 1.8 to $2.04 \times 10^{-8}/7.9 \times 10^{-12}$ at 2.	Direct fine mass scan, but it measures transient max-mix purification; the critical projector is singular and no stationary wall coefficients were saved.
Canonical CPU particle–Choi transfer, $16 \times (16, 24, 32)$, nine masses, $S = 10$	All 27 runs complete. At $N_y = 32$ the finite gap rises from 0.04936(560) at 1.8 to 0.08948(524) at 1.9; the finite sector is absent or severely censored thereafter.	Independent CPU confirmation of the crossover, but still a particle–Choi observable and not a thermodynamic tangent-gap determination.

window weights; do not select whichever individual QR vector looks most localized.

Controls, statistics, and gate. Compare aN_y^{-z} , a free-intercept power law, and a gapped constant plus corrections, using at least five sizes and bootstrap fits. Require the wall gap to have zero intercept, preferably $z = 1$, while the matched trivial control retains a positive intercept. Require the fixed- N_y temporal extrapolation above, initial-frame and rank convergence, near-degenerate-subspace interface weight above 70%, representation of both walls, and transverse localization satisfying W1.

Current legacy result. **Assessment: finite-size softening and wall localization are established as precursors, but a thermodynamic tangent-gap closing is not.** The strongest fully archived CPU result is the canonical perfect-correction campaign in `00_WORKSPACE/CURRENT/topological_frustration_diagnostics/results/campaigns/20260813_121340_Nx20_perfect_Ny20_22_24/`. At $N_x = 20$, $N_y = 20, 22, 24$, and $S = 25$, the mean final wall gaps are 0.11249(470), 0.10830(519), and 0.10142(464), whereas the uniform-control gaps are 0.89365(1721), 0.89638(1259), and 0.89600(916). The selected wall slow vector carries 0.99365(222), 0.99584(119), and 0.99222(333) of its norm in the interface region. This is a sharp geometry-resolved separation between a slow wall sector and an order-one bulk control. It is nevertheless a three-size raster pilot with $2N_y$ total cycles; its cumulative spectrum does not implement T1’s declared N_y tangent-frame alignment followed by a separate N_y accumulation. Moreover, the selected individual QR vector lies almost entirely on one wall, so it cannot replace the basis-independent low-subspace and two-wall test required

by the current estimator.

The broadest valid GPU precursor is the canonical campaign under `00_WORKSPACE/COLAB/colab_lyapunov/gpu_data/lyapunov_spectra/campaigns/N20_Ny30-50_nsh1_a1-1_S25_cyclesNy/`. For the support-terminated wall at $N_x = 20$, $N_y = 30, 40, 50$, $S = 25$, and $T = N_y$, its final mean one-body gaps are 0.02698(313), 0.02431(287), and 0.01968(225). The corresponding uniform controls remain at 0.72169(1097), 0.74833(863), and 0.76760(874). Directly on the two wall columns, the saved minimum-gap vector has weights 0.9326, 0.9349, and 0.9435; including the adjacent columns gives 0.9611, 0.9672, and 0.9708. These GPU data extend the softening trend to larger circumferences and independently confirm strong wall localization. They still use only $T = N_y$, the default initial state, one saved basis-dependent minimum-gap vector, and no alignment, rank, initial-frame, or temporal convergence repeats.

The canonical CPU CFT-extraction archive supplies the decisive finite-time warning. With ten trajectories per size and no burn-in, the $T/L = 2$ tangent gaps at $L = 20, 30, 40$ are 0.04511, 0.04002, and 0.02839; at $T/L = 5$ they are 0.05914, 0.04180, and 0.03651. The latter final-quarter tangent drifts remain 18.2%, 15.8%, and -3.0% . Even at $T/L = 20$, the $L = 20, 22, 24$ gaps are 0.08833, 0.07879, and 0.06195, with residual tangent drift of 2.1–3.3% and a 40% shift between free-intercept and through-origin scaling estimates. These runs therefore demonstrate that scaling observation time with circumference does not by itself deliver the fixed-size temporal limit. Their driver also failed to pass the declared sample seed into the canonical engine’s random initial state, so they are convergence diagnostics rather than replayable production

evidence.

The tangent implementation itself has a strong local check: the V0 five-scale comparison between the analytic tangent and finite differences reaches a minimum relative error of 7.37×10^{-11} . This validates the differential on the tested CPU cases, but no saved production campaign yet combines that validated map with T1's replay snapshots, independent-start and alignment protocol, five accepted sizes, rank/frame repeats, two-wall low-subspace localization, and $S = 25$. Table XI separates the physical precursors from this method-only validation.

Downgrade, figure, sharding, and legacy. If the intercept is unresolved, call the result finite-size softening rather than a thermodynamic closing. Results supply Fig. 5(b,c). Shard whole trajectories; never split one QR history across unordered shards. The valid legacy evidence summarized in Table XI establishes a wall-localized finite-size slow sector with a strongly gapped uniform control, but its size-dependent observation windows do not separate finite-time drift from a thermodynamic gap closing. The principal legacy figures are under `00_WORKSPACE/COLAB/colab_lyapunov/figs/`. T1 therefore remains planned pending the complete temporal, finite-size, rank/frame, and two-wall localization gates.

B. Campaign P2: max-mix purification and physical tangent scaling

Claim and status. Status PLANNED. The tangent singular spectrum quantitatively controls slow purification of a maximally mixed state at the wall.

Dependencies and run matrix. Depends on T1. Use the accepted N_x , Eq. (4), $S = 25$, wall and matched trivial controls, and a total duration of $2N_y$ cycles.

Initialization, duration, and observation. Set $C(0) = 1/2$. There is *no burn-in*: purification begins at $t = 0$. Save every early cycle and logarithmically spaced late times. Any stationary-window analysis is a separate post-purification operation.

Raw saves and estimator. Save global entropy, $\text{Tr}[C(\mathbf{1} - C)]$, purity gap, spatial entropy contour, Chern marker, charge cumulants, and matched physical-tangent singular data. For each trajectory propagate the *same* max-mix record and the full start-to- t physical tangent cocycle. Dense Choi covariance tracking is forbidden: production neither constructs a reduced Choi frame nor tests a Choi Gram identity. Instead compare per-record purification loss with the contemporaneous slow tangent rates, then compare the trajectory-bootstrap finite-size and finite-time scaling intervals. This empirical comparison does not identify the stationary pure-start T1 spectrum; compare the two only after their record, initialization, and time-window differences are controlled.

Use the entropy contour and the matched wall-off run to define a wall-excess residual $S_{\text{wall}}^{\text{res}}(t, L)$. Fit

$$S_{\text{wall}}^{\text{res}}(t, L) = L^y \mathcal{F}(t/L^z), \quad (51)$$

testing the per-length hypothesis $y = 1$ and a free y . A global entropy initialized from max-mix is not collapsed without this amplitude factor or an explicitly justified normalization. Fit z independently and compare it with the appropriate pair-sum tangent prediction; do not force agreement with the one-leg T1 gap.

Controls, statistics, and gate. Require a reproducible same-record association between slow physical-tangent rates and slow purification, compatible tangent and purification scaling intervals under the declared conversion, bulk purification faster than the wall sector, and a collapse stable to time-window and amplitude-normalization changes. Bootstrap whole trajectories and do not promote correlation alone to an exact operator identity.

Current legacy result. Assessment: rapid max-mix purification and a wall-localized late residual are established, but the matched same-record physical tangent campaign has not been run. The strongest direct purification archive is the canonical GPU campaign in `00_WORKSPACE/COLAB/colab_charge_fluctuations/gpu_data/purification_dynamics_maxmix/campaigns/N20_multi_geometry_nsh1_dwtrunc1_init-maxmix_S100_cycles-2Ny/`. It begins from max-mix and follows 100 perfect-correction trajectories at $N_x = 20$, $N_y = 30, 40, 50$ through $2N_y$ raster cycles. At $t = N_y$, the mean total entropies are 0.8591, 0.8571, and 0.8756, while $\text{Tr}[C(\mathbf{1} - C)]$ is 0.2552, 0.2532, and 0.2621. At $t = 2N_y$ these fall to entropies 0.2444(233), 0.2907(284), and 0.3313(271) and intrinsic charge variances 0.06647(802), 0.08464(991), and 0.09733(952). The corresponding final real-space Chern estimates are 0.9891(44), 0.9908(29), and 0.9879(49). Thus the purification clock is approximately organized by t/N_y over this range, but the nonzero and size-dependent residual at $2N_y$ is not an asymptotic collapse. This campaign saves streamed entropy, charge, and Chern observables, not covariance histories, branch records, purity gaps, entropy contours, or tangent data.

A canonical CPU mass sweep supplies the most useful control. In the same support-terminated slab geometry at $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, and $T = 2N_y$, the topological $\alpha_{\text{in}} = 1$ runs end with entropy 0.3459, 0.2076, and 0.1954 and $\text{Tr}[C(\mathbf{1} - C)] = 0.09898$, 0.05444, and 0.04793. For the geometrically matched trivial $\alpha_{\text{in}} = 3$ controller, entropy is already only $(1.01, 1.51, 2.02) \times 10^{-8}$ at $t = N_y$ and the variance is at or below 2×10^{-14} . The trivial sector therefore purifies parametrically faster on these sizes. The comparison has only ten trajectories, three circumferences, and $2N_y$ cycles, so it is a strong wall-versus-bulk precursor rather than a scaling determination.

A matched support-construction sweep strengthens this purification inventory. Under `00_WORKSPACE/COLAB/colab_charge_fluctuations/`, 27 untruncated-interface perfect-correction configurations keep `DW=True`, $\alpha_{\text{out}} = 30$, and `dw.truncation=False` at $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, and $T = 2N_y$. At $N_y = 32$, the untruncated final entropy is 0.10488, 0.26286, and 6.06×10^{-5} at $\alpha_{\text{in}} = 1, 1.5, 1.9$, while the matched support-terminated

TABLE XI. Completed valid legacy work relevant to T1. Parentheses give the standard error in the last quoted digits where it is available. The saved spectra are useful finite-time precursors and are not pooled into the preregistered T1 scaling fit.

artifact and coverage	strongest quantitative result	T1 implication and limitation
Canonical CPU wall and uniform spectra, $N_x = 20$, $N_y = 20, 22, 24$, $S = 25$, $T = 2N_y$	Wall gaps are 0.11249, 0.10830, and 0.10142, versus 0.89365, 0.89638, and 0.89600 for uniform controls. Interface weights are 0.99365, 0.99584, and 0.99222.	Strongest provenance-complete geometry control and localization result. Only three narrow sizes, raster order, and an individual QR vector are available; no T1 temporal, rank, frame, or two-wall-subspace gate was run.
Canonical GPU wall and uniform spectra, $N_x = 20$, $N_y = 30, 40, 50$, $S = 25$, $T = N_y$	Wall gaps fall from 0.02698 to 0.01968, while uniform gaps stay in $[0.72169, 0.76760]$. Direct wall-column weights rise from 0.9326 to 0.9435 and neighbor-inclusive weights from 0.9611 to 0.9708.	Broadest valid circumference trend and independent GPU confirmation. The observation window is short and size dependent, with no declared alignment, rank/frame repeats, or low-subspace localization.
Canonical CPU CFT-extraction histories, $S = 10$, $T/L = 2, 5, 20$	At $T/L = 5$, gaps are 0.05914, 0.04180, and 0.03651, but final-quarter drifts are 18.2%, 15.8%, and -3.0% . At $T/L = 20$, residual drift remains 2.1–3.3%.	Direct evidence that finite-time bias remains material. There is no declared alignment, localization save, or fully reproducible initial-state seed, and the long run spans only $N_y = 20, 22, 24$.
V0 analytic-tangent validation and cocycle infrastructure	The five-scale finite-difference test reaches relative error 7.37×10^{-11} ; CPU replay, resume, and serial-shard checks pass on validation sizes.	Supports the production estimator and save contract, but is method validation rather than evidence for a wall-localized thermodynamic near-marginal sector.

values are 0.18509, 0.45161, and 0.06416. The full-layer untruncated construction therefore retains a genuine spatial interface yet purifies more rapidly below the transition. Since the two arms contain different numbers of measured modes, raw totals and per-mode densities must be compared separately; neither substitutes for the matched same-record physical tangent analysis.

The spatial entropy-contour archive under `00_WORKSPACE/COLAB/colab_small_system_testing/gpu_data/purification_entropy_contours_maxmix/` localizes the residual directly. In the 16×32 , $n_{\text{shell}} = 1$, $S = 10$, $T = 100$ hard-wall run, the final entropy is 0.01934, of which 94.25% lies on the two wall columns and 97.46% lies within the wall columns and their nearest neighbors. The corresponding untruncated construction has final entropy 4.37×10^{-4} . An independent 30×40 , $S = 25$, $T = 100$ hard-wall run ends at entropy 0.16224, with 93.26% on the two wall columns and 96.30% including their nearest neighbors. A canonical CPU 20×20 , $S = 10$, $T = 40$ convergence study gives the same construction dependence: the final entropy is 0.3278(898) bits for the support-terminated wall and 0.01657(639) bits without support truncation. These archives demonstrate spatially concentrated slow purification, but the two constructions have different active supports and do not by themselves define the P2 wall-off subtraction.

Table XII records the resulting evidence hierarchy. No saved artifact combines the complete max-mix branch word with the start-to- t physical tangent cocycle, purity gap, and the same trajectory’s entropy spectrum. The available tangent-edge and flux-cocycle drivers implement useful pieces of the required record and stabilization machinery, but have no saved production outputs. Consequently a tangent-predicted purification exponent has no legacy numerical support. Historical Choi calculations are retained as proxy evidence only and are not part of

the production estimator.

Downgrade, figure, sharding, and legacy. If the tangent-purification association is stable but no universal collapse occurs, report the measured time dependence without a universal exponent. Results supply Fig. 5(d,e). Shard by complete trajectory. Table XII establishes rapid purification, a slower topological-wall sector, and a strongly wall-localized residual, but not the same-record tangent mechanism or a controlled universal exponent. P2 therefore remains planned pending the $2N_y$ wall/control family and complete physical-tangent save contract.

TABLE XII. Completed valid legacy work relevant to P2. Parentheses give the standard error in the last quoted digits where available. The purification archives are not pooled into a universal collapse because their sizes, durations, active supports, and saved objects differ.

artifact and coverage	strongest quantitative result	P2 implication and limitation
Canonical GPU max-mix wall dynamics, $N_x = 20$, $N_y = 30, 40, 50$, $S = 100$, $T = 2N_y$	At $t = N_y$, entropy and variance are nearly size independent at 0.857–0.876 and 0.253–0.262. At $2N_y$, entropy is 0.2444–0.3313, variance is 0.06647–0.09733, and the Chern estimate is 0.988–0.991.	Strongest ensemble purification and charge-sharpening result. It has no matched trivial arm, contour, purity gap, saved record, or tangent cocycle, and therefore lacks the complete P2 comparison contract.
Canonical CPU topological/trivial slab sweep, $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, $T = 2N_y$	The topological wall retains final entropy 0.3459, 0.2076, 0.1954 and variance 0.09898, 0.05444, 0.04793; the matched trivial controller is numerically pure by $t = N_y$.	Direct evidence that the wall sector purifies more slowly than the trivial bulk control, but only at three sizes and without matched tangent data or a long-time collapse.
Canonical GPU entropy contours, 16×32 and 30×40	The final hard-wall residual places 94.25% and 93.26% directly on the two wall columns, rising to 97.46% and 96.30% with nearest neighbors.	Establishes spatial localization of the late entropy. Sample counts are 10 and 25; the comparison arm changes support rather than turning the wall off at fixed support.
Canonical CPU construction comparison, 20×20 , $S = 10$, $T = 40$	Final entropy is 0.3278(898) bits with support truncation versus 0.01657(639) bits without it; late-cycle means are 0.4043 and 0.09011 bits.	Independent confirmation of delayed hard-wall purification, but a one-size construction test with unequal active supports and no spatial or tangent save.
Canonical CPU overlapping-interface sweep, $N_x = 16$, $N_y = 16, 24, 32$, $S = 10$, $T = 2N_y$	The 270-trajectory perfect-correction arm retains $\alpha_{\text{out}} = 30$ without support truncation. At $N_y = 32$, entropy falls from 0.26286 at $\alpha_{\text{in}} = 1.5$ to 6.06×10^{-5} at 1.9.	Strong multi-size evidence that the untruncated explicit interface purifies faster than the terminated slab below $\alpha = 2$; active supports differ and no recordwise tangent or purity-spectrum bundle was saved.
Same-record physical tangent comparison	No saved run contains both the max-mix record and the full start-to-time physical tangent products needed to compare tangent and purification scaling trajectory by trajectory.	The central P2 mechanism, tangent comparison, purity-gap gate, and universal collapse remain untested.

XII. EXACTLY THREE CHIRALITY ROUTES

The three routes use different clocks and prove different statements. Modular time probes the prepared state's entanglement generator; monitored circuit time probes physical Born-ensemble response; twist angle defines an auxiliary integer spectral-flow index. No one route substitutes for another, and none is called a quantized transport coefficient.

A. Campaign H1: modular charge drift

Claim and status. Status PLANNED. The wall-reduced modular generator is handed, with opposite signs on oppositely oriented walls.

Dependencies and run matrix. Depends on S1. Use $N_y = 32, 48, 64$, the accepted N_x , both oppositely oriented walls, the trivial control, and $S = 25$. A separately reversed-Chern controller is an escalation only if the two-wall orientation test is ambiguous; it is not a baseline trajectory family.

Initialization, duration, and observation. Use independent stationary snapshots from S1. Define the wall subsystem before examining the result and construct

$$h_{\xi,w} = \log \frac{1_w - C_{\xi,w}}{C_{\xi,w}}. \quad (52)$$

Launch identical localized packets and stop the modular-time fit before wraparound or loss of wall support.

Raw saves and estimator. Save modular spectra, clipping diagnostics, signed packet density, wall retention, width, periodic center of mass, and per-trajectory fitted velocity. Average signed densities and velocities only after per-trajectory construction.

Controls, statistics, and gate. Require opposite velocity signs with nonoverlapping 95% intervals, sign reversal with wall orientation, sustained wall localization, and no stable trivial-control drift. Vary the wall window, modular spectral cutoff, and fit interval.

Current legacy result. The exact handed target is established and trajectory-derived modular spreading is strong precursor evidence, but the H1 ensemble gate has not been run. The accepted B0 v2 campaign in `00_WORKSPACE/CURRENT/experiment_review/b0_exact_domain_wall/results/20260816_191957/` provides the cleanest deterministic calibration at $N_x = 20$, $N_y = 48$. For the coupled construction, the two endpoint-resolved modular velocities are $(-2.930, +4.536)$; for the hard-exterior construction they are $(-2.600, +4.406)$. In both cases the signs persist across all prespecified fit windows, the minimum three-column wall retention is 0.3245 and 0.3349, and the norm drift is below 4×10^{-15} . The v2 endpoint-resolved estimator supersedes the older centered-packet estimator, which mixed the two counterpropagating components and produced small same-sign slopes. This is an exact-state target with no trajectory distribution or trivial modular controller, not an H1 production result.

The strongest stochastic precursor starts from canonical GPU pure-state covariance snapshots under `00_WORKSPACE/COLAB/colab_small_system_testing/gpu_data/pure_state_covariance_snapshots/` and is analyzed in `00_WORKSPACE/COLAB/colab_small_system_testing/analysis_outputs/dynamic_modular_charge_spreading/`. It covers $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, ten trajectories, and snapshots at cycles 5, 10, 20, 50. The most useful product constructs and diagonalizes h_{ξ,y_0} separately for every trajectory and translated cut, evolves each packet, and only then averages its displacement. At cycle 50 and $N_y = 40$, $n_{\text{shell}} = 2$, the four endpoint packets have

$$\Delta y(t_{\text{mod}} = 32) = (8.821(120), -8.813(107), \quad (53) \\ 8.489(109), -8.520(131)),$$

ordered as the two longitudinal endpoints on the $x = 5$ wall followed by those on the $x = 11$ wall. Multiplying by the recorded wall orientations gives the expected opposite sign between physical walls for either endpoint convention. Charge drift is below 7×10^{-15} and the minimum retained window charge is 0.0297–0.0681 in this configuration. However, its quoted standard errors flatten ten independent trajectories and forty translated cuts into 400 observations. The cuts within one trajectory are correlated variance-reduction samples, so these errors are descriptive and are not the required trajectory-bootstrap confidence intervals. Moreover, about 77% of the modular eigenvalues are clipped at $\epsilon = 10^{-10}$ in this configuration.

A second product, `sample.y0_avg.hmod`, averages covariance-derived modular generators before packet evolution. Its cycle-50 $N_y = 40$ fits over $t_{\text{mod}} \in [0, 8]$ give the clean handed pair $(-2.769, +2.645)$ for $n_{\text{shell}} = 1$ and $(-2.138, +1.887)$ for $n_{\text{shell}} = 2$, with $R^2 = (0.846, 0.817)$ and $(0.860, 0.783)$, respectively. These are useful visualization and sign checks, but the nonlinear ordering violates H1's per-trajectory-first estimator and the remaining endpoint packets have low-quality fits. Finally, the older `notebooks/modular_charge_spreading/` archive evolves exact and eight flattened 20×40 constructions for modular time 40. It conserves charge to 4.4×10^{-7} or better, but all nine reported spectra saturate at ± 23.718998 because of the same cutoff; it is therefore construction calibration rather than an independent stochastic result.

Table XIII separates these estimator classes. None of the legacy artifacts provides the accepted-width $N_y = 32, 48, 64$, $S = 25$ trajectory family, a trivial modular control, trajectory-level velocity intervals, and simultaneous wall-window, cutoff, and fit-range sensitivity.

Downgrade, figure, sharding, and legacy. This proves modular handedness, not monitored-time transport. It supplies Fig. 3(a). Shard by complete circuit trajectory and retain all translated cuts within that shard. Table XIII establishes exact modular handedness and compatible legacy trajectory spreading, but not the production ensemble, trivial-control null, or trajectory-bootstrap intervals. H1 therefore remains planned.

TABLE XIII. Completed valid legacy work relevant to H1. Parentheses denote the reported standard error in the last digits. Translated cuts from one circuit trajectory are not counted as independent trajectories for the H1 acceptance gate.

artifact and coverage	strongest quantitative result	H1 implication and limitation
Accepted B0 v2 exact benchmark, $N_x = 20, N_y = 48$	Coupled velocities $(-2.930, +4.536)$ and hard-exterior velocities $(-2.600, +4.406)$; signs are window-stable, retention exceeds 0.32, and norm drift is below 4×10^{-15} .	Establishes the endpoint-resolved handed target and corrects the legacy estimator mismatch. It is deterministic and has neither a trajectory ensemble nor a trivial modular control.
Per-trajectory/per-cut GPU modular evolution, $N_x = 16, N_y = 30, 40, S = 10$	At $N_y = 40, n_{\text{shell}} = 2$, the four final displacements are $(8.821, -8.813, 8.489, -8.520)$ with nominal errors 0.107–0.131.	Strongest stochastic precursor and correct construction-before-averaging order. The 400 cut observations contain only ten independent trajectories, and spectral clipping affects about 77% of modes.
Pre-averaged modular generator, $N_x = 16, N_y = 40$, cycle 50	The clean endpoint pair has velocities $(-2.769, +2.645)$ for $n_{\text{shell}} = 1$ and $(-2.138, +1.887)$ for $n_{\text{shell}} = 2$ over $t_{\text{mod}} \in [0, 8]$.	Supports the sign pattern visually, but averaging before nonlinear evolution is not the H1 estimator and no trajectory confidence interval follows.
Exact and flattened legacy constructions, 20×40 , modular time 40	Charge drift is at most 4.4×10^{-7} , while all nine spectra terminate at the cutoff values ± 23.718998 .	Useful construction and conservation check; cutoff-saturated spectra and deterministic states prevent an ensemble handedness claim.
H1 production matrix and controls	No saved campaign combines accepted $N_x, N_y = 32, 48, 64, S = 25$, a trivial control, per-trajectory velocities, and the prescribed sensitivity grid.	The sign target is strongly motivated, but the stated H1 statistical and control gate remains open.

B. Campaign H2: physical Born response $\chi(k, \omega)$

Claim and status. Status PLANNED. The complete Born ensemble has a wall-localized causal response with a single low-energy velocity sign on each wall.

Dependencies and run matrix. Depends on W1 and V0 response validation. Use $N_y = 32, 48, 64$, $S = 25$, eight source positions and four stationary time origins per trajectory, and response horizon $N_y/2$. Only the production random-serial order is run.

Initialization, duration, and observation. Each of the 25 trajectories begins from an independent product-state draw and runs for $2N_y$ cycles with no separate physical burn-in. Apply the infinitesimal local density-potential kicks at the four preregistered late time origins whose response horizons fit inside that fixed trajectory. Sources and time origins inside one trajectory are correlated variance-reduction samples, not new trajectories.

Raw saves and estimator. For an observable O , evaluate the exact decomposition

$$\partial_\epsilon \mathbb{E}_{\xi, \epsilon}[O_{\xi, \epsilon}]|_0 = \mathbb{E}_\xi[\partial_\epsilon O_\xi] + \mathbb{E}_\xi[O_\xi \partial_\epsilon \log p_\xi]. \quad (54)$$

Save the conditional derivative, score term, total signed response, and cumulative score separately. Validate Eq. (54) against brute-force $\pm\epsilon$ ensembles on small systems. Compute

$$\chi_w(k, \omega) = \sum_{t, y} e^{i\omega t - iky} w_t \chi_w(y, t), \quad (55)$$

with a recorded window w_t . Zero padding is visual only; ridge fits use native resolution. Define in advance a low-energy wedge asymmetry between $\omega k > 0$ and $\omega k < 0$.

Controls, statistics, and gate. Bootstrap whole trajectories. Require a low-energy linear ridge, opposite slopes on the two walls, a stable sign across the declared random-serial ensemble, subdominant wrong-chirality weight, wall localization, and no corresponding trivial-control ridge. Both terms in Eq. (54) must be included in the headline panel.

Current legacy result. Only target calibration and conditional-response precursors exist; there is no saved full Born susceptibility. The accepted exact-domain-wall campaign under `00_WORKSPACE/CURRENT/experiment_review/b0_exact_domain_wall/results/20260816_191957/` supplies the clean Hamiltonian target. A localized density-phase impulse followed by central differencing produces paired physical velocities $(-0.8286, +0.8286)$, $(-0.8720, +0.8720)$, and $(-0.8947, +0.8947)$ at $N_x = 20$ and $N_y = 32, 48, 64$. At $N_y = 48$ the wavefront fits have $R^2 = 0.9938$, charge drift below 4×10^{-16} , and relative finite-difference error 5×10^{-7} . The archive also contains the corresponding momentum-frequency arrays. This is deterministic evolution under an exact Hamiltonian, so it calibrates orientation, Fourier conventions, and ridge extraction but contains neither adaptive records nor a likelihood-score term.

The strongest canonical-circuit precursor is the paired-reset benchmark under `00_WORKSPACE/CURRENT/topological_frustration_diagnostics/results/campaigns/20260816_233722_production/03_parallel_benchmark/response/`. It evolves eight 12×16 random-order trajectories after eight burn-in cycles and compares locally occupied and empty unit-cell resets for four response cycles. The diagonal wall velocities are $-0.252(322)$ and $-0.635(329)$, so the two walls do not have opposite signs. The paired branches share their visit schedule, but they do not share a realized outcome record. Moreover, an occupied-versus-empty reset is a finite injected charge, not the infinitesimal density-potential derivative in Eq. (54). This result is therefore a useful negative pilot for the current estimator, not evidence against Born-ensemble chirality.

The V1 smoke archive under `00_WORKSPACE/CURRENT/validation_campaigns/results/v0_v1_cpu_20260816_183244/smoke/v1/` makes the schedule problem explicit at 4×6 with only two trajectories per schedule. Random order gives two positive conditional velocities, 0.823 and 0.491; raster order gives 1.265 and -0.697 ; reverse raster gives means -0.747 and -0.560 , with intervals that include zero. The analysis classifies the result as schedule dependent, while its parent validation fails stationarity and explicitly defers the Born score to H2. A later 16×16 archive under `00_WORKSPACE/CURRENT/validation_campaigns/results/v1_cpu_N16x16_S10_20260816_225400/` completes all 30 trajectories: ten each for random, raster, and reverse-raster order over 64 cycles. Its conditional velocity means are respectively $(0.351, 0.423)$, $(0.352, 0.135)$, and $(0.256, -0.306)$, but every 95% interval contains zero. The reduced campaign therefore finds no resolved conditional-chirality sign and no categorical schedule change; it does not supply the Born likelihood-score term.

The score identity and record-Fisher increment are derived in the reference material under `tangent_edge_channel_note/`, while reusable tangent infrastructure exists under `00_WORKSPACE/CURRENT/tangent_edge_channel/`. Neither directory contains an H2 response run. In particular, no saved circuit artifact contains the conditional derivative, cumulative score, likelihood-score response, their sum, and $\chi_w(k, \omega)$ on the same trajectories. Table XIV records this limited evidence without promoting a finite-reset or fixed-record response to the physical Born susceptibility.

Downgrade, figure, sharding, and legacy. If only the fixed-record term is chiral, label it conditional response and do not claim physical Born-ensemble chirality. Results supply Fig. 3(b), with the two wall orientations shown inside one compound panel. Shard trajectories while keeping all sources together. Table XIV establishes the exact handed target and inventories the available conditional pilots, but no compatible legacy $\chi(k, \omega)$ result includes the score term. H2 therefore remains planned.

TABLE XIV. Completed valid legacy work relevant to H2. Parentheses give the standard error in the last quoted digits where available. Exact-Hamiltonian and conditional-response results are retained as calibration or pilot data, not as the full Born response.

artifact and coverage	strongest quantitative result	H2 implication and limitation
Accepted exact-domain-wall response, $N_x = 20$, $N_y = 32, 48, 64$ Canonical paired-reset circuit benchmark, 12×16 , $S = 8$	Opposite velocities evolve from ∓ 0.8286 to ∓ 0.8947 ; at $N_y = 48$, $v = \mp 0.8720$, $R^2 = 0.9938$, and charge and finite-difference checks pass. Random-order diagonal velocities are $-0.252(322)$ and $-0.635(329)$, so the opposite-wall test fails.	Strong deterministic target for wall orientation and ridge analysis. It has no adaptive Born record or score contribution. Strongest circuit response archive, but it uses a finite charge reset and separately sampled branches rather than an infinitesimal kick with full Born reweighting.
V1 conditional-response smoke, 4×6 , two trajectories per schedule Reduced V1 conditional response, 16×16 , ten trajectories per schedule	Random gives $(0.823, 0.491)$, raster gives $(1.265, -0.697)$, and reverse raster is unresolved; the saved classification is schedule dependent. All 30 trajectories complete. Random, raster, and reverse-raster means are $(0.351, 0.423)$, $(0.352, 0.135)$, and $(0.256, -0.306)$; every 95% interval crosses zero.	Exposes schedule and stationarity sensitivity. The score was explicitly deferred, and the sample count is diagnostic only. Completed finite-size schedule comparison with no categorical topology change, but the conditional-chirality sign is unresolved and the Born score remains absent.
Score-response implementation and H2 production data	The score increment is derived in reference notes, but no saved run contains both terms of Eq. (54) or a circuit $\chi_w(k, \omega)$.	The central physical-response estimator, trivial-control null, schedule reversal, size family, and trajectory-bootstrap gate remain untested.

C. Campaign H3: frozen-record integer twist flow

Claim and status. Status PLANNED. For one preregistered realized record, a continuous flux family has unit wall-resolved spectral flow; one matched-trivial record supplies the branch-level control. This representative-record test makes no claim about a Born-ensemble frequency.

Dependencies and run matrix. Depends on S1 and V0 twist validation. The promoted production replay uses $N_y = 24$, one preregistered frozen record per protocol, and

$$\phi_m = 2\pi m/32, \quad m = 0, \dots, 32. \quad (56)$$

Optional robustness extensions repeat 65 and 129 points at a representative size and move the twist seam; they are not part of the approved production scope below.

H3 uses record index zero from parent shard zero for both the explicit-interface and matched-trivial protocols, giving two complete 33-point circles in total. This is the approved production scope, not merely a timing stage. A 129-point grid, larger size, or additional record requires a separately reviewed campaign revision. A separate local CPU diagnostic uses one 20×24 record and a conservative nine-point circle to measure replay cost and numerical closure; it is not an additional production sample and is not pooled with the A100 calculation.

Initialization, duration, and observation. Sample each record physically at $\phi = 0$ from its own independent product-state start for a total of $2N_y$ cycles, with no separate physical burn-in. Freeze that entire discrete record, including the external ordered controller word and every Born outcome. Insert a one-particle $U(1)$ connection with holonomy $e^{i\phi}$ around the periodic y direction into every controller projector before support truncation, and replay the same word and outcomes at every ϕ . In a seam gauge this places the phase on controller-orbital components crossing one declared periodic seam; a uniformly distributed vector-potential gauge is equivalent and closes at 2π up to a large gauge transformation. Never draw an independent Hamiltonian, schedule, or outcome record at each twist. Appendix B gives the complete construction and explains why independently sampled records do not define a spectral-flow family.

Raw saves and estimator. Save bit-packed records, $p_\xi(\phi)$ in log form, purity gaps, wall entanglement eigenvalues and vectors, overlap matrices, tracking assignments, crossing tables, and 2π closure residuals. The replay covariance is used transiently to form these quantities at each twist point and is then released; it is not written to the production archive. Track

$$\varepsilon_a(\phi) = \log \frac{1 - \nu_a(\phi)}{\nu_a(\phi)} \quad (57)$$

by maximum-overlap subspace transport and count signed zero crossings separately at the two walls. Production evaluates one preregistered frozen record for each protocol. A complete twist circle is an internal deterministic descendant of one record and never increases the trajectory

sample count. No record-frequency or typicality estimator is formed.

Controls, statistics, and gate. Require nonzero branch weight for the full circle, closure up to the large gauge transformation, opposite signs on the two walls for the interface record, zero flow in the matched-trivial record, and stable mode tracking and reference-level choice inside the entanglement gap. Grid and seam variations remain optional robustness extensions rather than part of this representative-record production claim.

Downgrade, figure, sharding, and legacy. A record whose branch weight becomes exactly zero does not define this flux family and may not be replaced by another record. If either preregistered record fails, report that branch failure and any nonquantized spectral behavior without promoting a replacement. Results supply Fig. 3(c). Shard by frozen record, keeping its complete ϕ circle on one shard. The directory `tangent_cocycle_flux_snapshot/` is code reference only; no compatible saved production figure is identified.

XIII. TRAJECTORY-RESOLVED UNIVERSALITY

A. Campaign G5: exact fixed-word many-body singular spectrum

Claim and status. Status DERIVED ANALYSIS IMPLEMENTED; PARENT PRODUCTION PENDING. For each fixed Gaussian trajectory word $\hat{K}_\xi$, the complete normalized many-body singular spectrum is reconstructed exactly from an identity-input Gaussian calculation. This is not a replica spectrum, not the spectrum of a Hamiltonian, and not a new maximally mixed physical ensemble. It is a deterministic coordinate for the operator $\hat{K}_\xi \hat{K}_\xi^\dagger$.

Exact tangent-operator dictionary. Normalize the positive operator and diagonalize its number-conserving Gaussian one-particle correlation matrix:

$$\rho_R^K = \frac{\hat{K} \hat{K}^\dagger}{Z_K}, \quad Z_K = \text{Tr}(\hat{K} \hat{K}^\dagger), \quad \text{spec } C_R^K = \{\nu_i\}. \quad (58)$$

In its natural-orbital basis,

$$\rho_R^K = \prod_i \left[(1 - \nu_i)(1 - f_i^\dagger f_i) + \nu_i f_i^\dagger f_i \right], \quad (59)$$

so every normalized many-body eigenvalue and absolute singular value are

$$\lambda_n = \prod_i \nu_i^{n_i} (1 - \nu_i)^{1-n_i}, \quad \sum_n \lambda_n = 1, \quad (60)$$

$$\sigma_n(\hat{K}) = \sqrt{Z_K \lambda_n}. \quad (61)$$

The identity-input half-Jacobian singular value j_i determines the unordered pair $\{\nu_i, 1 - \nu_i\}$ exactly:

$$j_i = 2\sqrt{\nu_i(1 - \nu_i)}, \quad \mu_i^\pm = \frac{1 \pm \sqrt{1 - j_i^2}}{2}. \quad (62)$$

The identity probe retains the missing orientation by saving whether the cap is occupied or empty. If $n_i^{(0)} = \mathbf{1}_{\nu_i \geq 1/2}$ is the dominant occupation, flipping mode i changes relative charge by $q_i = 1 - 2n_i^{(0)}$. Its positive population and singular-amplitude gaps are, respectively,

$$\epsilon_i = \left| \log \frac{\nu_i}{1 - \nu_i} \right|, \quad \delta_i = \frac{\epsilon_i}{2} = \operatorname{arccosh} \frac{1}{j_i}. \quad (63)$$

Thus a density-matrix gap is twice a coherent singular-amplitude gap. Exact gain/loss caps have $j_i = 0$ and $\delta_i = \infty$ with a mandatory saved orientation.

For $N_{\text{orb}} = 2N_x N_y$ modes, replaying the word from $C_{\text{in}} = \mathbf{1}/2$ gives the branch probability

$$p_K(C_{\text{in}} = \mathbf{1}/2) = 2^{-N_{\text{orb}}} Z_K, \quad \log Z_K = N_{\text{orb}} \log 2 + \log p_K. \quad (64)$$

To prove the identification, introduce the normalized vectorization $|\Omega\rangle = 2^{-N_{\text{orb}}/2} \sum_{\mathbf{n}} |\mathbf{n}\rangle_R |\mathbf{n}\rangle_A$ only on paper. Then

$$\begin{aligned} |K\rangle &= (\hat{K} \otimes \hat{\mathbf{1}}) |\Omega\rangle, \\ \langle K|K\rangle &= 2^{-N_{\text{orb}}} Z_K, \\ \frac{\operatorname{Tr}_A |K\rangle \langle K|}{\langle K|K\rangle} &= \frac{\hat{K} \hat{K}^\dagger}{Z_K}, \\ &= \frac{\hat{K} (\mathbf{1}/2^{N_{\text{orb}}}) \hat{K}^\dagger}{p_K}. \end{aligned} \quad (65)$$

Equation (65) proves that normalized propagation of the identity input produces Eq. (58). Modewise Gaussian Schmidt decomposition then gives Eq. (59); the corresponding half-Jacobian has $j_i = 2\sqrt{\nu_i(1 - \nu_i)}$, completing the dictionary. No doubled Choi covariance is propagated: the auxiliary copy above is solely a proof device. This complete identity-input theorem is stronger than T1's pure-state tangent interpretation: the latter resolves projective one-particle-one-hole stability gaps but, by itself, neither fixes the binary cap orientations nor determines the complete factorized Fock spectrum or Z_K .

Dependencies and deterministic run matrix. Depends only on the completed primary T1 parent words. Consume the existing 02_pure_wall_master explicit-interface and matched-trivial archives at $N_x = 20$, $N_y = 20, 30, 40, 50, 60$, ten independent trajectories per size and arm. Replay each fixed word once from $C = \mathbf{1}/2$ through $2N_y$ and observe prefixes $N_y, 3N_y/2, 2N_y$. The support-terminated construction is a non-gating robustness extension. No parent seed, word, outcome, stochastic case, T1 product, or P2 product is changed. The derived loader searches the current output first and the completed production_10sample_v3_fixed_nx20_ny20_30_40_50_60 output second, requiring checksum-valid receipts and identical physical parent identities before accepting or deduplicating a case/shard pair.

At each prefix save ν_i , cap orientations, j_i , ϵ_i , δ_i , $\log Z_K$, and the leading normalized and absolute levels. Represent the complete spectrum by the factors in

Eq. (60); never materialize a $2^{N_{\text{orb}}}$ Fock-space matrix. A heap-ordered subset-sum reduction saves the lowest 128 singular-amplitude gaps in each relative-charge sector $q = -3, \dots, 3$, using ∞ when exact caps exclude a finite level.

Universal fits, controls, and statistics. Divide logarithmic amplitude gaps by the prefix cycle. At the final prefix fit the low charge-resolved levels to

$$\Delta_{q,n}(L) = \frac{2\pi v}{L} \left(\frac{q^2}{2k} + n \right) + O(L^{-2}). \quad (66)$$

The first neutral-current spacing fixes the common spectral scale v ; the $q = \pm 1$ primaries then test $k = 1$, and higher levels test integer descendant spacings and their partition degeneracies. For the absolute leading level define the matched-trivial, per-cycle, per-wall density

$$\begin{aligned} f_K(L) &= -\frac{1}{2L} \left[\frac{\mathbb{E}_\xi \log \sigma_{0,\xi}^{\text{wall}}}{T} - \frac{\mathbb{E}_\xi \log \sigma_{0,\xi}^{\text{triv}}}{T} \right], \\ f_K(L) &= f_\infty - \frac{\pi c_{\text{eff}} v}{12L^2} + O(L^{-4}). \end{aligned} \quad (67)$$

The factor two divides the two walls. Means are quenched, trajectory-first arithmetic means. Fixed-seed whole-trajectory bootstraps independently resample wall and matched control and report intervals for v , k , and c_{eff} ; prefix drift and deletion of the smallest size are mandatory sensitivity checks.

Small-Fock-space validation explicitly constructs $\hat{K}$ for invertible filters, rank-deficient projectors, and charge-changing gain/loss words. Direct singular values, the eigenvalues of Eq. (58), and the products in Eq. (60) must agree, including occupied and empty caps. Further tests enforce Eq. (64), the normalized sum rule, relative-charge assignment, the factor of two in Eq. (63), prefix determinism, and synthetic $U(1)$ towers with known v, c, k .

Current legacy result. Assessment: the exact bridge and reusable parent-word contract now exist, but no production G5 spectrum has yet been reported. The tangent machinery under 00_WORKSPACE/CURRENT/tangent_edge_channel/ supplies QR gauges and pure-state one-particle-one-hole gaps. The material under 00_WORKSPACE/COLAB/colab_regularized_choi_transfer_matrix/ is useful derivational history, not a production input: G5 propagates no Choi state. The active pure-wall master stores bit-packed outcomes and complete ordered site words, so its ten trajectories per size can be consumed without rerunning the Born ensemble. The derived implementation is 02_pure_wall_master/src/g5_manybody_spectrum_analysis.py; its NPZ products and adjacent receipts hash the parent archive, canonical GPU engine, and analysis source.

The older chain-rule record-entropy evidence remains scientifically admissible only as a supplemental large-deviation diagnostic. At $N_x = 20$, $N_y = 30, 40, 50$, $S = 100$, its late mean self-information per cycle divided by N_y is (2.9887, 2.9945, 2.9953), and the 20×40 wall/nonwall binary entropy is (0.1013, 0.0101) nats per

event. These facts diagnose a stationary extensive record background and wall-enhanced uncertainty; they do not determine an operator singular spectrum, k , or c_{eff} .

Downgrade, figure, and sharding. Results supply Fig. 4. Shard by whole parent archive and retain the parent sample ID. Failure of the exact small-system dictionary invalidates G5. If the dictionary passes but the charged tower is unstable, report the finite-word factorized singular spectrum and radial gaps without k . If the neutral scale or matched-control L^{-2} term is unresolved, do not quote c_{eff} . A gap from one frozen word is a representative branch result, not a universal trajectory-ensemble spectrum. Table XV records this boundary explicitly.

XIV. MEAN, POSTSELECTION, AND NOISE CONTROLS

A. Campaign M1/L1: exact mean channel and continuous-time generator

Claim, status, and narrative role. Status PRODUCTION-READY RECONSTRUCTION COMPLETE; PRODUCTION SWEEP NOT LAUNCHED. This is a deterministic CPU campaign, not a trajectory family and not a Colab job. It asks what survives after the measurement record has been discarded by comparing the exact autonomous finite-success averaged channel with its continuous-time weak-success generator. Both deterministic arms merit a compact main-text comparison: the finite channel is the channel-level object at nonzero success strength, while the generator exposes the controlled $p \rightarrow 0$ limit and its relaxation spectrum. Derivations, shell robustness, dephasing, and convention checks belong in an appendix. Neither arm is called a conditioned trajectory, and a mixed-state wall occupation flow is not described as preparation of a pure chiral critical state.

Four objects that remain distinct. Let $C_{ab} = \langle c_a^\dagger c_b \rangle$ and construct the canonical controller frames $W_{j,-}$ and $W_{j,+}$ for the target occupied and empty sectors. Define the positive one-body frame operators

$$\begin{aligned} V_- &= \sum_j |W_{j,-}\rangle \langle W_{j,-}|, \\ V_+ &= \sum_j |W_{j,+}\rangle \langle W_{j,+}|, \\ D &= n_a V_- + (1 - n_a) V_+. \end{aligned} \quad (68)$$

The canonical continuous weak-success model is

$$\dot{C} = n_a V_- - \frac{1}{2} \{D, C\}. \quad (69)$$

Its stationary covariance solves the Lyapunov equation $DC_{\text{ss}} + C_{\text{ss}}D = 2n_a V_-$. This is the primary continuous deterministic arm.

For finite success strength p , production does not Euler-step Eq. (69). With

$$G_p = e^{-pn_a V_-/4}, \quad L_p = e^{-p(1-n_a)V_+/2}, \quad (70)$$

define exact Gaussian gain and loss subchannels

$$\Phi_{g,p/2}(C) = \mathbf{1} + G_p(C - \mathbf{1})G_p, \quad \Phi_{\ell,p}(C) = L_p C L_p, \quad (71)$$

and the completely positive symmetric composition

$$\Phi_p = \Phi_{g,p/2} \circ \Phi_{\ell,p} \circ \Phi_{g,p/2}. \quad (72)$$

At fixed physical time T , $\Phi_p^{T/p}$ converges to Eq. (69) with the prospective second-order Strang error; no clock conversion is fitted. The implementation evaluates the affine power by binary composition, preserving the exact channel form while reducing the cost from $O(T/p)$ to $O(\log(T/p))$ batched matrix products.

Two historical models remain separately labelled. The bath-density gain/loss generator and its optional quartic-dephasing two-point closure in `scripts/legacy/CI.Lindblad.DW.py` use older wall and covariance conventions and are robustness/provenance controls. The realized conditioned adaptive trajectories are the fourth object and retain their own $S = 25$ Born ensemble. None of these objects is renamed to make their spectra agree.

Immutable hybrid CPU production matrix. The main grid uses complex128 NumPy/SciPy CPU execution at $N_x = 20$. The complete static scan is performed only at the publication circumference,

$$\begin{aligned} N_y &= 64, & \alpha_{\text{out}} &= 30, \\ \alpha_{\text{in}} &= 1, 1.5, 1.75, 1.875, \\ &2.125, 2.25, 2.5, 3, \\ n_{\text{shell}} &= 1, 2, \text{None}, & n_a &= \frac{1}{2}, \\ \text{dw.truncation} &= \text{True}. \end{aligned} \quad (73)$$

The response endpoint $(\alpha_{\text{in}}, \alpha_{\text{out}}) = (1, 30)$ additionally uses $N_y = 32, 48, 64$, all three shell choices, and both values of `dw.truncation`. Domain-wall truncation is canonical; the untruncated-domain arm is an endpoint robustness check. The union contains 24 static cases and 15 additional endpoint size/truncation cases, hence 39 unique primary wall cases. An accepted-width decision may replace $N_x = 20$ only through a validated JSON override.

The canonical wall and $C(0) = \frac{1}{2}\mathbf{1}$ are the baseline. There is no random-pure sample arm: the exact one-body mean of the half-filled random-pure ensemble is already $\frac{1}{2}\mathbf{1}$. Representative empty and filled starts test convergence to the same stationary solution. Uniform $\alpha = 1$ and 30 systems receive the same response probe at $N_y = 64$ as null controls. Filling, initialization, wall-width, uniform-phase, and labelled dephasing controls remain at representative sizes rather than being promoted to a second circumference matrix.

There is no burn-in. The declared window is $T = 2N_y$, with observations at $t/N_y = 0, 0.25, 0.5, 0.75, 1, 1.5, 2$. The finite-channel grid is $p = 1, 1/2, 1/4, 1/8, 1/16$, always compared at fixed physical time. Controls compare the legacy and canonical wall widths; $n_a = 0.45, 0.50, 0.55$;

TABLE XV. Legacy inputs and present implementation status for Campaign G5. Record self-information remains supplemental and is not renamed as an operator spectrum.

source	available object	G5 use and limitation
Pure-wall master archives	Ten ordered words for each primary wall/control size once Bundle 02 completes	Exact deterministic parents for the identity-input replay; no new Born sampling.
Tangent and regularized-Choi studies	Pure-state tangent gaps, QR infrastructure, and historical endpoint proxies	Supply algebra and validation ideas; the Choi covariance is not propagated in G5.
Sequential probability maps	$\hat{H}_{\text{chain}}/N_y = (2.9887, 2.9945, 2.9953)$ and wall-enhanced uncertainty	Supplemental record large-deviation evidence only; not the many-body singular spectrum.
Derived G5 stage	Factorized spectra, $q = -3, \dots, 3$ low levels, $\log Z_K$, receipts, and bootstrap fits	Implementation complete; numerical claims remain pending until all ten wall/control groups pass checksum and finite-size validation.

uniform topological and trivial systems; empty, filled, and max-mix starts; and representative full-dephasing calculations at $n_{\text{shell}} = 1, 2, \text{None}$. The dephasing closure is explicitly labelled and never enabled silently in the main gain/loss matrix.

Wall-resolved directional density response. For each wall, apply the local density-phase pulse to both orbitals at one transverse source position,

$$C_{\pm}(0) = U_{\pm\epsilon} C_{\text{ss}} U_{\pm\epsilon}^{\dagger},$$

$$\chi(t) = \frac{C_{+}(t) - C_{-}(t)}{2\epsilon}, \quad \epsilon = 10^{-3}. \quad (74)$$

Translation invariance removes source-position sampling. The continuous generator is evaluated through $t = N_y/2$ at spacing 0.25; the exact finite channel is evaluated at the same final physical time for $p = 1, 1/2, 1/4, 1/8, 1/16$. Low-rank momentum-block propagation evolves only the kicked vectors and never materializes a full covariance response history.

All shell choices save real-space wall-projected $\chi_w(y, t)$, wall retention and norm, the unwrapped absolute-response peak as a diagnostic, positive- and negative-lobe centers, the positive-lobe velocity fitted over $2 \leq t \leq 3N_y/8$, signed directionality, pulse-linearity checks at $\epsilon/2, \epsilon, 2\epsilon$, and the finite-channel response error. The signed lobe estimator is essential: the absolute response is left-right symmetric for this gain/loss probe, while its odd signed lobes carry the wall orientation. The result is called *wall-resolved directional density response*, not quantized transport. No modular-Hamiltonian evolution is part of M1/L1.

Momentum-resolution rule. Only $n_{\text{shell}} = \text{None}$ may calculate, archive, plot, or support an interpretation of k_y -resolved occupation spectra, wall-branch occupations, k_y -resolved wall weights, or branch slopes. This arm saves the paired wall flow about $\nu = 1/2$, half-occupation and bulk purity gaps, extensive entropy, charge variance, relaxation rates, convergence arrays, and numerical residuals.

For $n_{\text{shell}} = 1, 2$, translation symmetry may accelerate internal linear algebra but may not leak into a momentum-resolved scientific product. These arms save only real-space or momentum-integrated occupation histograms, half-occupation and bulk purity gaps, wall-

localization profiles, entropy, charge variance, relaxation rates, channel-continuous convergence, and residuals. Their schemas reject any selected-field name containing **ky**, **momentum**, or **branch**; the analysis and figure builder independently enforce the same prohibition.

Storage, execution, and validation. No trajectory, schedule, covariance matrix, final covariance, or covariance history is a permanent product. Covariance blocks exist transiently in memory only while compact observables are formed. Each case writes an atomic selected-observable NPZ and JSON receipt with configuration and solver hashes, timings, CPU allocation, BLAS-thread count, Hermiticity and occupation-bound checks, stationary residuals, and checksums. Independent cases are process-parallel, while BLAS defaults to one thread per worker to prevent oversubscription. The runner is resumable at case granularity.

Small-system tests compare canonical frame operators and generators directly with `scripts/legacy/CI.Lindblad_DW.py`; require Hermiticity, complete positivity, $0 \leq C \leq 1$, stationary residual convergence, and initialization independence; and verify decreasing fixed-time error with fitted order near two. Tests also compare low-rank response propagation with brute-force real-space evolution, check pulse linearity and translated-source covariance, require the uniform null, and compare both domain-truncation frame constructions directly with the canonical CPU OW engine. The hybrid primary matrix contains 39 cases. Wall/filling/uniform/initialization controls add 21 cases and the full-dephasing matrix adds six, for 66 deterministic cases in the immutable configuration.

Legacy summary and present deterministic reconstruction. The legacy source already contains useful frame construction, partial- y Fourier operators, affine vectorization, full-dephasing closure, and pulse-response routines. Its rendered wall spectra and current-like summaries are secondary evidence because their raw data spine and conventions are incomplete. The sampled comparison in `figs/corr_y_profiles/` independently establishes $|\bar{C}|^2 \neq |\bar{C}_{\xi}|^2$, with saved ratios 0.185–0.772 across the transverse positions, but it is not an exact deterministic-channel output. Historical conditioned Choi/tangent

products are likewise not mean-channel spectra.

Atlas Result 24 is the hybrid deterministic reconstruction. At $N_y = 64$ the canonical untruncated arm has two wall-selected occupation branches with slopes ± 0.949605 and intercept 0.5 to numerical precision. At the response endpoint, the canonical $N_y = 64$ positive-lobe velocities are $(+0.077455, -0.077455)$ with $R^2 > 0.9975$; the signs are stable at $N_y = 32, 48, 64$. Turning off domain-wall truncation preserves the signs and yields $(+0.038030, -0.038030)$ at $N_y = 64$. Both uniform controls give the exact response null. The finite-channel response converges to the continuous result with order 2.0008, and the largest stationary residual is 3.91×10^{-15} . The earlier ± 0.646804 branch slope is retained only as the no-domain-truncation endpoint value, not the canonical static result. These numbers support a channel-level interface result but do not turn the mixed steady state into a pure chiral CFT or define a conserved conductance.

Figure and downgrade rule. The recommended main-text element gives equal emphasis to the $N_y = 64$ untruncated wall occupation bands and the two-wall signed density-response heatmaps. Its finite-size panel contains only velocity and directionality versus $1/N_y$, distinguishing domain truncation on and off; its last panel shows finite-channel response convergence to the continuous generator. Entropy density, branch slopes, localization weights, and residuals are tabulated diagnostics. A separate appendix static scan at $N_y = 64$ contains purity gaps, wall localization, entropy density, and relaxation versus α_{in} , plus shell, filling, initialization, wall-width, uniform, and dephasing controls.

Opposite wall signs together with a uniform-system null are the chirality criterion. If the response becomes symmetric in the signed estimator or exists only with domain truncation, downgrade it to appendix evidence. If the mixed-state interface sector survives but conditioned-trajectory critical observables differ, state that contrast directly; do not assign an effective temperature, quantized transport coefficient, or a pure-state central charge to the mean channel.

B. Campaign M2: postselection and imperfect feedback

Claim and status. Status PLANNED. The new wall-CFT, chirality, tangent, and record signatures occur in the physical Born ensemble rather than only in a forced favorable branch; separately, a declared interval of feedback imperfection tests their stability.

Dependencies and run matrix. At $N_y = 32, 48, 64$, compare perfect correction with the declared correction-error grid

$$\begin{aligned} \epsilon_{\text{corr}} &= 0.01, 0.025, 0.05, 0.10, \\ &\quad 0.20, 0.40, 0.50, \\ p_{\text{gain}} &= p_{\text{loss}} = 1 - \epsilon_{\text{corr}}, \end{aligned} \quad (75)$$

and with a forced postselected endpoint. Here the imperfect points remain physical stochastic Born trajectories; the error parameter modifies the feedback success probability rather than forcing measurement outcomes. Every stochastic point uses $S = 25$ independent trajectories. The deterministic postselected endpoint is a distinct conditioned control and is not padded with duplicate samples.

Initialization, saves, estimator, and controls. Use the same observable definitions as P1, S1, and T1. Store branch probabilities for postselection and never describe a zero-event record as a stochastic ensemble.

Current legacy result. Assessment: several canonical GPU and CPU controls are substantial, but they span three inequivalent protocol axes and do not execute the M2 robustness grid. The most direct interpolation is the canonical GPU campaign under `00_WORKSPACE/COLAB/colab_partial_post-select/gpu_data/`. It uses $N_x = 20$, $N_y = 40$, $C = 40$, $S = 10$ for $p < 1$, $\alpha_{\text{in}} = 1$, $\alpha_{\text{out}} = 30$, and

$$p = 0, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95, 0.99, 1. \quad (76)$$

For $p < 1$ the engine still runs the physical perfect-correction protocol, but an independent site coin forces the favorable postselected outcome with probability p . The saved forced-site fractions agree with their declared values to within 4.9×10^{-4} . The $p = 1$ endpoint instead invokes the deterministic postselection branch and therefore contains one actual trajectory.

At the final cycle, the trajectory-mean real-space Chern markers for the eight stochastic points in increasing p order are

$$\begin{aligned} &0.9808(114), 0.9582(161), 0.9596(218), 0.9252(240), \\ &0.9276(245), 0.9475(182), 0.9194(260), 0.9283(204), \end{aligned} \quad (77)$$

where parentheses are trajectory standard errors. The deterministic $p = 1$ endpoint gives 0.7691 without an ensemble error bar. A descriptive final-cycle fit of each saved entropy curve to

$$S(A_y) = a \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + b, \quad 1 \leq A_y \leq 20, \quad (78)$$

gives $a = 0.4398(61)$ at $p = 0$, $0.4126(38)$ at $p = 0.5$, and $0.3860(95)$ at $p = 0.99$; the single $p = 1$ curve gives $a = 0.3769$. This mild, nonmonotonic drift is a finite-geometry diagnostic, not evidence for a postselection transition or a precision CFT coefficient. The campaign also saves squared correlators, charge fields, branch probability maps, and measurement-frustration fields, but no covariance histories from which the P1 chirality and T1 tangent observables can be reconstructed.

Larger canonical GPU postselection controls under `00_WORKSPACE/COLAB/colab_charge_fluctuations/gpu_data/` cover $N_y = 30, 40, 50$ for $C = 2N_y$. The default-initialized deterministic branches give final real-space Chern markers 0.8731, 0.7877, and 0.8821, respectively. Although their metadata request $S = 100$, each

correctly records one actual trajectory. The corresponding max-mix postselected runs approach a Chern marker of one and total charge variance near 0.5; they are purification controls with a different initialization and are not combined with the default-initialized values.

The one genuine imperfect-feedback legacy point is the matched CPU activity pilot under `00_WORKSPACE/CURRENT/topological_frustration_diagnostics/results/campaigns/`. At 12×16 , $S = 16$, $C = 64$, and burn-in 32, the mean defect rate is 0.04055 in the wall geometry versus 0.00742 in the uniform control. Within the wall run it is 0.08017 in the interface region and 0.00885 in the interior. This establishes a strongly wall-localized cost of imperfect correction, but supplies neither an imperfection-strength interval nor the state, chirality, and tangent observables required by M2.

Finally, the no-feedback/perfect-correction GPU sweep already summarized in Table X supplies an endpoint comparison for a conditioned particle–Choi transfer gap. For example, at $\alpha_{\text{in}} = 1.9$ its no-feedback mean gap decreases from 0.01215(229) to 0.00249(90) between $N_y = 20$ and 40, while the perfect-correction values remain near 0.10. This is evidence that feedback changes the conditioned transfer sector, not a measurement of stability across a continuously tunable feedback error.

Acceptance and downgrade. The physical Born-ensemble wall claim requires the perfect-correction point only. Robustness is claimed solely over the tested imperfect interval. Put this comparison in the supplement unless it is indispensable to a mechanism. Table XVI records the available controls without conflating their protocol axes; none supplies the declared imperfect-feedback size grid, so M2 remains planned.

C. Campaign M3: noise and the BPJ bulk transition

Claim and status. Status PLANNED. This campaign tests, but does not assume, bulk–boundary locking under the noise perturbation used for the BPJ domain-wall-free bulk transition.

Dependencies and run matrix. First run the domain-wall-free geometry at $N = 16, 20, 24, 28, 32$, $S = 25$, on the declared grid

$$\sigma = 0, 0.025, 0.050, \dots, 0.700. \quad (79)$$

Locate and bootstrap the bulk transition using the topological marker, purity gap, and bulk tangent diagnostic. The wall campaign does *not* repeat this complete dense grid. After the bulk gate, select a preregistered bracket consisting of the nearest points below, at, and above the inferred bulk crossing (expanded by one neighboring point on each side when the uncertainty interval requires it), and run only that bracket over the primary N_y sequence. This staged rule prevents spending the wall GPU budget before the bulk transition has been located.

Initialization, saves, estimator, and controls. Use independent initial states and a total of $2N_y$ cycles at every noise value. Save all P1, S1, and T1 observables and the exact noise convention. Estimate bulk and boundary pseudocritical points independently in each bootstrap resample.

Current legacy result. **Assessment: the external BPJ baseline fixes the perturbation and expected bulk scales, but the present repository contains no admissible run of that noise protocol.** BPJ inserts, between complete adaptive cycles, the onsite charge-preserving unitary

$$U_\sigma(\theta) = \prod_{\mathbf{r}, \mu} \exp \left[4\pi i \theta_{\mathbf{r}, \mu} \left(c_{\mathbf{r}, \mu}^\dagger c_{\mathbf{r}, \mu} - \frac{1}{2} \right) \right], \quad (80)$$

$$\theta_{\mathbf{r}, \mu} \stackrel{\text{iid}}{\sim} \text{Uniform}[0, \sigma),$$

with a fresh independent spatial field at each application. In the domain-wall-free $\alpha = 1$ system, BPJ reports loss of the trajectory-resolved topological phase near $\sigma_{c,1} \simeq 0.2$ and a distinct closing of the trajectory-averaged occupation spectral gap, accompanied by a change of its regularized Chern marker, near $\sigma_{c,2} \simeq 0.55$ [1]. Those values calibrate the M3 scan; they are not imported as boundary critical points.

The closest local notebook is under `notebooks/test_coherent_term_markov_circuit.ipynb`. Its historical coherent branch used 10×10 , one trajectory, 20 cycles, a domain wall, and the deterministic full domain-wall Hamiltonian step $G \mapsto e^{-iH_{\text{DW}}} G e^{iH_{\text{DW}}}$ after every cycle. This is neither the independent onsite phase field in Eq. (80) nor a noise-strength scan. The current notebook explicitly states that the canonical CPU driver no longer accepts `coherent_term`; it retains only a baseline smoke run and stale rendered cells that refer to a no-longer-defined coherent output. It therefore supplies implementation history, not M3 data.

Other superficially nearby assets vary different physical axes. The dephasing terms in `CI_Lindblad_DW.py` act in a deterministic trajectory-averaged generator rather than through recordwise coherent phases. The click sequences under `00_WORKSPACE/CURRENT/topological_frustration_diagnostics/results/local_gain_loss_clicks/` resolve the intrinsic gain/loss feedback record at zero external noise. The mass sweeps and no-feedback controls change the target Hamiltonian or correction law, respectively. None can be placed on the BPJ σ axis by a clock rescaling or parameter relabeling.

Acceptance gate and downgrade. Bulk–boundary locking is accepted only if extrapolated critical noise values have compatible joint confidence intervals. A preemptive boundary crossover or transition is explicitly allowed and must not be forced to coincide with the bulk point. This campaign belongs in the supplement unless it reveals a central new phenomenon. Table XVII records the completed path audit: no saved repository artifact matches both the BPJ observable and Eq. (80), so M3 remains planned.

TABLE XVI. Admissible legacy evidence relevant to Campaign M2. Forced favorable outcomes, imperfect correction, and absent feedback are distinct controls and are not placed on one unlabelled interpolation axis.

legacy source	available result	M2 interpretation
GPU partial-postselection sweep	20×40 , $S = 10$, eight stochastic $p < 1$ points; final Chern marker 0.919–0.981	Strong forced-branch interpolation with entropy, correlator, charge, probability, and frustration fields; one size and no P1/T1 data.
GPU deterministic postselection	Default-initialized $N_y = 30, 40, 50$ endpoints; one actual trajectory; final marker 0.8731, 0.7877, 0.8821	Valid favorable-branch controls without stochastic error bars; the max-mix family is kept separate.
CPU imperfect-feedback activity	12×16 , $S = 16$; wall/uniform defect rates 0.04055/0.00742 and interface/interior rates 0.08017/0.00885	Genuine imperfect correction and matched spatial control, but only one small-geometry activity point.
GPU no-feedback transfer sweep	$N_y = 20, 30, 40$, $S = 10$; no-feedback and perfect-correction particle–Choi gaps	Endpoint sensitivity of a conditioned transfer construction, not a declared imperfection interval or the physical covariance tangent.
Missing M2 production products	No matched $N_y = 32, 48, 64$, $S = 25$ grid with P1, S1, and T1 observables	No robustness threshold, common finite-size interval, or equivalence between Born and postselected ensembles can be claimed; M2 remains planned.

TABLE XVII. Legacy evidence relevant to Campaign M3. Only Eq. (80) defines the BPJ coherent-noise axis used by this campaign.

legacy source	available result	M3 interpretation
Published BPJ bulk study	Uniform $\alpha = 1$ circuit; trajectory and averaged transitions near $\sigma \simeq 0.2$ and 0.55	Defines the perturbation, observables, and scan scale; external baseline rather than a saved repository campaign.
Historical coherent-term notebook	10×10 , $S = 1$, $C = 20$ domain-wall smoke comparison using deterministic $e^{-iH_{DW}}$	Protocol mismatch with no σ grid, uniform control, finite-size analysis, or current canonical entry point.
Lindblad and click-record controls	Deterministic dephasing and zero-external-noise gain/loss activity	Different generator and record axes; neither realizes onsite random phases between cycles.
Mass and feedback sweeps	Existing topological-to-trivial and perfect/no-feedback comparisons	Useful clean-system controls only; changing α or feedback is not coherent noise.
Missing M3 production products	No uniform $N = 16, 20, 24, 28, 32$ σ grid, purity gap, bulk tangent, or matched noisy wall families	No local estimate of either bulk threshold or bulk–boundary locking; M3 remains planned.

XV. FIGURE ARCHITECTURE

A. Main figures

Figure 1: inherited BPJ platform and validated wall implementation. (a) Vector schematic of the four overcomplete-Wannier occupation measurements and the outcome-conditioned ancillary fSWAP; (b) explicit controller-interface and support-terminated walls, two wall orientations, periodic direction, and flux seam; (c) representative prepared Chern profile; (d) topological and trivial finite-size indices; (e) transverse-converged comparison of wall implementations. The figure validates the inherited topological platform and fixes the two microscopic wall implementations on which the new universal claims are built.

Figure 2: universal state data. (a) entropy versus log chord with a $q = 1, 2, 3$ coefficient inset; (b) quantum interval variance and k , with $\delta_{SF}(N_y)$ and its zero target in the inset; (c) the unnormalized trajectory-averaged entanglement contour; (d) max-mix wall-on/wall-off pu-

rification; (e) transverse profile of the residual entropy or purity defect. The squared correlator, small- k structure factor details, and topological sweep move to the supplement. Exact-Hamiltonian calibration appears only as a visually distinct gray reference.

Figure 3: chirality in three clocks. (a) opposite modular drifts, with the modular clock marked in the panel; (b) total Born $\chi(k, \omega)$ on both walls, including a score/conditional decomposition inset; (c) frozen-record spectral flow with an integer histogram inset. These are exactly three primary panels. A compact three-clock key may sit above them, but is not a fourth result panel. The caption states what each route proves and explicitly denies a quantized transport claim.

Figure 4: fixed-word many-body singular spectrum. (a) the exact map from the identity-input occupations ν_i and tangent singular values j_i to the factorized normalized spectrum; (b) representative charge-resolved low singular-amplitude levels at the three declared prefixes; (c) the five-size conformal tower collapse using the neutral spacing to determine v ; (d) the $q = \pm 1$ estimate of k and descendant-degeneracy residuals; (e) the matched-trivial per-wall L^{-2}

leading amplitude fit for c_{eff} . Every panel identifies fixed-word singular amplitudes, not Hamiltonian eigenvalues or a Choi-state spectrum.

Figure 5: tangent edge channel and purification. (a) tangent-cocycle construction with independent starts and tangent-frame alignment; (b) wall/trivial gap scaling; (c) globally normalized low-mode localization; (d) max-mix purification collapse; (e) same-record purification rate versus physical tangent rate; optional inset comparing trajectory and mean covariance.

B. Supplement and schematic requirements

Supplement figures contain the exact benchmark contact sheet, transverse gate, fixed-window transient diagnostics, fit-window sensitivity, update-order and n_{shell} controls, conditional and score response pieces, twist closure, tangent rank/null diagnostics, postselection, mean-channel comparisons, and noise transition. Every caption states S , the independent sampling unit, initialization, total trajectory window, estimator order, uncertainty type, and fit window.

All conceptual schematics should be vector/TikZ or native illustration assets rather than screenshots. The key sketches are: the actual measurement-and-feedback tile; the two-wall cylinder with entanglement interval, response source, and twist seam; a taxonomy from one record to C_ξ , p_ξ , and $D\Phi_\xi$ versus $\bar{C}$; and the three-clock chirality diagram. A generic brick-wall circuit is not an adequate substitute for the implemented overlapping-orbital schedule.

XVI. PRODUCTION ORDER, DETERMINISTIC CPU WORK, AND A100 RESOURCE STAGING

Execute in the following stop/go order:

1. V0 small-system validation.
2. B0 exact benchmark.
3. P1 bulk topology.
4. W1 transverse gate and selection of the actual production N_x .
5. M1/L1 exact finite-channel and continuous-generator CPU campaign at the accepted width; its deterministic cases may run independently of the Born sample queue.
6. S1 state CFT, including the matched entropy-variance spectral test.
7. T1 and P2, with tangent-frame rank pilots before full sizes.

8. The explicit pure/max-mix and support-terminated pure S2 topological sweep, after the P1, S1, and T1 launch gates pass; run P2 first when practical to exercise the common physical-tangent analysis.
9. B2 mechanism analysis, reusing the late-time activity records and the completed T1 and P2 outputs.
10. H1, H2, and H3 as three separate datasets.
11. G5 deterministic identity-input replay and factorized-spectrum analysis after the primary pure-wall words are complete.
12. M2–M3 stochastic controls.
13. Optional B1 controller-frame surrogate-attraction diagnostic, executed last and incapable of blocking any preceding campaign.

Each stochastic stage begins with one parameter point and four samples, then a complete $S = 25$ pilot at the smallest accepted size, then the full family. M1/L1 instead begins with the small-system CPU parity and physicality tests, then one complete three-shell point, then the immutable deterministic matrix. GPU batch size is autotuned against the 40 GB A100 budget with at least 20% headroom, but a trajectory is never divided in a way that changes its ordered updates or random stream. Colab notebooks import the canonical GPU source as a separate file and end by disconnecting the runtime.

A. Colab bundle and Drive-staging contract

Production artifacts are organized by shared trajectory family rather than by manuscript claim. The intended Colab packages are:

1. **validation**: V0 engine, branch normalization, frozen replay, observer, device-parity, tangent-QR, and compact-archive checks;
2. **bulk_width_gate**: P1 and W1, executed before accepting a production width;
3. **pure_wall_master**: the common S1, T1, and B2 observer run, followed shard-locally by H1 modular observables and H2 response continuations while the required parent states remain live; it emits the reconstructible parent IDs, pre-run RNG states, and ordered records consumed by H3 and G5; after the five primary wall/control sizes complete, its deterministic G5 stage reconstructs factorized many-body singular spectra with independent receipts. Its existing R1 SCGF merger remains an optional supplemental record large-deviation diagnostic;
4. **chirality_replay**: H3 frozen-record twist circles attached to parent sample IDs, together with exact replay recovery when a completed compact parent archive is resumed in a new runtime;

5. **maxmix_master**: P2, the max-mix S1 observables, and same-record physical tangent products; and
6. **scans_and_controls**: the explicit pure/max-mix S2 grids, their pure support-terminated mirror and CPU entanglement-transition analysis, and M2–M3, launched only after the corresponding core gates remain viable; max-mix S2 carries purification and physical tangent products with dense Choi tracking disabled; and
7. **b1_controller_frame**: the optional final 20×48 interface/control surrogate-attraction test, with no downstream gate.

The deterministic M1/L1 campaign is not a Colab package; it lives under `00_WORKSPACE/CURRENT/mean_channel_lindblad_cpu_campaign/`. Two or three stochastic claims are bundled only when they consume the same trajectory family or a deterministic descendant of it. Bundling never licenses pooling different initial ensembles, controllers, schedule laws, or noise values. The **maxmix_master** and **scans_and_controls** runners also require a checksummed schema-versioned launch-decision file. P2 cannot launch until T1 is marked viable; each S2 or M2–M3 case checks only the P1, S1, or T1 decisions on which that case actually depends. A template remains **pending** by default, so copying a later bundle to Drive cannot silently bypass the campaign order.

The three production lanes contain 148, 50, and 677 shards respectively, for 875 before strict reuse; **scans_and_controls** contributes 653. The speed-optimized execution unit is one live five-trajectory shard. S1, T1, and B2 are streamed during its single physical evolution; H1 consumes the declared late covariances before they are released, and H2 forks its finite-difference continuations from the four live origin states instead of replaying the common prefix for every source. Each stage writes an atomic compact receipt, so a later-stage failure does not invalidate completed parent products. H3 is intentionally not fused into this all-or-nothing invocation: its 65–129 full twist replays dominate runtime and use independent restart checkpoints based on the compact RNG state and frozen record.

Google Drive is a staging area rather than the permanent raw-data archive. At most one geometry-size-control arm is live at a time. The $S = 25$ sample list is assigned before execution to deterministic five-trajectory shards. A shard runs first in Colab local storage, computes all streaming observables, writes an atomic manifest and checksums, and only then copies one compact archive to Drive. The permanent shard contains ensemble-resolved observables, replay-capable ordered controller/outcome records, the pre-run RNG state, provenance, and replay hashes. Per-event float64 probabilities are used live where H2 requires them but are reduced to per-cycle sums before Drive archival. A dense parent covariance may exist only as local in-memory or scratch state while observables or immediate descendants are evaluated. It is never an

ensemble-level archival product: descendants that cannot run in the same process regenerate the parent exactly from the pre-run RNG state and frozen record and must pass replay-residual and checksum tests. Dense covariance histories are never saved merely for convenience.

The 15 GB Drive allocation may be filled deliberately: after one experiment arm is complete, its checksummed archives are downloaded to the cluster and the staging area is cleared before the next arm. Compact replay records are therefore retained whenever they make an experiment independently reproducible; Drive scarcity is not used to weaken a record. The routine ceiling for any one staged arm or live auxiliary bank is 12 GiB, with 12–14 GiB allowed only as an explicitly reported edge case. A preflight above 14 GiB must split the arm or immediate descendant stages. These storage limits are separate from the 40 GB A100 device-memory contract and do not license sample covariance archives.

The canonical engine has $N_{\text{layer}} = 2N_x N_y$. At $N_x = 20, N_y = 80$, one dense $N_{\text{layer}} \times N_{\text{layer}} = 3200 \times 3200$ complex128 one-particle covariance therefore occupies 163.84 MB (decimal units), not 41 MB. Three such checkpoints for 25 trajectories require 12.288 GB for one arm before records, tangents, or controls; even exact Hermitian-triangle storage requires approximately 6.146 GB. Consequently wall and matched-control arms are processed separately, and a five-trajectory shard is reduced to compact observables before the next shard is admitted. Parent covariances are not copied to Drive, including as a nominally temporary production artifact. H3 keeps every record’s complete twist circle on one shard; T1 keeps every ordered QR history on one trajectory shard.

The implementation exposes these contracts directly in the canonical GPU engine. **record_observer** receives the ordered controller word, Born outcomes, branch probabilities, feedback targets, and cycle/update/sample identifiers without changing the dynamics. **frozen_schedule** together with **frozen_outcomes** replays that exact conditional branch. **set_controller_twist(phi,gauge,seam,y)** applies an absolute rather than cumulative controller twist, with seam and uniform gauges used as mutual controls. **onsite_phase_noise_sigma** inserts the M3 onsite phase kick once after each complete adaptive cycle, while **noise_observer** records per-cycle moments and a SHA-256 digest of the realized phase field. Exact regeneration uses the pre-run RNG state, engine hash, and configuration; the full field is not copied to Drive. Production archives save these compact records and API conventions, not samplewise covariance matrices.

M1/L1 has no fixed-schedule descendant, trajectory shard, CUDA branch, Drive archive, or Colab runtime. Its CPU runner constructs the exact canonical frame operators, solves the continuous Lyapunov problem, and applies exact finite Gaussian affine channels by binary composition. Independent cases are process-parallel with explicit CPU affinity and one-thread BLAS by default. The selected-observable schema writes no covariance, and

it permits momentum-resolved arrays only for $n_{\text{shell}} = \text{None}$.

B. Audited legacy implementation lineage

Tables XVIII and XIX identify the actual runners and observer packages behind every substantial legacy result used above. An output directory by itself is not treated as an implementation specification. The production bundle inherits a legacy component only when the runner, canonical engine call, observable definition, and output schema have all been identified. *Rebuild* below means that the legacy artifact supplies a diagnostic or formal target but no conforming production runner.

TABLE XVIII. Core trajectory and state-data implementation lineage. Paths are repository-relative.

campaign	substantial legacy artifact	audited runner or observer	production use
B0; exact H1/H2	00_WORKSPACE/CURRENT/ experiment_review/b0_exact_ domain_wall/results/ 20260816_191957/	00_WORKSPACE/CURRENT/ experiment_review/b0_exact_ domain_wall/run_b0_campaign.py with 00_WORKSPACE/CURRENT/ experiment_review/b0_exact_ domain_wall/b0lib.py and locked 00_WORKSPACE/CURRENT/ experiment_review/b0_exact_ domain_wall/campaign_config.v2. json	Keep as the deterministic CPU calibration and V0 reference. Its entropy, modular, response, and twist routines provide exact tests, not a stochastic Colab runner.
B1	00_WORKSPACE/CURRENT/ experiment_review/b1_ controller_frame/	00_WORKSPACE/CURRENT/final_ production_ready_figure_ scripts/06_b1_controller_frame/ calling classA_U1FGTN_gpu.run_ markov_circuit	Preserve the stopped CPU pilot as partial provenance. Production uses the self-contained A100 bundle, two 20×48 arms, scalar manifold-distance and Ky Fan histories, and no charge-sector weights or activity-prediction fit.
P1, W1, T1	00_WORKSPACE/COLAB/colab_ lyapunov/gpu_data/lyapunov_ spectra/campaigns/N20_ Ny30-50_nsh1_a1-1_S25_ cyclesNy/	00_WORKSPACE/COLAB/colab_ lyapunov/notebooks/data_ generation/run_lyapunov_ spectra_N20_Ny30-50_S25_ cyclesNy.ipynb with 00_ WORKSPACE/COLAB/colab_lyapunov/ src/lyapunov_observables_gpu.py	Reuse the canonical A100 launch, streamed Chern estimator, charge observer, and tangent QR plumbing in <code>bulk_width_gate</code> and <code>pure_wall_master</code> . Replace the single selected QR vector by the declared aligned low-subspace products.
B2 and H2	00_WORKSPACE/CURRENT/ topological_frustration_ diagnostics/results/ campaigns/20260813_121340_ Nx20_perfect_Ny20_22_24/ and 00_WORKSPACE/CURRENT/ topological_frustration_ diagnostics/results/ campaigns/20260812_233722_ production/03_parallel_ benchmark/response/	00_WORKSPACE/CURRENT/ topological_frustration_ diagnostics/launch_campaign.py dispatching 00_WORKSPACE/CURRENT/ topological_frustration_ diagnostics/run_cpu.py modes activity, spectral, and response	Port the eventwise X_j, Y_j , branch-probability, activity-profile, and paired-response observers. Rebuild H2's likelihood-score term on GPU; the legacy response runner contains only the conditional/reset precursor.
S1	00_WORKSPACE/COLAB/colab_ large_entanglement_scaling_ N20/ and 00_WORKSPACE/COLAB/ colab_charge_fluctuations/ gpu_data/streaming_ covariance_observables/	00_WORKSPACE/COLAB/colab_large_ entanglement_scaling_N20/ notebooks/run_slope_vs_system_ size.ipynb with 00_WORKSPACE/COLAB/colab_large_ entanglement_scaling_N20/src/ strip_entropy_streaming_gpu.py; and 00_WORKSPACE/COLAB/colab_ charge_fluctuations/notebooks/ data_generation/run_streaming_ covariance_observables_multi_ geometry.ipynb with 00_ WORKSPACE/COLAB/colab_charge_ fluctuations/src/streaming_ covariance_observables_gpu.py	This is the primary streaming-observer skeleton for <code>pure_wall_master</code> . Preserve the sample axis and add matched interval-charge spectra and the three declared late checkpoints. The retired TMI estimator under 00_WORKSPACE/LEGACY/ <code>sample_average_testing/</code> is provenance only and contributes no production computation.
P2	00_WORKSPACE/COLAB/colab_ charge_fluctuations/gpu_ data/purification_dynamics_ maxmix/campaigns/N20_multi_ geometry_nsh1_dwtrunc1_ init-maxmix_S100_ cycles-2Ny/	00_WORKSPACE/COLAB/colab_ charge_fluctuations/notebooks/ data_generation/run_ purification_dynamics_maxmix_ multi_geometry.ipynb with 00_ WORKSPACE/COLAB/colab_charge_ fluctuations/src/purification_ dynamics_observables_gpu.py	Use as the <code>maxmix_master</code> base. Add the matched trivial arm, spatial contour, purity-gap output, realized record, and same-record physical tangent products.
H1	00_WORKSPACE/COLAB/colab_ small_system_testing/gpu_ data/pure_state_covariance_ snapshots/ and 00_WORKSPACE/COLAB/colab_ small_system_testing/ analysis_outputs/dynamic_ modular_charge_spreading/	00_WORKSPACE/COLAB/colab_small_ system_testing/notebooks/data_ generation/run_pure_state_ markov_snapshots.ipynb followed by 00_WORKSPACE/COLAB/colab_small_ system_testing/src/dynamic_ modular_charge_spreading.py	Reuse the snapshot validation and modular-Hamiltonian construction in <code>chirality_replay</code> . Perform packet evolution and velocity fitting per trajectory; do not reuse the legacy product that averages modular generators before evolution.

TABLE XIX. Derived, record, scan, and control implementation lineage.

campaign	substantial legacy artifact	audited runner or observer	production use
H3	<code>tangent_cocycle_flux_snapshot/</code> and the exact twist validation in B0	<code>tangent_cocycle_flux_snapshot/run_flux_cocycle_snapshot.py</code> calling the canonical CPU engine; exact reference routines in <code>00_WORKSPACE/CURRENT/experiment_review/b0_exact_domain_wall/b0lib.py</code>	Port record freezing, replay checksums, twist checkpointing, and large-gauge closure to <code>chirality_replay</code> . Rebuild the controller-projector twist before support truncation and add wall entanglement-spectrum subspace tracking. The existing tangent flux snapshot is not the H3 estimator.
G5	<code>00_WORKSPACE/CURRENT/tangent_edge_channel/</code> , <code>00_WORKSPACE/COLAB/colab_regularized_choi_transfer_matrix/</code> , and the ordered-word products of <code>02_pure_wall_master</code>	<code>02_pure_wall_master/src/g5_manybody_spectrum_analysis.py</code> , generated from <code>final_production_ready_figure_scripts/_shared_src/g5_manybody_spectrum_analysis.py</code>	Replay each checksum-verified primary wall/control word from $C = 1/2$, with <code>track_choi=False</code> ; save natural occupations, oriented caps, tangent/radial gaps, $\log Z_K$, factorized levels, and the lowest 128 levels in $q = -3, \dots, 3$. Keep the old SCGF merger as a separately named record diagnostic.
S2	<code>00_WORKSPACE/COLAB/colab_no_feedback_alpha_sweep_transfer/</code>	<code>00_WORKSPACE/COLAB/colab_no_feedback_alpha_sweep_transfer/notebooks/data_generation/run_no_feedback_alpha_sweep_dwtrunc1_N20_Ny20-30-40_S10_cycles2Ny_gpu.ipynb</code> with <code>00_WORKSPACE/COLAB/colab_no_feedback_alpha_sweep_transfer/src/choi_transfer_observables_gpu.py</code>	Reuse the A100 configuration-grid, manifest, resume, and matrix-free near-gap machinery in <code>scans_and_controls</code> . Replace the legacy no-feedback emphasis by paired $S = 25$ perfect-correction random-pure and max-mix physical-tangent S2 grids; attach the early/logarithmic purification checkpoints to the max-mix arm.
T1/P2 method checks	<code>00_WORKSPACE/CURRENT/tangent_edge_channel/</code> and <code>00_WORKSPACE/COLAB/colab_regularized_choi_transfer_matrix/</code>	<code>00_WORKSPACE/CURRENT/tangent_edge_channel/run_tangent_edge_channel.py</code> ; and <code>00_WORKSPACE/COLAB/colab_regularized_choi_transfer_matrix/notebooks/data_generation/run_particle_choi_transfer_gap_N20_Ny30-50_S25_cycles2Ny_gpu.ipynb</code> with <code>00_WORKSPACE/COLAB/colab_regularized_choi_transfer_matrix/src/choi_transfer_observables_gpu.py</code>	Reuse positive-diagonal QR gauges, scale-separated products, and rank diagnostics from the tangent path. The Choi material is retained as historical proxy evidence only and is not executed by a production bundle.
M1	<code>00_WORKSPACE/CURRENT/mean_channel_lindblad_cpu_campaign/</code> , <code>scripts/legacy/CI_Lindblad_DW.py</code> , <code>figs/corr_y_profiles/</code> , and atlas Result 24	<code>mean_channel_lindblad_cpu.py</code> is the exact finite-channel/continuous-generator CPU solver; <code>CI_Lindblad_DW.py</code> is the parity reference and historical bath/dephasing implementation	Run the immutable 123-case CPU configuration. Keep finite-shell products integrated, reserve k_y spectra for the untruncated arm, and archive only compact observables, manifests, timings, hashes, and residuals.
M2	<code>00_WORKSPACE/COLAB/colab_partial_post-select/gpu_data/</code>	<code>00_WORKSPACE/COLAB/colab_partial_post-select/notebooks/run_partial_postselection_sweep_N20x40_gpu.ipynb</code> with <code>00_WORKSPACE/COLAB/colab_partial_post-select/src/streaming_covariance_observables_gpu.py</code>	Reuse the complete Colab sweep, streamed observables, and partial-postselection bookkeeping. Replace its pilot grid by the declared M2 configurations and $S = 25$ independent trajectories at every stochastic point.
M3	<code>notebooks/test_coherent_term_markov_circuit.ipynb</code>	No conforming runner: rendered cells refer to a removed <code>coherent_term</code> argument, and <code>scripts/legacy/CI_Lindblad_DW.py</code> implements a different noise model	Rebuild. Add the BPJ onsite random-phase step to the canonical GPU class, validate it in V0, and only then create the M3 portion of <code>scans_and_controls</code> .

At this audit revision, the bundled `classA_U1FGTN_gpu.py` files in `00_WORKSPACE/COLAB/colab_lyapunov/`, `00_WORKSPACE/COLAB/colab_large_entanglement_scaling_N20/`, and `00_WORKSPACE/COLAB/colab_charge_fluctuations/` are byte-identical to the canonical source. The copies in `00_WORKSPACE/COLAB/colab_small_system_testing/`, `00_WORKSPACE/COLAB/colab_no_feedback_alpha_sweep_transfer/`, `00_WORKSPACE/COLAB/colab_regularized_choi_transfer_matrix/`, and `00_WORKSPACE/COLAB/colab_partial_post-select/` match as well. The canonical file is `src/fgtn/classA_U1FGTN_gpu.py`, and the common SHA-256 is

$$h_{\text{GPU}} = 19\text{b}200\text{dce}7\text{b}5\text{fd}004\text{a}627770492\text{d}32\text{c}573 \quad (81)$$

$$8\text{ef}19\text{ff}8\text{f}94066\text{eebad}5\text{ad}66\text{d}9\text{f}23\text{d}.$$

This hash is an audit fact, not a permanent pin: every new bundle copies the then-current canonical source, records its hash, and refuses to run if its observer expects an incompatible engine signature.

XVII. CLAIM-DOWNGRADE MATRIX

TABLE XX. Mandatory interpretation when a gate fails.

observation	licensed wording
c fit works but charge/correlator fails	entropy is consistent with a $c = 1$ wall; no full CFT identification
c and k fit separately but the matched $\pi^2/3$ relation fails	report separate entropy and current coefficients; no minimal $U(1)_1$ identification
wall tangent gap decreases but intercept unresolved	finite-size softening and wall-localized slow mode
modular drift only	handed modular flow, not physical chiral propagation
conditional response only	fixed-record response, not Born-ensemble response
χ chiral but twist not robust	chiral response without an extracted integer wall index
bulk Bott is quantized but $n_{\text{shell}} = 1$ marker does not converge	topological index with nonquantized local-marker finite-range corrections
identity-input dictionary fails an exact Fock test	no G5 operator-spectrum claim
factorized spectrum passes but charged towers drift	report finite-word singular levels, not k
neutral scale or matched-control Casimir fit is unresolved	do not report c_{eff} from G5
one frozen-word gap only	representative branch gap, not a universal operator spectrum
explicit-interface and support-terminated walls disagree	two distinct programmable interfaces
bulk and boundary noise points differ	preemptive boundary crossover/transition, not bulk-boundary locking

XVIII. DEFERRED WORK

The core campaign excludes a Chern number assigned to an ad hoc quenched-log tangent Hamiltonian, polar-unitary alignment of independently sampled states, a full scattering network, regularized-Choi endpoint spectroscopy, an absorbing-transition phase diagram, high-rank many-body reconstruction, a broad non-Hermitian survey, and a full Lindblad phase diagram. It also excludes promoting the natural-orbital coarse graining in `notebooks/flattened_hamiltonian_analysis/gaussian_entanglement_rg/` to a MERA result. That legacy analysis usefully tests whether locally mixed modes retained under quasi-one-dimensional Gaussian coarse graining move toward the walls, but its isometries are scale dependent, its repeated-tensor fits have order-one mismatch, and purified variants explicitly project the coarse covariance back to a Slater projector. It becomes main-text relevant only after a repeated finite-depth tensor has controlled discarded-mode error, stable state fidelity or correlators, and a nontrivial prediction beyond re-extracting $c \simeq 1$. These topics are follow-up projects unless a core gate specifically requires one.

Appendix A: Overlap-section winding and the occupied Bloch bundle

This appendix fixes the formal meaning and numerical implementation of the OW overlap winding used in the B1 legacy summary. Consider the two-band Bloch Hamiltonian

$$\begin{aligned} h(\mathbf{k}) &= \mathbf{n}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \\ \mathbf{n} &= (\sin k_x, \sin k_y, \alpha - \cos k_x - \cos k_y), \end{aligned} \quad (\text{A1})$$

with lower-band projector $P_- (\mathbf{k}) = [\mathbf{1} - \hat{\mathbf{n}}(\mathbf{k}) \cdot \boldsymbol{\sigma}]/2$. The fixed X -trial spinor is $|\tau_X\rangle = (1, 1)^T/\sqrt{2}$. For no truncation take $|\phi_{\text{none}}(\mathbf{k})\rangle = |\tau_X\rangle$. A square finite-shell field is

$$\begin{aligned} |\phi_n(\mathbf{k})\rangle &= \sum_{\mathbf{r} \in [-n, n]^2} e^{-i\mathbf{k} \cdot \mathbf{r}} C_{\mathbf{r}} |\tau_X\rangle, \\ C_{\mathbf{r}} &= \int_{\text{BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\mathbf{q} \cdot \mathbf{r}} P_-(\mathbf{q}). \end{aligned} \quad (\text{A2})$$

Let $L \rightarrow T^2$ be the occupied line bundle. On momentum patches U_i , choose normalized frames satisfying $|u_j\rangle = e^{i\chi_{ij}} |u_i\rangle$ on $U_i \cap U_j$. The local overlap representatives

$$f_{n,i}(\mathbf{k}) = \langle u_i(\mathbf{k}) | \phi_n(\mathbf{k}) \rangle \quad (\text{A3})$$

then satisfy $f_{n,j} = e^{-i\chi_{ij}} f_{n,i}$ and therefore define a section of L^* .

Here is a direct proof of the winding relation, requiring only Stokes' theorem. Remove a small counterclockwise disk D_a around every isolated zero $\mathbf{k}_a$ and call the remaining punctured Brillouin zone $M = T^2 \setminus \bigcup_a D_a$. The

normalized projected trial state is well defined and single valued on M :

$$\begin{aligned} |v_n(\mathbf{k})\rangle &= \frac{P_-(\mathbf{k})|\phi_n(\mathbf{k})\rangle}{\|P_-(\mathbf{k})|\phi_n(\mathbf{k})\rangle\|} \\ &= e^{i\vartheta_i(\mathbf{k})} |u_i(\mathbf{k})\rangle, \\ \vartheta_i &= \arg f_{n,i}. \end{aligned} \quad (\text{A4})$$

Although $|u_i\rangle$ and ϑ_i separately depend on the patch gauge, their product does not: $\vartheta_j = \vartheta_i - \chi_{ij}$ precisely cancels $|u_j\rangle = e^{i\chi_{ij}} |u_i\rangle$. Define $A_i = -i\langle u_i | du_i \rangle$ and $A_v = -i\langle v_n | dv_n \rangle$. Equation (A4) immediately gives

$$A_v = A_i + d\vartheta_i, \quad dA_v = dA_i \equiv F. \quad (\text{A5})$$

Because A_v is a single connection one-form throughout M , Stokes' theorem gives

$$\begin{aligned} 2\pi C &= \lim_{D_a \rightarrow 0} \int_M F = \lim_{D_a \rightarrow 0} \oint_{\partial M} A_v \\ &= - \sum_a \lim_{D_a \rightarrow 0} \oint_{\gamma_a} (A_i + d\vartheta_i) = -2\pi \sum_a \nu_a. \end{aligned} \quad (\text{A6})$$

The minus sign is purely an orientation sign: each hole is traversed clockwise as part of ∂M , whereas $\gamma_a = \partial D_a$ is counterclockwise in the definition of winding. The integral of the smooth local A_i vanishes with the disk radius, leaving $\oint_{\gamma_a} d\vartheta_i = 2\pi\nu_a$. Therefore

$$\boxed{\sum_a \nu_a = \sum_a \frac{1}{2\pi} \oint_{\gamma_a} d \arg f_{n,i} = -C.} \quad (\text{A7})$$

This is the zero-index statement specialized and proved for the present overlap problem. Consequently, every smooth global trial field has total overlap vorticity $-C$ against a Chern band, even when that trial field spans a trivial bundle. In particular, $n_{\text{shell}} = 0$ has compact-spinor Chern number zero but its overlap still has total winding -1 when the target band has $C = +1$. In a trivial target band, isolated zero-antizero pairs may occur, but their total index must be zero.

The zero search is gauge independent because it begins from

$$|f_n(\mathbf{k})|^2 = \langle \phi_n(\mathbf{k}) | P_-(\mathbf{k}) | \phi_n(\mathbf{k}) \rangle. \quad (\text{A8})$$

The implementation evaluates Eq. (A8) on a periodic 181×181 grid, selects all discrete local minima, refines the lowest 32 by periodic two-dimensional Nelder-Mead minimization, and merges points separated by less than 10^{-5} in wrapped momentum distance. A residual below 10^{-6} is declared a zero. For each such point, choose the reference spinor $|r_a\rangle$ maximizing $\|P_-(\mathbf{k}_a)|r_a\rangle\|$ from a fixed four-spinor set, and hold it fixed around the loop. This defines

$$|u_a^{\text{loc}}(\mathbf{k})\rangle = \frac{P_-(\mathbf{k})|r_a\rangle}{\|P_-(\mathbf{k})|r_a\rangle\|}. \quad (\text{A9})$$

On the radius-0.02 circle $\mathbf{k}(\theta) = \mathbf{k}_a + 0.02(\cos \theta, \sin \theta)$, 721 equally spaced angles are used to compute

$$\nu_a = \text{round} \left\{ \frac{1}{2\pi} [\text{unwrap arg} \langle u_a^{\text{loc}}(\mathbf{k}(\theta)) | \phi_n(\mathbf{k}(\theta)) \rangle]_{\theta=0}^{2\pi} \right\}. \quad (\text{A10})$$

The same 181×181 mesh independently computes the target-band FHS number using normalized links $U_\mu(\mathbf{k}) = \langle u(\mathbf{k}) | u(\mathbf{k} + \hat{\mu}) \rangle / |\langle u(\mathbf{k}) | u(\mathbf{k} + \hat{\mu}) \rangle|$ and

$$C_{\text{FHS}} = \frac{1}{2\pi} \sum_{\mathbf{k}} \text{Arg}[U_x(\mathbf{k})U_y(\mathbf{k} + \hat{x})U_x(\mathbf{k} + \hat{y})^*U_y(\mathbf{k})^*]. \quad (\text{A11})$$

This independent surface calculation gives $C = +1$ at $\alpha = 1, 1.5$ and $C = 0$ at $\alpha = 3, 30$. The corresponding overlap windings and positive trivial-phase gaps are reported in Eq. (19). The critical value $\alpha = 2$ is excluded because the Bloch gap closes and the isolated-band Chern number is undefined.

Appendix B: Frozen-record $U(1)$ twist family and wall spectral flow

This appendix defines the branchwise twist used in Campaign H3. It is an auxiliary spectral-flow construction for a single realized Born record. It is not a physical monitored-time pump, and its integer is not called a quantized transport coefficient.

1. Boundary holonomy and twisted controller projectors

Let $y \in \mathbb{Z}_{N_y}$ denote the periodic direction and let ϕ be the flux through the cylinder. A one-particle $U(1)$ connection $A_{ij}^{(\phi)}$ is chosen so that the oriented product of Peierls phases around one complete y cycle is

$$\prod_{(ij) \in S_y^+} e^{iA_{ij}^{(\phi)}} = e^{i\phi}. \quad (\text{B1})$$

In a seam gauge all links are untwisted except those crossing a declared bond $y = N_y - 1 \rightarrow 0$, which carry $e^{+i\phi}$ in the positive orientation and $e^{-i\phi}$ in the reverse orientation. A uniform gauge distributes the same holonomy as a phase $e^{i\phi/N_y}$ per positive- y step. The two descriptions are related by a one-particle gauge transformation U_ϕ and must give identical spectra.

For a normalized controller orbital

$$|\chi_j(0)\rangle = \sum_i w_{ji} |i\rangle, \quad P_j(0) = |\chi_j(0)\rangle \langle \chi_j(0)|, \quad (\text{B2})$$

parallel transport each component from the declared controller center to site i using $A^{(\phi)}$. This defines

$$|\chi_j(\phi)\rangle = \sum_i e^{i\vartheta_{ji}(\phi)} w_{ji} |i\rangle, \quad P_j(\phi) = |\chi_j(\phi)\rangle \langle \chi_j(\phi)|. \quad (\text{B3})$$

The phase difference $\vartheta_{ji} - \vartheta_{jk}$ equals the line integral of the connection between the two components. This prescription matters for a controller whose finite support straddles the periodic seam: multiplying the whole orbital by one common phase would cancel from P_j and would not insert a twist. Construct the full twisted controller frame first and only then apply the support mask. Twisting after truncation can omit seam-crossing components and defines a different protocol.

At $\phi = 2\pi$ the seam-gauge projectors close directly. In a uniform gauge they obey

$$P_j(2\pi) = U_{2\pi} P_j(0) U_{2\pi}^\dagger, \quad (\text{B4})$$

where $U_{2\pi}$ is the large gauge transformation. Every covariance and overlap closure test must therefore be performed in one common gauge, applying $U_{2\pi}$ where required. Moving the seam is a gauge-relocation control rather than a new physical flux family.

In a translation-invariant equilibrium ribbon, the same holonomy shifts the allowed momenta as $k_y \mapsto k_y + \phi/N_y$. Every bulk band is periodic over the Brillouin zone, so after 2π flux the bulk spectrum returns to itself as a set; this fact alone does not mean that each eigenstate returns to the same label. A chiral interface branch connects the valence and conduction continua. Under one flux quantum its levels undergo a cyclic permutation across the reference energy: the branch need not close onto itself before its label is exchanged with the neighboring level. This spectral flow, rather than a failure of bulk cosine periodicity, is the equilibrium analogue of the wall-resolved entanglement flow measured here. The two oppositely oriented interfaces carry opposite permutations, so their net flow on the periodic sample vanishes.

2. One record, one continuous conditional branch

At $\phi = 0$, independently prepare and evolve trajectory s and save its complete record

$$\xi_s = [(j_1, m_1), (j_2, m_2), \dots, (j_M, m_M)], \quad (\text{B5})$$

including the external controller/channel word j_e , every Born outcome m_e , the pre-run RNG state (or equivalent counter-based initial-state seed map), and the replay hash. Feedback targets and correction rules remain those of the physical protocol. For the twisted projectors define the unnormalized branch operator and branch weight

$$K_{\xi_s}(\phi) = K_{m_M|\xi_s, < M}(\phi) \cdots K_{m_1}(\phi), \quad (\text{B6})$$

$$p_{\xi_s}(\phi) = \text{Tr}[K_{\xi_s}(\phi) \rho_0 K_{\xi_s}^\dagger(\phi)]. \quad (\text{B7})$$

Whenever $p_{\xi_s}(\phi) > 0$, normalize this same branch and obtain $C_{\xi_s}(\phi)$. The physical sampling occurs only once, at $\phi = 0$; all other twist points are deterministic replays of that record. The resulting family asks how one conditional branch changes under boundary holonomy. It does not

assert that $p_{\xi_s}(\phi)$ is uniform or that the frozen record is typical of the Born ensemble at nonzero ϕ .

Drawing a new outcome record at each ϕ would instead produce unrelated random covariances $C_{\xi(\phi)}(\phi)$. Sorting their spectra can manufacture apparent jumps or crossings, but there is no continuous eigenbundle and hence no branchwise spectral-flow integer. Likewise, the external random-serial word is frozen along with the outcomes; a new word at every ϕ changes the microscopic conditional map. If $p_{\xi_s}(\phi) = 0$ anywhere on the circle, the normalized branch is undefined there and record s does not define the required flux family. Such a failure is saved and counted; the record may not be replaced after its twist behavior is known.

3. Entanglement flow, wall resolution, and crossing count

For the preregistered wall subsystem A_w , diagonalize the restricted covariance at every twist point,

$$C_{\xi_s, A_w}(\phi) u_{sa}(\phi) = \nu_{sa}(\phi) u_{sa}(\phi), \quad \varepsilon_{sa}(\phi) = \log \frac{1 - \nu_{sa}(\phi)}{\nu_{sa}(\phi)}. \quad (\text{B8})$$

The reference level $\varepsilon = 0$ is equivalent to $\nu = 1/2$. Between adjacent twist points, form the overlap matrix

$$\mathcal{O}_{ab}^{(m)} = \langle u_{sa}(\phi_m) | u_{sb}(\phi_{m+1}) \rangle. \quad (\text{B9})$$

Track isolated levels by maximum absolute overlap. For a near-degenerate cluster, track the entire subspace by its singular values and polar transport before assigning individual branches; an arbitrary basis rotation inside the cluster is not a physical crossing. Save every overlap matrix, assignment, minimum tracking singular value, and ambiguity flag.

Assign a tracked branch to wall w using its preregistered wall-window weight or a projected transverse-position operator. A crossing between ϕ_m and ϕ_{m+1} contributes the sign of the tracked spectral slope. The wall-resolved flow is

$$\nu_{\text{flow}, w}(\xi_s) = \sum_{a, c \in w} \text{Sgn}[\varepsilon_{sa}(\phi_{m_c+1}) - \varepsilon_{sa}(\phi_{m_c})], \quad (\text{B10})$$

where c runs over unambiguous zero crossings of tracked branch a assigned to wall w . The global sign convention is fixed once using the B0 orientation labels. Opposite walls of the periodic domain-wall geometry should carry opposite flow, so the target is $(-1, +1)$ up to the declared orientation convention and the net flow is zero.

Repeat the crossing count after changing the twist grid, moving the seam, varying the tracking tolerance, and moving the reference level within the entanglement gap. A valid integer cannot depend on any of these choices. Verify

$$C_{\xi_s}(2\pi) \simeq U_{2\pi} C_{\xi_s}(0) U_{2\pi}^\dagger \quad (\text{B11})$$

and the corresponding subspace closure. Failure of closure, an unresolved overlap assignment, or a zero branch weight is not silently repaired by smoothing the spectrum.

Production uses $S = 1$ preregistered Born-sampled record per protocol at $\phi = 0$. Each record and its complete twist circle form one branch-level diagnostic unit. Report its signed wall flows, branch-weight and closure diagnostics, and ambiguity flags directly; do not report a success fraction or infer typicality from $S = 1$. The 65 twist points, the two walls, and multiple reference levels are correlated deterministic descendants and do not increase S .

Appendix C: Quenched finite-time tangent spectra and localization modes

This appendix fixes the tangent-product convention used in Campaigns T1 and P2. It also separates three objects that are easily conflated: the quenched Lyapunov spectrum, the eigenmodes of a trajectory-averaged finite-time log-strain operator, and an annealed second-moment spectrum.

1. Cycle Jacobian, product, and time normalization

Along realized trajectory s , let $\Phi_{s,t}$ denote one complete normalized circuit cycle and let

$$\delta C_{t+1} = J_{s,t} [\delta C_t], \quad J_{s,t} = D\Phi_{s,t}|_{C_{s,t}}, \quad (\text{C1})$$

after restriction to the declared physical tangent space. The T -cycle product is

$$M_s(T) = J_{s,T-1} \cdots J_{s,1} J_{s,0}. \quad (\text{C2})$$

Here T is the number of complete circuit cycles. The finite-time singular exponents are

$$\lambda_{sa}(T) = \frac{1}{T} \log \sigma_a[M_s(T)]. \quad (\text{C3})$$

Equivalently, on the nonsingular physical subspace define the right log-strain operator

$$L_{s,T}^R = \frac{1}{2T} \log [M_s^\dagger(T) M_s(T)]. \quad (\text{C4})$$

Its eigenvalues are $\lambda_{sa}(T)$ and its eigenvectors are the right finite-time singular vectors, namely input perturbations at the left endpoint of the window. The analogous left operator

$$L_{s,T}^L = \frac{1}{2T} \log [M_s(T) M_s^\dagger(T)] \quad (\text{C5})$$

has the same finite eigenvalues and output singular vectors. The normalization is multiplication by $1/(2T)$, not division by $1/(2T)$. Exact null directions have $\log 0 = -\infty$

and are recorded in a null mask rather than replaced by an arbitrary finite floor.

The reported unit is inverse circuit cycles. The scaled variable T/N_y may be used to display convergence, and the size dependence may subsequently be fitted to $\Delta(N_y) \sim N_y^{-z}$, but N_y is not inserted into Eq. (C3). Normalizing by elementary update count or hardware layer depth defines a different clock and must be reported as a separate conversion.

2. Stabilized QR accumulation and finite-time singular vectors

The dense product in Eq. (C2) is not formed at production sizes. Strong contraction, exact projector nulls, and expanding directions make it ill-conditioned. Starting from an orthonormal rank- r physical frame $Q_{s,0}$, iterate

$$Y_{s,t} = J_{s,t} Q_{s,t}, \quad Y_{s,t} = Q_{s,t+1} R_{s,t}, \quad (\text{C6})$$

with a fixed diagonal-sign convention for $R_{s,t}$. After the declared alignment interval, the QR estimator is

$$\ell_{sa}(T) = \frac{1}{T} \sum_{t=0}^{T-1} \log |(R_{s,t})_{aa}|. \quad (\text{C7})$$

The asymptotic limit gives the Oseledec exponents represented by the retained frame. QR vectors themselves are backward Lyapunov vectors and need not equal covariant Lyapunov vectors or finite-window singular vectors, particularly inside a near-degenerate band.

To reconstruct finite-window singular modes, save block- R factors together with their diagonal log scales and the input/output frames. A stabilized SVD of the coefficient-space product

$$\mathcal{R}_s(T) = R_{s,T-1} \cdots R_{s,0} = \tilde{U}_s \Sigma_s \tilde{V}_s^\dagger \quad (\text{C8})$$

gives retained input and output singular vectors

$$v_{sa}^{\text{in}} = Q_{s,0} \tilde{V}_{sa}, \quad v_{sa}^{\text{out}} = Q_{s,T} \tilde{U}_{sa}. \quad (\text{C9})$$

The rank- r result is exact only within the propagated subspace. Rank doubling and independent initial frames test whether the near-zero band has been captured.

3. Quenched spectrum, mean log strain, and annealed spectrum

The primary quenched spectrum takes the logarithm trajectory by trajectory and then averages continuously tracked indices,

$$\bar{\lambda}_a^q(T) = \frac{1}{S} \sum_{s=1}^S \lambda_{sa}(T) = \frac{1}{ST} \sum_{s=1}^S \log \sigma_a[M_s(T)]. \quad (\text{C10})$$

The gap is formed only after this indexwise average. Means and medians of the per-trajectory minimum gap

are retained as distinct sensitivity statistics. Averaging $M_s(T)$ before taking singular values is neither this quenched estimator nor the tangent spectrum of a typical realized record.

If all right log-strain operators have been embedded or parallel transported into one declared common physical tangent space, one may also define

$$\bar{L}_T^R = \frac{1}{S} \sum_{s=1}^S L_{s,T}^R. \quad (\text{C11})$$

This is a useful mean finite-time log-strain operator. Its eigenvectors answer which common directions are slow or expanding under the averaged logarithmic strain. In general, however,

$$\text{eig}(\bar{L}_T^R) \neq \frac{1}{S} \sum_s \text{eig}(L_{s,T}^R), \quad (\text{C12})$$

because the sample log-strain operators do not commute. Consequently there is no unique single vector whose eigenvalue is the indexwise quenched average in Eq. (C10). The eigenmodes of $\bar{L}_T^R$ may be plotted, but they are labelled “mean quenched log-strain modes,” not “the quenched Lyapunov modes.” Since physical tangent spaces at different covariances need not coincide, the embedding or transport convention and its residual are part of this definition.

For comparison, the annealed second-moment operator is

$$L_T^{\text{ann}} = \frac{1}{2T} \log \left[\frac{1}{S} \sum_s M_s^\dagger(T) M_s(T) \right]. \quad (\text{C13})$$

It places enhanced weight on rare records with large stretch and is not substituted for the quenched spectrum. Equations (C10), (C11), and (C13) are saved and named separately whenever more than one is reported.

4. Localization of a nonunique slow band

For an isolated nondegenerate exponent one may plot its input or output singular-vector density, provided the endpoint convention is stated. Near degeneracy, individual vectors rotate under numerical noise, QR gauge, and sample-to-sample wall exchange. The stable trajectory-level object is the projector onto the declared slow band $\mathcal{L}_s$,

$$P_{\mathcal{L},s} = \sum_{a \in \mathcal{L}_s} |v_{sa}\rangle \langle v_{sa}|, \quad (\text{C14})$$

$$\rho_s(x, y) = \frac{1}{r_s} \sum_{a \in \mathcal{L}_s, \mu} |v_{sa, \mu}(x, y)|^2.$$

Average $P_{\mathcal{L},s}$ only after expressing it in the same ambient real-space basis; otherwise average the gauge-invariant density ρ_s and wall-window weights. Resolve the two

walls by diagonalizing the projected transverse-position operator inside $\mathcal{L}_s$ or by fixed left/right windows. Do not average signed eigenvectors, select the most localized QR column, or count a basis rotation inside a degenerate subspace as motion of a physical mode.

When the tangent action factorizes as a one-leg congruence,

$$\delta C \mapsto K_s(T) \delta C K_s^\dagger(T), \quad (\text{C15})$$

and $K_s(T)$ has one-leg exponents ℓ_{sa} , the full covariance-tangent exponents are pair sums

$$\lambda_{s,ab}^{(C)} = \ell_{sa} + \ell_{sb}, \quad (\text{C16})$$

subject to the physical Hermiticity/projector tangent constraints. A purification rate must therefore be compared with the appropriate covariance pair-sum sector, not automatically with the smallest one-leg gap.

For every trajectory save the QR log increments, block- R factors, frame ranks, null and gauge masks, input/output endpoint convention, slow-band membership, projector density, wall weights, orthogonality residuals, and replay hash. The independent sample count is the number of Born trajectories S ; tangent-frame repeats, QR checkpoints, and multiple vectors within one slow band are correlated numerical diagnostics and do not increase S .

Appendix D: Charge full counting statistics and the level-one current algebra

For a Gaussian trajectory with subsystem correlation matrix $C_{\xi,A}$, the quantum FCS of Q_A is

$$\begin{aligned} \mathcal{Z}_{\xi,A}(\lambda) &= \det[\mathbf{1}_A - C_{\xi,A} + e^{i\lambda} C_{\xi,A}], \\ \mathcal{K}_{\xi,A}(\lambda) &= \log \mathcal{Z}_{\xi,A}(\lambda). \end{aligned} \quad (\text{D1})$$

With $\kappa_{n,\xi}^q = (-i\partial_\lambda)^n \mathcal{K}|_{\lambda=0}$,

$$\kappa_{1,\xi}^q = \text{Tr } C_{\xi,A}, \quad (\text{D2})$$

$$\kappa_{2,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})], \quad (\text{D3})$$

$$\kappa_{3,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})(\mathbf{1}_A - 2C_{\xi,A})], \quad (\text{D4})$$

$$\kappa_{4,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1}_A - C_{\xi,A})(\mathbf{1}_A - 6C_{\xi,A} + 6C_{\xi,A}^2)]. \quad (\text{D5})$$

The state-CFT observable is $\mathbb{E}[\kappa_{n,\xi}^q]$. It is distinct from the FCS of the Born mixture,

$$\mathcal{Z}_A^{\text{mix}}(\lambda) = \mathbb{E}_\xi[\mathcal{Z}_{\xi,A}(\lambda)], \quad (\text{D6})$$

whose logarithm includes classical trajectory fluctuations. In particular,

$$\text{Var}_{\text{mix}}(Q_A) = \mathbb{E}_\xi[\kappa_{2,\xi}^q] + \text{Var}_\xi(\kappa_{1,\xi}^q). \quad (\text{D7})$$

Both terms in Eq. (D7) are saved and plotted separately.

For a chiral $U(1)$ current, the normalization convention is fixed by

$$\begin{aligned} J(z)J(0) &\sim \frac{k}{z^2}, \\ [J_m, J_n] &= k m \delta_{m+n,0}. \end{aligned} \quad (\text{D8})$$

A free chiral complex fermion has $k = 1$. A single wall contributes

$$\begin{aligned} X_L(\ell) &= \log \left[\frac{N_y}{\pi a} \sin \left(\frac{\pi \ell}{N_y} \right) \right], \\ \kappa_2^{(1w)}(\ell) &= \frac{k}{2\pi^2} X_L(\ell) + b + \dots, \end{aligned} \quad (\text{D9})$$

whereas two well-separated, oppositely oriented walls contribute

$$\kappa_2^{(2w)}(\ell) = \frac{k}{\pi^2} X_L(\ell) + b + \dots \quad (\text{D10})$$

The common correlation-spectrum derivation of the matching entropy coefficients and the $\pi^2/3$ relation is given in Appendix E. The charged primaries obey $h_q = q^2/(2k)$, giving Eq. (66) at $k = 1$ for the fixed-word operator spectrum. The analysis must declare whether a subsystem contains one or two walls before converting a fit coefficient to k or c_{edge} . The coefficient k identifies the current algebra; by itself it does not determine chirality, which is tested only by Campaigns H1–H3.

Signed feedback-transfer FCS is a different classical observable. If $Y_e \in \{-1, 0, 1\}$ is the system-layer charge transfer at event e , save

$$\begin{aligned} Q_{\text{rec}}(R, T) &= \sum_{e \in R \times T} Y_e, \\ \theta_R(s, T) &= \frac{1}{T} \log \mathbb{E}_\xi[e^{-s Q_{\text{rec}}}], \end{aligned} \quad (\text{D11})$$

The empirical finite-time θ_R is not called a tilted-channel eigenvalue. Use log-sum-exp stabilization, trajectory bootstrap intervals, and a reweighting effective sample-size flag. Record FCS and quantum subsystem FCS may be compared only after their distinct definitions are displayed.

Appendix E: Common correlation-spectrum origin of entropy and charge fluctuations

This appendix fixes the exact Gaussian relation between interval entanglement and quantum charge fluctuations and then states the stronger CFT test used in Campaign S1. It applies to a number-conserving Gaussian reduced state whether or not the reduced density matrix is pure. It does not identify the within-trajectory quantum variance with classical trajectory-to-trajectory charge wandering.

1. Spectral calculus of the restricted covariance

Let

$$C_{\xi,A} u_{\xi a} = \nu_{\xi a} u_{\xi a}, \quad 0 \leq \nu_{\xi a} \leq 1, \quad (\text{E1})$$

and define the positive charge-susceptibility operator

$$\mathcal{V}_{\xi,A} = C_{\xi,A}(\mathbf{1}_A - C_{\xi,A}), \quad q_{\xi a} = \nu_{\xi a}(1 - \nu_{\xi a}) \in [0, 1/4]. \quad (\text{E2})$$

The interval quantum variance and von Neumann entropy are

$$F_{\xi,A}^q = \text{Tr } \mathcal{V}_{\xi,A} = \sum_a q_{\xi a}, \quad (\text{E3})$$

$$\begin{aligned} S_{\xi,A} &= -\text{Tr}[C_{\xi,A} \log C_{\xi,A} + (\mathbf{1}_A - C_{\xi,A}) \log(\mathbf{1}_A - C_{\xi,A})] \\ &= \sum_a h(\nu_{\xi a}) = \text{Tr } g(\mathcal{V}_{\xi,A}), \end{aligned} \quad (\text{E4})$$

where

$$\begin{aligned} h(\nu) &= -\nu \log \nu - (1 - \nu) \log(1 - \nu), \\ g(q) &= h\left(\frac{1 + \sqrt{1 - 4q}}{2}\right). \end{aligned} \quad (\text{E5})$$

The branch of the square root is immaterial because $h(\nu) = h(1 - \nu)$. More generally,

$$\begin{aligned} S_{\xi,A}^{(n)} &= \text{Tr } g_n(\mathcal{V}_{\xi,A}), \\ g_n(q) &= \frac{1}{1 - n} \log[\nu(q)^n + (1 - \nu(q))^n], \end{aligned} \quad (\text{E6})$$

with $\nu(q) = (1 + \sqrt{1 - 4q})/2$. Thus entropy and variance are different spectral functionals of the same operator. The scalar $F_{\xi,A}^q$ alone does not determine $S_{\xi,A}$: distinct multisets $\{q_{\xi a}\}$ can have the same first moment. The full spectrum of $\mathcal{V}_{\xi,A}$ does determine every Gaussian Rényi entropy.

The corresponding single-particle modular Hamiltonian is

$$h_{\xi,A}^{\text{mod}} = \log[(\mathbf{1}_A - C_{\xi,A})C_{\xi,A}^{-1}], \quad \nu_{\xi a} = \frac{1}{1 + e^{\epsilon_{\xi a}}}, \quad (\text{E7})$$

with zero and unit occupations understood by continuity. In modular-energy variables,

$$q(\epsilon) = \frac{1}{4 \cosh^2(\epsilon/2)}, \quad (\text{E8})$$

$$s(\epsilon) = \log\left[2 \cosh\left(\frac{\epsilon}{2}\right)\right] - \frac{\epsilon}{2} \tanh\left(\frac{\epsilon}{2}\right). \quad (\text{E9})$$

If $\rho_{\xi,A}(\epsilon) = \sum_a \delta(\epsilon - \epsilon_{\xi a})$, then

$$F_{\xi,A}^q = \int d\epsilon \rho_{\xi,A}(\epsilon) q(\epsilon), \quad S_{\xi,A} = \int d\epsilon \rho_{\xi,A}(\epsilon) s(\epsilon). \quad (\text{E10})$$

Both kernels select entanglement levels with $|\epsilon| = O(1)$, and

$$\int_{-\infty}^{\infty} d\epsilon q(\epsilon) = 1, \quad \int_{-\infty}^{\infty} d\epsilon s(\epsilon) = \frac{\pi^2}{3}. \quad (\text{E11})$$

Consequently a common slowly varying, logarithmically growing low-modular-energy density produces $S = (\pi^2/3)F^q + O(1)$. Equation (E11) explains the spectral origin of the coefficient but is not an exact finite-size equality for an arbitrary covariance spectrum.

2. CFT coefficient and level-one test

For the interval convention of Appendix D, one chiral wall contributes

$$S_n^{(1w)}(\ell) = \frac{c}{12} \left(1 + \frac{1}{n}\right) X_L(\ell) + b_{S_n} + \dots, \quad (\text{E12})$$

$$F_q^{(1w)}(\ell) = \frac{k}{2\pi^2} X_L(\ell) + b_F + \dots, \quad (\text{E13})$$

whereas two well-separated oppositely oriented walls contribute twice these logarithmic coefficients,

$$S_n^{(2w)}(\ell) = \frac{c}{6} \left(1 + \frac{1}{n}\right) X_L(\ell) + b'_{S_n} + \dots, \quad (\text{E14})$$

$$F_q^{(2w)}(\ell) = \frac{k}{\pi^2} X_L(\ell) + b'_F + \dots. \quad (\text{E15})$$

The geometry factor cancels in the ratio. Denoting the corresponding log-chord slopes by a_{S_n} and a_F gives

$$\frac{a_{S_n}}{a_F} = \frac{\pi^2 c}{6k} \left(1 + \frac{1}{n}\right), \quad \frac{a_S}{a_F} = \frac{\pi^2 c}{3k}. \quad (\text{E16})$$

For a charge-one chiral complex fermion, $c = k = 1$, and therefore

$$a_S - \frac{\pi^2}{3} a_F = 0, \quad a_{S_n} - \frac{\pi^2}{6} \left(1 + \frac{1}{n}\right) a_F = 0. \quad (\text{E17})$$

Equivalently,

$$S(\ell) - \frac{\pi^2}{3} F^q(\ell) = \text{constant} + \text{finite-size corrections} \quad (\text{E18})$$

has no logarithmic term. This is a leading-scaling statement, not a claim that the two spectral kernels agree mode by mode.

Under a unitary chiral-CFT interpretation and a fixed microscopic normalization of charge, the $U(1)$ current generates a $c = 1$ Sugawara sector. The combination $c \simeq 1$, $k \simeq 1$, Eq. (E17), and an independent H1–H3 chirality test supports a minimal $U(1)_1$ charged wall without an additional neutral gapless sector. Entropy and variance alone do not determine the sign of the chiral central charge or the direction of propagation.

3. Cross-boundary coherence and intrinsic mixedness

For a pure Slater trajectory the full correlation matrix is a projector. With the spatial block decomposition

$$C_\xi = \begin{pmatrix} C_{\xi,A} & C_{\xi,AB} \\ C_{\xi,BA} & C_{\xi,B} \end{pmatrix}, \quad C_\xi^2 = C_\xi, \quad (\text{E19})$$

the AA block of the projector identity gives

$$\mathcal{V}_{\xi,A} = C_{\xi,A}(\mathbf{1}_A - C_{\xi,A}) = C_{\xi,AB}C_{\xi,BA}. \quad (\text{E20})$$

Hence the $q_{\xi a}$ are the squared singular values of the cross-boundary correlation block,

$$F_{\xi,A}^q = \|C_{\xi,AB}\|_F^2, \quad S_{\xi,A} = \text{Tr } g(C_{\xi,AB}C_{\xi,BA}). \quad (\text{E21})$$

The variance is the total squared cross-boundary coherence, whereas the entropy retains the complete singular-value distribution.

For a globally mixed Gaussian state, $0 \leq C_\xi \leq \mathbf{1}$ but $C_\xi^2 \neq C_\xi$, and the exact decomposition becomes

$$\mathcal{V}_{\xi,A} = C_{\xi,AB}C_{\xi,BA} + (C_\xi - C_\xi^2)_{AA}. \quad (\text{E22})$$

Both terms are positive. The first measures cross-boundary coherence and the second is intrinsic mixedness already present in the global state. Their traces add in the charge variance, but the entropy is not generally the sum of separate functions of the two terms because they need not commute. Equation (E22) is the appropriate diagnostic during max-mix purification; Eq. (E20) is used only after the pure-trajectory gate passes.

The shared natural orbitals also define matched real-space contours,

$$\begin{aligned} f_{\xi,A}(i) &= \sum_a |u_{\xi a}(i)|^2 q_{\xi a}, \\ s_{\xi,A}(i) &= \sum_a |u_{\xi a}(i)|^2 h(\nu_{\xi a}), \end{aligned} \quad (\text{E23})$$

with the exact sum rules

$$\sum_{i \in A} f_{\xi,A}(i) = F_{\xi,A}^q, \quad \sum_{i \in A} s_{\xi,A}(i) = S_{\xi,A}. \quad (\text{E24})$$

Comparing these unnormalized contours tests whether the same wall-localized natural orbitals carry both logarithms. Each contour is averaged over trajectories only after its per-trajectory construction.

4. Full charge statistics and ensemble decomposition

Writing $z = e^{i\lambda}$, the interval FCS polynomial factorizes as

$$\chi_{\xi,A}(z) = \det[\mathbf{1}_A - C_{\xi,A} + zC_{\xi,A}] = \prod_a (1 - \nu_{\xi a} + z\nu_{\xi a}). \quad (\text{E25})$$

Its zeros obey

$$z_{\xi a} = -\frac{1 - \nu_{\xi a}}{\nu_{\xi a}} = -e^{\epsilon_{\xi a}}. \quad (\text{E26})$$

Thus the exact full interval-charge distribution, whose probabilities are the coefficients of $\chi_{\xi,A}(z)$, determines the single-particle entanglement spectrum up to permutation

and hence every Gaussian entropy. Numerical inversion of high-degree noisy counting polynomials is ill conditioned and is not a production gate; direct covariance eigenvalues remain the primary simulated estimator.

Finally, if one first samples a trajectory and then measures its interval charge, the law of total variance is

$$\text{Var}_{\text{total}}(N_A) = \mathbb{E}_\xi[\text{Tr } \mathcal{V}_{\xi,A}] + \text{Var}_\xi[\text{Tr } C_{\xi,A}]. \quad (\text{E27})$$

Only the first term is the quantum interval variance governed by the spectrum in Eq. (E2) and used to infer k . Boundary localization of the second term is meaningful dynamical information about the trajectory ensemble, but it is not substituted for the current-algebra coefficient.

5. Production estimator and acceptance rule

Campaign S1 computes $S_{\xi,A}$, $F_{\xi,A}^q$, their contours, and the $q_{\xi a}$ spectrum from the same restricted covariance for every declared interval. Origin and stationary-snapshot averages are internal to a trajectory and are performed identically for both observables. The common linear fit then produces $m_{1,\xi}$, f_ξ , and the paired difference $\delta_{SF,\xi}$ in Eq. (39). Whole trajectories are resampled jointly, preserving the covariance between the two slopes and among all interval lengths.

The primary test is the thermodynamic extrapolation of $\delta_{SF}^{\text{Born}}$ and the equivalent residual curve in Eq. (40). The ratio a_S/a_F and the Rényi relations in Eq. (41) are secondary. Repeat the complete paired analysis after each preregistered fit-window deletion and on the exact-Hamiltonian, matched trivial, and one-versus-two-wall controls. A nonzero residual logarithmic coefficient forbids the minimal $U(1)_1$ identification even if entropy and variance separately admit logarithmic fits. It does not erase the separate estimates of c and k , and it says nothing by itself about chirality.

Legacy comparison. The canonical pure-state covariance archive under `00_WORKSPACE/COLAB/colab_small_system_testing/gpu_data/pure_state_covariance_snapshots/` does contain the information needed for matched full-width interval entropy and within-trajectory quantum charge variance. A direct cycle-50 reduction at $N_x = 16$, $N_y = 30, 40$, $n_{\text{shell}} = 1, 2$, and $S = 10$ gives $c_{\text{eff}} = 1.032\text{--}1.086$ and $k_{\text{eff}} = 1.033\text{--}1.085$, with the four matched values tracking one another. This is a genuine legacy point in the leading $\pi^2/3$ relation, but it is not the production test: only two sizes and ten trajectories are present, the Rényi slopes and one-wall charge decomposition are absent, and no thermodynamic residual extrapolation can be performed. The max-mix campaign's total charge variance during purification remains a different observable.

Appendix F: Circuit depth, Lu bound, wall hybridization, and fixed duration

For a product input and a depth- D local circuit, the rank-entropy growth across region A obeys the Lu–Lessa–Kim–Hsieh bound [2]

$$S_0(A, T) - S_0(A, 0) \lesssim D |\partial A|. \quad (\text{F1})$$

For the full-width strip $A = [0, N_x] \times [0, \ell]$, $|\partial A| \sim 2N_x$. A two-wall CFT contribution therefore requires parametrically

$$\frac{1}{3} \log N_y \lesssim \kappa D N_x. \quad (\text{F2})$$

Here D counts layers of mutually nonoverlapping local maps, not circuit sweeps. The production random-serial schedule visits overlapping orbitals sequentially, so one cycle has a size-dependent implementation depth. For $n_{\text{shell}} = 1$, an update centered at $u = (x, y)$ has the periodic 3×3 cell support

$$\mathcal{S}_u = \{u + (\delta_x, \delta_y) : \delta_x, \delta_y \in \{-1, 0, 1\}\}. \quad (\text{F3})$$

The conflict graph joins u and v when $\mathcal{S}_u \cap \mathcal{S}_v \neq \emptyset$, equivalently when their cyclic separations obey

$$\begin{aligned} d_{N_x}(x, x') &\leq 2, & d_{N_y}(y, y') &\leq 2, \\ d_L(a, b) &= \min_{m \in \mathbb{Z}} |a - b + mL|. \end{aligned} \quad (\text{F4})$$

It has maximum degree at most 24 and hence at most 25 colors by a greedy bound. If $q_x \mid N_x$, $q_y \mid N_y$, and $q_x, q_y \geq 3$, the explicit residue coloring

$$c(x, y) = (x \bmod q_x, y \bmod q_y) \quad (\text{F5})$$

uses $q_x q_y$ colors and is proper: distinct equal-color centers have separation at least three in one coordinate. A commensurate 3×3 coloring therefore uses nine colors. For every size in Eq. (4), the conservative 4×4 coloring wraps cleanly and gives $\chi_{\text{conf}} \leq 16$.

Let $d_{\text{inst}} = O(1)$ denote the physical layer depth of one occupation-measurement-feedforward-fresh-ancilla-fSWAP instrument block. Keeping the four controller channels as separate blocks gives

$$d_{\text{cyc}} \leq 4\chi_{\text{conf}} d_{\text{inst}} \leq 64d_{\text{inst}}, \quad D = d_{\text{cyc}} T_{\text{cyc}}. \quad (\text{F6})$$

This is an $O(1)$ bound on the physical depth of one colored sweep, not on the total preparation depth. The BPJ bulk can converge in $O(1)$ sweeps while the additional critical wall requires a cycle count that grows with N_y ; the two statements concern different target entanglement structures. Equation (F2) gives only

$$T_{\text{cyc}} \gtrsim \frac{\log N_y}{3\kappa d_{\text{cyc}} N_x}. \quad (\text{F7})$$

The unknown coefficient makes this too weak to choose 10 versus 50 versus $2N_y$ cycles. The production duration

is therefore a declared numerical choice rather than a consequence of this bound.

This coloring gives an optional quantum-simulation compilation. Within a color, disjoint local instruments commute, so simultaneous hardware readout and sequential numerical evaluation define the same joint Born instrument even though its outcomes need not be statistically independent. Reordering different color blocks is not a parallel reinterpretation of a saved random-serial trajectory because overlapping conditional maps generally do not commute. The production campaign therefore makes no constant-depth-per-sweep claim for its random-serial word; it reports Eq. (F6) only as a realizable resource bound for the corresponding color-block hardware schedule.

The circuit-specific operator-Schmidt-rank bounds are

$$r(a_\chi), r(a_\chi^\dagger) \leq 2, \quad r(n_\chi), r(\mathbf{1} - n_\chi) \leq 4. \quad (\text{F8})$$

Along record ξ , a sharper bookkeeping bound is

$$\Delta S_0^\xi \leq 2 \log 2 N_{=}^\partial(\xi) + \log 2 N_{\neq}^\partial(\xi), \quad (\text{F9})$$

with the two counts distinguished according to the boundary-crossing branch type. These bounds assume an initially product state. A random Slater determinant is already entangled and cannot be used to validate a product-input depth bound. The maximally mixed state is treated separately as a purification problem beginning at $t = 0$.

Finite transverse width produces an independent infrared scale. If the opposing-wall hybridization behaves as

$$m(w) \sim m_0 e^{-w/\xi_\perp}, \quad (\text{F10})$$

then one requires

$$m(w) \ll \frac{2\pi v}{N_y}, \quad w \gtrsim \xi_\perp \log N_y \quad (\text{F11})$$

up to nonuniversal constants. B0 determines $\xi_\perp$ and W1 verifies the same condition in the stochastic circuit.

Every production trajectory begins from its own independently sampled declared initial state and runs for exactly $2N_y$ complete cycles. There is no separate physical burn-in, no shared equilibrated parent state, and no cloning of one evolved covariance into 100 continuations. Time-resolved bulk markers, entropy coefficients, charge statistics, wall profiles, purity diagnostics, and successive-covariance changes are retained as transient checks inside this fixed window.

The three pure-state checkpoints used by S1 are $t = N_y, 3N_y/2, 2N_y$. Agreement among them is a sensitivity diagnostic, not a device for converting cycle snapshots into independent samples. If the late checkpoints are incompatible beyond the preregistered tolerance, the relevant stationarity or temporal gate fails. Production is not silently extended beyond $2N_y$; a longer campaign requires a versioned amendment and a new output directory.

Max-mix purification likewise starts at $C = 1/2$ at $t = 0$ and retains its complete $2N_y$ clock. T1 initializes its physical trajectory and tangent frame at $t = 0$, uses the first N_y cycles for tangent-frame alignment only, resets only the QR log accumulator, and accumulates exponents over the final N_y cycles. The physical trajectory is never reset and the alignment interval is not called burn-in. This procedure is tailored to the production serial implementation and must be re-evaluated if the actual overlap-conflict schedule changes.

Appendix G: Campaign-card checklist

Before a campaign is marked complete, its final note must answer all of the following:

1. Was the canonical GPU entry point recorded?
2. Were all 25 trajectories independently initialized and run for the declared total duration without cloning an evolved state?
3. Were max-mix and tangent clocks treated according

to Appendix F?

4. Can every plotted scalar be reconstructed from per-trajectory raw data?
5. Is the order of nonlinear evaluation, time/source averaging, and trajectory averaging explicit?
6. Are fit windows, excluded points, and correction models preregistered and saved?
7. Do paired controls share declared random streams without being called the same record?
8. Did every shard pass manifest, seed, replay/restart, and numerical-residual checks?
9. Does the caption state sample count, independent unit, uncertainty, and geometry?
10. If the gate failed, was the prescribed downgrade used without changing the estimator?

[1] A. Bhuiyan, H. Pan, and C.-M. Jian, [Physical Review Research](#) **8**, 023147 (2026), [arXiv:2507.13437 \[quant-ph\]](#).

[2] T.-C. Lu, L. A. Lessa, I. H. Kim, and T. H. Hsieh, [PRX Quantum](#) **3**, 040337 (2022).